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Academic English Readings

Advanced B

The English Unit

2026

כל הזכויות שמורות למכללה האקדמית ע"ש סמי שמעון.
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Busting the Boom

A new type of aircraft will break the sound barrier without battering your eardrums. Has the supersonic dream finally come true?

Paragraph 1 - Anyone who is among the super-rich these days has a personal jet parked at the local airport. Money can't buy you love but it can get you a much smaller plane. A piece of advice, though, to anyone thinking of mortgaging their penthouse for one of those status symbols: wait a while. In the labs of Boeing and Lockheed, engineers and designers are drawing up blueprints for the next generation of luxury jets. And for "luxury," read speed. Supersonic executive jets are about to speed people from London to Rome in just 30 minutes and to Hong Kong in time for lunch after a morning start.

Paragraph 2 - What was wrong with Concorde planes?, you might ask. Concorde is fine for shooting across the Atlantic, but you can forget about flying across continents in it or around the Pacific Rim – it was too noisy and didn't have the range. Moreover, powerful jet engines of this type release vast amounts of nitrogen oxides, which wreak havoc* with the ozone layer at the high altitudes at which such plane must cruise. Finally, the noise produced by a Concorde could be heard at a great distance from its actual path. For every 300 meters of altitude, a supersonic jet's shock waves spread to about 1 kilometer either side of the flight path at ground level. This shock can often show up on seismographs up to 5000 kilometers away.

Paragraph 3 - The sonic boom that Concorde created when it punched through the sound barrier meant that it was banned from supersonic flight over land. Its sonic booms exceed almost all possible regulations - they could even shatter glass or damage buildings. Sonic booms result from a sudden change in air pressure as a supersonic craft passes overhead. The aircraft is travelling so fast that the sound waves it produces pile up in front to form shock waves. Each aircraft forms two conical shock waves that spread out behind it: one starting at the nose and one at the tail. These are called a bow shock and a tail shock waves.

Paragraph 4 - Today's supersonic aircraft engineers believe they have come up with a cost-effective way not only to make jet engines more efficient, but also to solve the problem of noise during the supersonic flight. They have found all kinds of designs to achieve their aim. Back in the 1970s, Richard Seebass at the University of Colorado calculated that simply blunting the aircraft's nose would virtually eliminate the boom. A blunt nose would raise the air-pressure ahead of the aircraft, increasing air temperature and stretching the bow shock out in front of the aircraft. Then in 1998 they found a better way to achieve this effort. It involved a flat flap that extends forwards from the aircraft's nose like the drop nose on Concorde. When this flap tilts upwards a few degrees, it acts just like Seebass's blunt nose, stretching the bow shock wave and reducing its peak pressure.

Paragraph 5 - This design also creates lift, helping to cut fuel consumption. Over the ocean, where noise isn't a problem, it can be flicked down to its "off" position. The tail, too, would be equipped with a flap to soften the tail shock wave.

Paragraph 6 - One of the most radical - if speculative - proposals comes from H.K.Cheng, a retired engineer formerly at the University of California. "With a laser you could probably heat up the gas in certain parts of the shock wave," he says. Just like the blunt nose, it would reduce pressure ahead of the craft.

Paragraph 7 - To conclude, while shorter journeys may help eliminate the dangers of deep vein thrombosis and even calm potential air ragers, there's one thing to bear in mind before you head for check-in. These airliners might carry you twice the speed of sound, but they will never outrun those interminable delays that keep your feet stuck firmly in the departure lounge.

* - wreak havoc – cause destruction

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Busting the Boom - Questions

Answer the following question in English:

1. Give three reasons why Concorde planes are problematic.

Check [v] TRUE or FALSE for each statement. Only if a statement is FALSE then:

- a. Copy the incorrect word and correct the false statements. (use one word only)
- b. Do not use the word 'NOT' as a correcting word.
- c. If both statements are marked as TRUE, none will be accepted.

1. The higher the plane is, the further from the source the shock waves can be heard at ground level.

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

2. The thing that Seebass's blunt nose and flat flap have in common is that they both act by contracting the bow shock wave in front of the aircraft.

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

Check [v] the best answer, according to the information in the text.

1. We can understand from the subheadings that
 - a. a deafening noise is an insurmountable barrier
 - b. the supersonic dream has reached its end
 - c. the sonic boom will cease to be a problem
 - d. it is true that some supersonic dreams are prophetic
2. Which sentence is LEAST relevant to paragraph 1?
 - a. A personal jet is considered a status symbol
 - b. Small planes more than compensate for the loss of affection
 - c. It would be unwise to rush into buying a personal jet
 - d. The super-rich are finding new ways to set themselves apart
3. The author advises the reader to "wait a while" (par. 1) because
 - a. planes will be much faster in the future
 - b. money can't buy you love
 - c. planes will be much smaller in the future
 - d. mortgaging a penthouse takes time

4. Concord's speed is limited over land because
 - a. its engines produce nitrogen oxides which interfere with the ozone layer
 - b. for every 300 meters of altitude the shock waves spread out 1 km at ground level
 - c. the shock waves do not show up on seismographs
 - d. the sonic boom exceeds allowable regulations
5. The reason the sound waves pile up in front is because
 - a. the aircraft is traveling very fast
 - b. the shock waves are conical
 - c. the air pressure changes
 - d. the supersonic craft passes overhead
6. "... this effort" (par. 4) refers to
 - a. a blunt nose
 - b. a flat flap
 - c. eliminating the boom
 - d. increasing the air temperature at the back of the aircraft
7. We can infer from paragraph 4 that
 - a. the sonic boom was the bone of contention throughout the years
 - b. engineers were particularly engaged in reducing the sonic boom
 - c. the sonic boom has helped engineers take a backward step in planning aeronautical designs
 - d. engineers have overlooked the significance of the sonic boom
8. In paragraphs 4 and 5 the author is primarily interested in discussing
 - a. the effects of the aircraft's blunt nose design on today's supersonic engineers
 - b. how the reduction of peak pressure causes the flap to move upwards
 - c. the contribution of calculations to the design tricks and temperature fluctuations
 - d. the relationship between the aircraft's nose design and air-pressure
9. We can infer that the plane's nose can be flicked down to its "off" position over the ocean because (par. 5)
 - a. a sonic boom does not occur over the ocean
 - b. the noise regulations that apply over land do not apply over the ocean
 - c. it will soften the tail shock wave
 - d. the flat flap acts just like Seebass's blunt nose

10. Which of the following statements is incorrect?
- a. One of the differences between the 1998 design and Seebass's design is that the former can soften the shock wave both at the front and at the back of the plane
 - b. Both the 1998 design and the blunt nose design enhance cost-effectiveness
 - c. No matter how fast your jet plane flies, your flight will inevitably be subject to delay
 - d. The intensity of the sonic boom increases when the bow and tail shock are stretched

Check the best answer, according to the text.

- 1. The word "mortgaging" (par.1, line 3) is used as
 - a. a noun
 - b. a verb
 - c. an adjective
 - d. an adverb
- 2. The word "morning" (par.1, line 7) is used as
 - a. a noun
 - b. an adjective
 - c. an adverb
 - d. a verb
- 3. The word "it" (par.3, line 1) refers to
 - a. sonic boom
 - b. sound barrier
 - c. shock
 - d. Concorde
- 4. The word "its" (par.4, line 10) refers to
 - a. Seebass's blunt nose
 - b. the drop nose on Concorde
 - c. the bow shock wave
 - d. peak pressure
- 5. The connector "if" (par.6, line 1) indicates a relationship of
 - a. contrast
 - b. comparison
 - c. condition
 - d. exemplification

6. We can infer from the last paragraph that deep vein thrombosis may be caused by
- a. potential air ragers
 - b. long distance flights
 - c. shorter journeys
 - d. check-in
7. The word "outrun" (par.7, line 4) means
- a. run faster
 - b. surpass
 - c. fail
 - d. run away

How Bridges Fail

Paragraph 1 - In physical terms, bridges deteriorate because of weather and traffic. Water corrodes steel and can scour away bridge foundations. Meanwhile every car and truck that passes over a bridge causes it to flex. Excessive loads can cause cracks that destroy the structure's integrity.

Paragraph 2 - Although engineers agree on the general mechanisms of bridge failure, the details are not well understood. Bridge structures are too complex for complete computer analysis and so simulations require a host of simplifying assumptions. Consequently, the results seldom match behavior in the field.

Paragraph 3 - Once a bridge has begun to deteriorate, the process of decay accelerates. The portions of metal beams that are under the most stress corrode more rapidly and stress concentrations increase as the thickness of sound metal decreases. Similarly, damaged structural members have reduced load-bearing capacity and are thus more vulnerable to the effects of heavy traffic.

Paragraph 4 - Simple problems, if allowed to progress unchecked, can lead to severe damage. Debris on a bridge deck, for example, may block drains, causing water to accumulate. In cold regions, water may freeze inside the deck, cracking it. Deicing compounds, however, form salt solutions that rapidly corrode reinforcing bars and other structural members. When the salt content reaches a critical level, the concrete must be replaced even though the cement and aggregate are still sound.

Paragraph 5 - Draining the water poses its own problems. Engineers have learned at great expense to direct water away from structural elements and bearing surfaces, where the combination of salt and stress can destabilize a bridge in a few years. Yet complex drainage systems are expensive and require periodic maintenance to keep them from clogging.

Paragraph 6 - Bridge substructures face problems similar to those of the superstructure and deck. In addition, they are often much more difficult to inspect and to repair. The Schoharie Creek Bridge in upstate New York, for example, collapsed in 1987 because flowing water had scoured away its foundation. Since then, states have begun employing

scuba divers to inspect waterways—often in murky water where they must work entirely by feel. If a diver finds damage, workers must erect a cofferdam to make repairs.

Paragraph 7 - One failure mechanism over which engineers have little control is accident. According to Issam E. Harik and his colleagues at the University of Kentucky, collisions caused 42 of 79 nationally reported bridge failures between 1951 and 1988. Almost half of the collisions involved ships that rammed bridge supports; the Huey R Long Bridge near New Orleans suffered two strikes in five years, as did the Sunshine Skyway Bridge near Tampa.

Paragraph 8 - Finally, there are the collapses that defy statistical analysis—such as the failure of the Buckman Bridge near Jacksonville, Fla., in the spring of 1970. Engineers filled the bridge's hollow pilings with river water before sealing them. Anaerobic bacteria feasted on the cardboard lining of the pilings and generated enough methane to rupture the pilings and bring down part of the bridge.

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How Bridges Fail - Questions

I. Answer in English.

- 1) What measure has been taken following the collapse of Schoharie Creek Bridge?

II. True or False? Correct the False statements

- 1) Damaged structural members are more vulnerable to heavy traffic.

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

- 2) Salt solutions resulting from deicing compounds slow down the corrosion of reinforcing bars.

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

III. Circle the best answer according to the text.

- 1) The main idea of paragraph 1 is that

- a) traffic leads to an accumulation of corrosion on bridge foundations
- b) flexing enables the bridge to bear excessive loads
- c) the combination of water and traffic damage bridge integrity
- d) water can scour away bridge foundations

- 2) Paragraph 2 makes the point that

- a) bridge structures should be simplified
- b) engineers have learned how to successfully handle bridge failure
- c) computer analysis of bridge structure is very advanced
- d) engineers lack expertise in the details of bridge failure

- 3) The process of bridge decay accelerates as

- a) the thickness of sound metal results in stress
- b) metal becomes thinner and thus there is more stress on it
- c) stress concentration increases in parts where the metal is thick
- d) load-bearing capacity damages structural members

- 4) The "debris" detail illustrates
 - a) the blockage of drains
 - b) the accumulation of water
 - c) the deterioration of a simple problem
 - d) the damage caused by water

- 5) We can infer that deicing compounds
 - a) are meant to protect the bridge deck but fail to do so
 - b) cause cracks in the bridge decks
 - c) stabilize the bridge's structure over the years
 - d) are used to maintain stable salt levels

- 6) What is the main idea of paragraph 5?
 - a) draining water is a complex problem
 - b) systems for draining water are expensive
 - c) engineers now know how to drain water
 - d) directing water away has become an affordable process

- 7) The superstructure is the part of the bridge that is
 - a) submerged in water
 - b) the most difficult to repair
 - c) similar to the substructure
 - d) the least problematic

- 8) What is the main problem introduced by paragraph 6?
 - a) A problem similar to those of the superstructure and deck
 - b) The difficulties involved in inspecting and repairing the superstructure and deck
 - c) The difficulty imposed by the need to carry out inspection and repair under water
 - d) The Schoharie bridge collapse

- 9) Paragraphs 7 and 8 present
 - a) solutions for preventing bridge failures
 - b) a statistical analysis of collisions into bridges
 - c) examples of accidents that caused bridge failures
 - d) different reasons for the collapse of bridges

- 10) We can understand from the last paragraph that
 - a) sometimes bridges can collapse without a reason
 - b) many bridge collapses are caused by anaerobic bacteria
 - c) the collapse of Buckman Bridge was caused by something unexpected
 - d) filling the bridge's hollow pilings with river water protected the bridge

IV. Circle the best answer according to the text.

- 1) What is the main relationship in paragraph 2?
 - a) cause and effect
 - b) contrast
 - c) comparison
 - d) illustration

- 2) The word "form" in paragraph 4, line 4 is used as
 - a) a noun
 - b) an adjective
 - c) a verb
 - d) an adverb

- 3) The word "sound" in paragraph 4, line 6 means
 - a) loud
 - b) undamaged
 - c) corroded
 - d) voice

- 4) The word "those" in paragraph 6, line 1, refers to
 - a) substructures
 - b) superstructure and deck
 - c) bridges
 - d) problems

- 5) The word "since" in paragraph 6, line 4 indicates a relationship of
 - a) contrast
 - b) time
 - c) cause and effect
 - d) comparison

- 6) The word "did" in paragraph 7, line 5 refers to
 - a) suffered
 - b) rammed
 - c) involved
 - d) collided

- 7) The word "lining" (paragraph 8, line 4) is used as
 - a) an adjective
 - b) an adverb
 - c) a verb
 - d) a noun

The Leaning Tower of Pisa

Paragraph 1 - The leaning tower has never really been straight. Not long after work began in 1173, the foundation settled unevenly and the tower started inclining toward north, when construction continued, after a nearly 100-year hiatus, the building shifted again, so that by 1272, it was visibly leaning south. Today the top of the tower is 5.227 meters off-center, tilting southward.

Paragraph 2 - Throughout the monument's history, architects and engineers have attempted to halt the lean, but since regular monitoring began in 1911, the offset at the top of the tower has increased at a fairly consistent rate of about 1.2 millimeters each year. Fears about the safety of the landmark became acute when a similarly constructed bell tower at the Cathedral of Pavia collapsed suddenly in 1989. Shortly thereafter, the tower at Pisa was closed to visitors.

Paragraph 3 - In 1990, a special commission, composed of Italian and foreign experts in the fields of structural engineering, geotechnical engineering, history of art and restoration of monuments, was brought together by the Italian government to determine new ways to save the tower. The group I work with, the Consorzio Progetto Torre di Pisa (the Tower of Pisa Project Consortium), has supervised several projects that have stabilized the structure and slowed the rate of its incline.

Paragraph 4 - Initial efforts focused on the exterior of the tower, but in the next few months, we plan to try other, more radical techniques to halt the tower's lean. These methods will be applied directly to the soil modifying the monument's footing. Large-scale field trials are now under way at the Piazza dei Miracoli (Miracle Square), where the tower stands, but all work is being done far from the monument itself, because we must guard against the possibility that altering the ground too close to the tower could eventually damage the building.

Paragraph 5 - Our ultimate goal is not to straighten the tower. Because the structure tilted in different directions during the early stages of construction, it became curved—like a banana and will never stand truly upright. Instead, we hope to ease its top back some 10 or 20 centimeters. With luck, our efforts will keep the landmark standing into the next century, when a new generation of scientists will tackle the 800-year-old problem of the leaning tower of Pisa.

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The Leaning Tower of Pisa - Questions

Answer the following questions in English.

2. What has the monitoring of the tower shown since 1911?

3. What prompted the authorities to close the tower in 1989?

4. Why does the writer tell us that the field trials are being carried out far from the monument itself?

5. Why does the author not aim to straighten the tower?

6. From the context we can understand that "hiatus" means

7. The word "It" (par. 1, line 3) refers to
 - a. the 100-year hiatus
 - b. construction
 - c. the building
 - d. the top of the tower
8. Two positive things that the team of engineers has already achieved are
_____ and _____.
9. "These methods" (par. 4, line 3) refers to _____.
10. The author's hope is that in the _____, a new generation of
_____ will be able to stabilize the structure.

Find synonyms to these words.

1. tilting - _____
2. gap, break - _____
3. fell down - _____
4. to decide on - _____
5. first - _____
6. to bring to a stop - _____
7. altering - _____
8. final - _____
9. to move carefully into place - _____
10. circular shape - _____

The Electric Plastics Show

Paragraph 1 - The man from Disney World wanted to buy some immediately. His dancers wore clothes that lit up: they had to have 200-volt battery packs strapped to their backs and he wanted to lighten their load if he could. He was calling Frank Karasz, a materials scientist at the University of Massachusetts In Amherst, who had just announced that he had made a piece of plastic that could shine blue when only 6 volts was applied to it. Karasz had to explain that his light-emitting plastic, with a working life of only a few seconds, was nowhere near commercial quality. But while the blue polymer isn't ready for the market, the 60 phone calls Karasz received in a single week show that the market is ready for it.

Paragraph 2 - Producing a plastic that can emit light would have seemed outlandish just two or three years ago. Even the idea that plastics can conduct electricity comes as a surprise to most people. After all, plastics are familiar as insulating sheaths round wires and as the casings around all manner of electrical appliances. But in the mid-1970s, a group of scientists discovered that plastics do not have to be insulators, when they stumbled across a polymer that behaved as a semiconductor. This discovery opened up the prospect of materials that might combine the electrical properties of metals with the ease of processing and versatility that have made plastics a runaway success over the past half-century. These pioneers predicted that a world of plastic wires and batteries was just around the corner.

Paragraph 3 - The predictions have not yet come true. The plastics that conducted electricity were hard to fabricate and they were pretty poor conductors when compared with metals. But interest in these materials continued to grow. The past few years have seen the long-predicted technology creeping closer to reality, as scientists have learnt how to make plastics that can conduct electricity as well as metals. Along the way, researchers have learnt how to make these plastics emit light and the materials have been seized on by people who see in them a way of making plastic television screens and computer displays. There are still plenty of obstacles blocking the road that could, one day, lead to these applications, but chemists, physicists and engineers are daily learning much more about how the materials are made, the way they work and strategies for improving them.

Paragraph 4 - Like conducting plastics, the new light-emitting materials are conjugated polymers. A polymer is a giant molecule that is formed from hundreds or even thousands of smaller molecules called monomers. The archetypal polymer is polyethylene (polythene), which consists of long chains of carbon atoms, with two hydrogen atoms attached to each carbon. The molecules are held together by the sharing of the electrons that belong to their constituent atoms.

Paragraph 5 - A carbon atom has four electrons available for bonding and a hydrogen atom has one. When a carbon atom bonds to hydrogen, one of its electrons starts to do double duty by orbiting both atoms. The electron from the hydrogen atom also orbits both partners. The two shared electrons between them constitute a single covalent bond.

Paragraph 6 - In a polyethylene chain, each carbon atom shares one of its four electrons with each of the carbons next to it and one each with the two hydrogens attached to it. It is these bonds that hold the polymer together, but the electrons that form them are fixed. They cannot leave their prescribed orbits and because an electric current is nothing more than the free movement of electrons, polyethylene cannot conduct electricity.

Paragraph 7 - The key to making conducting polymers is to provide electrons which are not fixed in their positions. Remove one hydrogen from each carbon and you leave the atoms with an extra electron to play with. The extra electron will form a second bond with just one of the neighbouring carbons on the chain, creating a double bond with that atom, while the other neighbouring carbon atom remains attached by a single bond. The result is a sequence of alternating single and double bonds all the way down the carbon backbone. In this regular "conjugated" sequence, the extra electrons can be induced to move along the polymer chain, forming an electric current, which is why conjugated polymers conduct electricity.

Paragraph 8 - But the first such material, discovered by the pioneering group at the University of Pennsylvania in 1975, had an electrical resistance several million times as high as that of copper. Moreover, the material was brittle and difficult to process using standard production techniques, in which the polymer is dissolved and then spun to form fibres or cast into a thin film. But the pioneers persevered: Alan MacDiarmid, along with Hideki Shirakawa, now at the University of Tsukuba in Japan and Alan Heeger, now at the University of California in Santa Barbara, kept working with the new material until they discovered that by "doping" the polymers—adding a small quantity of another material—they could increase the conductivity.

Paragraph 9 - MacDiarmid likens doping a polymer to adding a single drop of red dye to a pale cake mix. "The whole cake becomes pink," he says. "The one drop of red food colouring is a tiny, minuscule volume of the whole cake, but it has enormous effect on one property, the property of colour. The red dye is now an intrinsic part of the cake, you cannot undye it."

Changing properties

Paragraph 10 - Similarly, it is possible to change a particular property of a conducting polymer by doping it. Through trial and error, the group discovered that soaking the polymer in a solution of iodine removes about 10 percent of the conducting electrons, leaving a polymer with some positive charges and negatively charged iodide ions lying alongside. With fewer electrons in the polymer, those that remain can move more freely—just as a few marbles will roll freely around a box, while marbles jam-packed into a full box will barely move at all.

Paragraph 11 - Doping raised polymer conductivity by a factor of a million or so, but the polymers were still nothing like as good as copper at conducting electricity. Two things are essential for good conductivity: having a lot of movable "carriers" in the form of electrons (or their opposite numbers, holes in an array of electrons) and making sure that they can travel through the polymer easily. The number of carriers did not pose a problem, because conducting polymers have thousands of times as many available electrons as familiar inorganic semiconductors such as gallium arsenide. On the other hand, there was plenty of room to improve mobility.

Paragraph 12 - Polymer chains typically coil up like a mass of tangled string. Electrons travel along the molecules more easily than they travel between the chains of molecules, so straightening out each skein provides the electrons with an easier path. To create such a path, the researchers focused on ways of increasing molecular order. They found that one simple way of doing this was to stretch the polymers out into a thin sheet, which tended to pull the molecules into alignment with each other.

Paragraph 13 - In 1992, MacDiarmid and his team discovered an additional strategy: with the skeins straightened out, packing them together as tightly as possible further increases conductivity. So they increased the polymer's crystallinity, because two molecules can be no closer than when they are part of a crystal. Clearly, a densely packed polymer would allow electrons to jump almost effortlessly from chain to chain. Such insights brought this polymer's conductivity up, though still behind copper's conductivity at 600,000 Siemens per centimetre.

Paragraph 14 - Then, about a year ago, MacDiarmid's group, in a single step, managed to improve the conductivity of their polymer a thousand-fold, bringing its conductivity into the range typical of metals—though still lagging behind copper by a factor of 10 000. MacDiarmid's trick was to change the solvent used to dissolve the polymer when forming it into a thin plastic sheet. Instead of chloroform he used fluorophenol, which worked almost as a "secondary dopant" to improve the polymer's conductivity. At the time, the precise mechanism was a mystery, but MacDiarmid has since come up with a theory. He thinks that the solvent coats the negative iodide ions created by the primary dopant, shielding the molecules of the polymer backbone from their influence. In the absence of the neutralising effect of the negative ions, the positive molecules begin to repel each other, though they remain bound together. Under these constraints, the only way they can move farther apart is by straightening themselves out.

Paragraph 15 - Despite their imperfections as conductors, conjugated polymers have from the outset inspired interest in possible commercial applications. MacDiarmid holds a patent for plastic rechargeable batteries and other people are studying the possibility of using the material to ward off static. Normally, the static charge that builds up on medical equipment is conducted away to earth as the user makes contact from time to time with a metal earthing plate. A conducting polymer shield could continuously drain away electric charge, maintaining a static-free environment.

Paragraph 16 - Electrical conductivity is not the only application of conjugated polymers. Certain materials can change their properties—such as transparency—when a voltage is

applied. A plastic that can be darkened electrically could be used on windows to keep heat out when the sun shines brightly on hot days and made transparent when the weather is colder. Medical scientists are also considering the polymers as a way of releasing controlled quantities of a drug. The idea would be to infuse a polymer with a particular drug and attach it to a patient's skin. Applying a small voltage would change the polymer's conductivity, releasing a precisely measured quantity of the drug.

Paragraph 17 - Despite this plethora of possibilities, most of the recent interest in conjugated polymers can be attributed to just one thing: the possibility that they could be used to manufacture light-emitting diodes (LEDs) out of plastic. A major driving force for this research is the dream of building ultra-slim screens, made entirely of plastics, for TVs and computers.

Paragraph 18 - The first steps on the road to this goal came in 1990, when a multidisciplinary group at the University of Cambridge worked out how a semiconducting polymer could be made to give off light. To make the plastic LED, the researchers sandwiched a 200-nanometre film of semiconducting polyphenylene vinylene (PPV), between a transparent positive electrode made of indium/tin oxide and a negative electrode made of aluminium. The negative electrode injects electrons into the PPV film, while the positive electrode pulls them out—a process equivalent to injecting positively charged holes. When an electron and a hole meet, they give off energy, which is transferred to the polymer chain. This extra energy kicks an electron into what chemists call an "excited" or high-energy state. It does not stay there long, however, but quickly releases its energy and returns to its original ground state.

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The Electric Plastics Show - Questions

Answer the questions in English.

1. Why did a Disney World man want to have glowing plastics?

2. The materials scientist was unprepared for the Disney World agent demand. Why?

Find the synonyms to the words given below (par. 2)

1. to make, produce light _____
2. far-fetched, unrealizable _____
3. coverings _____
4. equipment _____
5. came across, found by chance _____

Complete the sentences, according to par. 2.

1. Although it had been supposed that plastics were _____, it was discovered that certain _____ behaved as _____.
2. The advantages of electric plastics are that they both have _____ properties and are easy to _____.

True or False? Correct the false statements (use one word only).

1. In the past few years it has been revealed that plastic conductors were hard to produce.
☐ TRUE ☐ FALSE
a. Copy the incorrect word: _____
b. Write the correction: _____
2. The primary purpose of the scientists working on plastics was to make them light-emitting and the secondary aim was to reach high conductivity in them.
☐ TRUE ☐ FALSE
a. Copy the incorrect word: _____
b. Write the correction: _____

3. The way to using plastics in TV and computer screens is long and full of barriers.

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

4. In polymers, the same electrons belong to several atoms, thus holding the molecule together.

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

Choose the best answer (par.6)

1. Polymers are non-conductors usually because

- a. they don't leave their orbits
- b. the molecules stick too closely together
- c. there is no supply of free electrons in them
- d. hydrogen and carbon molecules share some electrons

Complete these sentences in English.

1. The plastic invented by Frank Karasz could _____.

2. The word "although" could be used to replace the word _____ in paragraph 1.

3. Plastics are usually known as materials that do not _____ or _____.

4. The fact that plastics can be used in a variety of ways is expressed by the word _____ in paragraph 2.

5. Polythene cannot normally conduct electricity, because its _____ cannot _____.

6. The bonds of a "conjugated" sequence alternate between _____ and _____.

7. The conduction of the first material was much _____ than that of copper.

8. The word "compare" may be used to substitute the word _____ in the last paragraph of the unit.

Say if True or False. Correct the False sentences. (use one word only)

1. Karasz's polymer can already be marketed.
☐ TRUE ☐ FALSE
a. Copy the incorrect word: _____
b. Write the correction: _____

2. The possibility that plastics could be conductors was discovered by chance.
☐ TRUE ☐ FALSE
a. Copy the incorrect word: _____
b. Write the correction: _____

3. Polymer molecules are held together by sharing the atoms of their electrons.
☐ TRUE ☐ FALSE
a. Copy the incorrect word: _____
b. Write the correction: _____

4. The electrons in a polymer do not move freely.
☐ TRUE ☐ FALSE
a. Copy the incorrect word: _____
b. Write the correction: _____

5. Free electrons in a polymer result in "conjugating" its molecular sequence.
☐ TRUE ☐ FALSE
a. Copy the incorrect word: _____
b. Write the correction: _____

6. The "dope" added to a material becomes an inseparable part of it.
☐ TRUE ☐ FALSE
a. Copy the incorrect word: _____
b. Write the correction: _____

Solvents Get the Big Squeeze

Paragraph 1 - For those who like their coffee strong, but don't want to stay awake all night suffering from the side effects, decaffeinated coffee has been a familiar sight on supermarket shelves for decades. But when in the early 1960s, the German chemist Kurt Zosel hit on a novel way of removing caffeine from coffee beans, he could not have foreseen how the same technique would become the basis of a method for destroying toxic waste, the inspiration for ways of making industrially important chemicals without conventional, toxic solvents and a bone of contention for real ale aficionados.

Paragraph 2 - Behind all these possibilities is the extraordinary behaviour of some very ordinary fluids. If you place certain gases under enough pressure and at the same time heat them beyond a certain temperature, then, provided they do not decompose first, they become "supercritical": they enter the no-man's-land between a liquid and gas, where they are neither one nor the other. The same thing happens to liquids heated to this critical point under high enough pressure. In the 19th century, scientists treated this phenomenon as a curiosity. Only later did chemists discover its practical uses.

Paragraph 3 - When Zosel passed a stream of supercritical carbon dioxide through wet coffee beans he was already familiar with the way chemists used supercritical fluids (SCFs) to purify mixtures of chemicals by dissolving out unwanted components. Some inspired lateral thinking led him to the idea that an SCF might also dissolve and remove caffeine from coffee. Until then the only alternatives were organic solvents such as aromatic hydrocarbons or benzene derivatives which could leave toxic residues.

Paragraph 4 - More recently, the ability of supercritical fluids to dissolve things that will not dissolve in everyday solvents has attracted the attention of Britain's brewing industry. In a bid for consistency that will make drinkers of real ale turn pale, some brewers have begun to use a similar technique to "clean up" hops for the brewing industry. Originally a German idea, it was taken up by a company called English Hops to remove some of the natural flavours and colours from hops and so give a more consistent taste to beer. The food industry uses SCFs in similar ways. Universal Flavours of Slough, which extracts oily essences for use in everything from celery soup to aerosol sprays, uses SCFs to dissolve out the chemicals responsible for flavour from vegetable matter. SCFs are used similarly by analytical chemists to separate tricky mixtures of chemicals. Now, thanks to recent developments in chemists' understanding of supercritical fluids, researchers are extending their repertoire even further.

Paragraph 5 - Water becomes supercritical at temperatures above 374 °C and pressures above 218 atmospheres. Though it still looks like a liquid, many of its other properties have changed radically. It mixes with oily materials, but will not dissolve ordinary table salt. It will still dissolve oxygen, though, a property that means it is able to support a flameless variety of combustion. This is being explored as a means of "clean" incineration

of organic toxic waste. Steve Rich at Sandia National Laboratories in New Mexico is investigating the use of SCFs to destroy a stockpile of artillery shells at McAlester Army Ammunition Unit in Oklahoma, while General Atomics, a company in San Diego, is considering their potential for destroying chemical weapons. Other chemists are exploiting the solvent properties of SCFs to make new types of polymer and other molecules that could function as industrial catalysts without the need for the potentially harmful solvents in which much conventional industrial chemistry is carried out.

Paragraph 6 - An ideal solvent for a chemical reaction is an inert fluid that acts as a medium for carrying molecules of reactants. The reaction occurs as dissolved molecules collide. For a solvent to do its job, its molecules have to approach solute molecules closely enough to interact with them. This is why water is such a good solvent for some substances. Water molecules are "polar": there is a slight negative electric charge on the oxygen atom, which is balanced by slight positive charges on the hydrogen atoms. So a water molecule will interact electrostatically with ions such as chloride, which is why water will dissolve sodium chloride, common salt, a substance made up exclusively of ions. The general rule is that like dissolves like: that is, most organic chemicals, which are nonpolar or "oily", are not soluble in polar solvents like water, but dissolve in solvents such as hexane, chlorine-containing hydrocarbons and ethers, which are themselves oily.

Paragraph 7 - But such conventional solvents have disadvantages. Most organic solvents are toxic, some are flammable or even explosive and all of them cause pollution if not properly disposed of. Some may decompose in the atmosphere to produce, among other things, reactive forms of chlorine that damage the ozone layer. They can be recovered at the end of a reaction and reused, but there are limits to how many times this can be done. The infamous CFCs* are inert to most chemicals and would be ideal for many applications but for the damage they do to the ozone layer.

Paragraph 8 - Many fluids can be made supercritical, but three in particular—water, carbon dioxide and xenon—have the useful qualities for a solvent of being inert and nontoxic. Some reactions that require high temperatures when carried out in conventional industrial solvents will take place almost at room temperature in supercritical carbon dioxide, which becomes supercritical at 31°C. Also, SCFs flow more easily than other solvents and have different electrostatic properties, which help them to cluster round reactant molecules more effectively. The result is that less energy is needed to make these molecules collide, which is why reactions happen at lower temperatures.

* - chlorofluorocarbon

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Solvents Get the Big Squeeze - Questions

I. Answer the following question in English.

1. According to paragraph 4, what are the different uses of SCFs in industries other than that of the brewing of beers?

II. Check [✓] TRUE or FALSE for each statement. Only if a statement is FALSE then:

- a. Copy the incorrect word and correct the FALSE statement (use one word only).
- b. Do not use the word 'NOT' as a correcting word.
- c. If both statements are marked as TRUE, none will be accepted.

1. Hexane, chlorine-containing hydrocarbons and ethers are polar solvents.

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

2. Reactions with supercritical fluids taking place at close to room temperatures result from the special electrostatic attributes of SCFs.

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

III. Check [✓] the best answer, according to the text.

1. Zosel's method for coffee decaffeination had a great effect on the chemical industry because it
 - a. was replaced in the early 1960s by newer modes of extraction procedures
 - b. dispenses with the need for toxic solvents
 - c. removed caffeine from coffee beans
 - d. is the basis for making toxic solvents
2. Supercritical fluids are created by
 - a. heating gases or liquids under pressure to a critical temperature
 - b. decomposing gases at a critical temperature
 - c. mixing the gas and the liquid phases for practical uses
 - d. changing the densities of the liquid and gas phases in the atmosphere
3. Before Zosel, chemists had used SCFs to
 - a. pass moisture through coffee beans in order to decaffeinate them
 - b. dissolve carbon dioxide
 - c. purify benzene derivatives
 - d. remove undesirable components from chemicals

4. Zosel's idea was based on
 - a. a completely novel way of thinking
 - b. the result of his experiments with aromatic hydrocarbons
 - c. an application of existing knowledge concerning SCFs
 - d. mathematical calculations which chemists used to use
5. The English Hops company treats its hops through SCF-inspired techniques
 - a. to create flavourless beer
 - b. to standardize the flavour of beer
 - c. to give beer a more natural flavour
 - d. to turn the colour of beer paler
6. SCFs are used in various industries
 - a. to extend their use
 - b. for extracting oils from celery soup
 - c. as aerosol sprays
 - d. because of their dissolving properties
7. Which of these statements about supercritical water is **INCORRECT**?
 - a. It can burn without any flame
 - b. Salt may be dissolved in it
 - c. Poisonous organic waste may be burned with it
 - d. Oils may be dissolved in it
2. What is the topic of paragraph 5?
 - a. The unusual properties of supercritical water
 - b. "Clean" incineration of organic toxic waste
 - c. Potential uses of supercritical water and other SCFs
 - d. The harmful properties of conventional industrial solvents
3. Salt dissolves in water because
 - a. water cannot react with the chloride ions
 - b. of the positive charge of its hydrogen atoms
 - c. the oxygen atom has a negative charge
 - d. of the polarity of water molecules
4. The main idea of paragraph 7 is
 - a. organic solvents can be flammable
 - b. conventional solvents are problematic in their use
 - c. CFCs are ideal for many applications
 - d. conventional solvents are limited to the number of times they can be used

IV. Check [✓] the best answer.

1. According to paragraph 2, ____ certain gases do not break down first under pressure and heat, they become supercritical.
 - a. unless
 - b. if
 - c. whereas

2. The "no-man's-land" (par. 2, line 4) is in this case
 - a. an in-between state of matter
 - b. both a liquid and a gas
 - c. neither a liquid nor a fluid

3. According to paragraph 3, benzene derivatives are given as an example of
 - a. aromatic hydrocarbons
 - b. mixtures of chemicals
 - c. organic solvents

4. From "like dissolves like" (par. 6, line 9) we can understand that
 - a. similar substances repel each other unless they are themselves oily
 - b. a substance dissolves in a substance which is similar to it
 - c. polar solvents are able to dissolve oily solvents within them

5. The word "They" (par. 7, line 4) refers to
 - a. organic solvents
 - b. disadvantages of conventional solvents
 - c. reactive forms of chlorine

6. The word "recovered" (par.7, line 4) means
 - a. covered again
 - b. got better
 - c. regained

7. The word "them" (par. 8, line 6) refers to
 - a. other solvents
 - b. SCFs
 - c. electrostatic properties

High–Power Electronics

Paragraph 1 - Consumers of electricity are demanding customers. The silicon chips that now pervade daily life, bringing sophisticated behavior to everything from toasters to machine tools, are highly vulnerable to irregularities in their electrical demands. A loss of power for a single cycle of alternating current, one sixtieth of a second, can make computer screens go blank or interrupt other sensitive electronic equipment. At the same time that users of electric power demand quality, they also want more power. As a result, transmission networks are being pressed closer to their operating limits. Yet, a range of problems limits expansion and power transfers from one part of the country to another challenge the network's adequacy. All these factors increase the risk of disruption and even blackout.

Paragraph 2 - To avoid such problems, the engineers who manage the transmission of electric power must act with extreme caution. They operate the power grid well below its theoretical maximum capacity. The strategy reduces the possibility that a sudden, unforeseen increase in demand or loss of capacity might cause overloads that would ripple outward until they engulf a significant portion of the nation's transmission network. On the infamous night of November 9, 1965, for example, a blackout struck most of the northeastern U.S. and parts of Canada. Utilities must also maintain large reserves of generating capacity as a safety margin against such contingencies.

Paragraph 3 - The traditional equipment does not make the job any easier. When engineers switch in additional generating equipment or otherwise cope with failures or changing demand, they rely on massive electromechanical switches—essentially enormous cousins of the household circuit breaker—that take several AC cycles to turn on or off. These switches introduce their own electrical noise and potential instabilities into the system. They cannot be used to make the continuous, fine adjustments that full utilization of power transmission capacity would require.

Paragraph 4 - The development of high-voltage silicon switches may provide utilities with a technology that enables them to cope effectively with economic constraints as they meet the needs of their customers. These switches are the basis of control systems that can guide the flow of megawatts as rapidly and efficiently as integrated circuits handle microwatts. They can fend off cascading power interruptions and significantly increase the usable capacity of many transmission lines. Indeed, they open up avenues for controlling the distribution of power that could not be exploited by their electromechanical predecessors. Utilities using them will be able to deliver more power of better quality while reducing transmission losses and thus the amount of energy they must generate.

Paragraph 5 - The troubles that power transmission systems experience today go back to the roots of the industry and the conflict that raged a century ago between Thomas Edison and George Westinghouse: direct versus alternating current. In DC systems,

electrical charge flows in only one direction: from producer to consumer. In AC systems, the flow of charge reverses itself many times each second; a terminal that had a positive potential a fraction of a second ago may now have a negative potential (and will have a positive one in another fraction of a second). A simple generator, consisting of a coil rotating in a magnetic field, will produce alternating current because the relative orientation of the coil and the magnetic field reverses with every half rotation.

Paragraph 6 - Edison chose direct current for his first power plant, built to light lower Manhattan in 1882. DC at low voltage was safe, reliable and easy to control. It was also very inefficient: low voltage meant high current and resistive losses increase as the square of the current transmitted. As a result, power plants could not serve customers who lived or worked more than a few miles away.

Paragraph 7 - These limitations prompted Westinghouse to push for alternating current, which could be converted by transformers to high voltages for efficient transmission and then back to lower levels for safe use. (A transformer consists of two interwound coils; an alternating current of one voltage passing through the primary coil generates an alternating current of another voltage in the secondary coil; the ratio between the two voltages depends on the number of turns in each coil.)

Paragraph 8 - Edison warned against the dangers of alternating current, which can be more hazardous than equivalent levels of direct current. The advantages of alternating current became clear in 1896, however, when a high-voltage transmission line began to carry current to Buffalo, N.Y., from the first hydroelectric power plant at Niagara Falls. Other high-voltage links between widely separated load centers and power plants followed. Today most of the U.S. and Canada has been tied to one of several highly interconnected AC power systems.

Paragraph 9 - Opponents of alternating current did have a point. AC power is more difficult to control than is DC. Line voltages tend to rise or fall in complicated ways as loads change. Voltage instabilities are more likely to get out of hand. These problems arise because AC generates changing electric and magnetic fields around the lines that carry it; these fields in turn affect the flow of current. Ideally, the power flowing through an AC line consists of a series of waves of rising and falling voltage and current that reverses polarity 120 times a second, in perfect sinusoidal rhythm.

Paragraph 10 - When the sine waves representing voltage and current at any given point in the transmission line rise and fall in synchrony, they are said to be "in phase"—that is, the phase angle between them is zero. When a load is imposed at one end of a line, current flows through the load (doing work and creating heat in the process); this flow of electrical charge causes a slight drop in the line voltage and so power flows from the higher-voltage end of the line to the lower-voltage one.

Paragraph 11 - In a real power transmission system, these factors grow even more complicated. The impedance along any given path depends on the interaction between the cables and other equipment and the changing electric and magnetic fields engendered by the power flow. Transformers and motors create inductive impedance, whereas capacitors and long, lightly loaded transmission lines create capacitive impedance. As

power flows change in response to these distortions, however, the magnitude of capacitive and inductive effects can change with them, thus altering the milliseconds-earlier balance and causing power flows to change yet again.

Paragraph 12 - Controlling impedance is crucial to the proper distribution of power, but it is also very difficult. Once a transmission line, transformer or motor has been built, its impedance is fixed and cannot easily be changed. The impedance of transmission lines can be changed in large steps by inserting either series capacitors or inductors in the line, but this process is time-consuming and does not lend itself to the rapid control of electrical flows.

Paragraph 13 - Taken together, the various kinds of high-power silicon devices offer utilities unprecedented control over power transmission and distribution. The economic benefits of the technology should be substantial for both utilities and their customers. Reduction of sags, momentary outages and subharmonic resonances will save some of the billions that those problems cost every year, as well as the billions now spent on uninterruptible power supplies. Moreover, "smart" power devices in utility lines consume only about 1 percent of the power they deliver, instead of the 10 to 20 percent that existing devices take.

Paragraph 14 - In addition, solid-state controllers could increase the overall capacity of today's transmission network by 20 percent or more. Engineers have estimated, for example, that adding thyristor-controlled series capacitors to two 500-kilovolt lines serving Florida could increase the power transfer capacity from 2,000 megawatts to 3,000 megawatts. Such an investment could pay for itself in less than a year. Nationwide, utilities could save \$6 billion compared with the cost of adding the same capacity with new lines.

Paragraph 15 - Even more importantly, increased flexibility in distribution will make it easier to transfer power from one region to another on demand, thus permitting individual utilities to reduce their generating reserve margins. Smaller reserve margins, in turn, will enable utilities to defer construction of new power plants. If thyristor-based devices make it possible to reduce margins from the current 20 percent to 15 percent, \$50-billion worth of power plants could be left unbuilt over the next 25 years.

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High-Power Electronics – Questions

Questions to be answered orally

Each student in the class is to be allocated a paragraph to read, translate and understand. The student will then be required to answer the questions set for that paragraph.

1. What is the main idea of paragraph 1?
2. What is the blackout of 1965 an example of? (par.2)
3. Define and explain the problem mentioned in paragraph 3.
4. What are cascading power interruptions? (par.4)
5. Give a definition of AC. (par.5)
6. Give a definition of DC. (par.6)
7. What is a transformer used for and why? (par.7)
8. Why are Niagara Falls mentioned in paragraph 8?
9. What is the drawback of AC? (par.9)
10. What is the “phase angle”? (par.10)
11. Explain the content of paragraph 11.
12. What does the insertion of capacitors and inductors achieve? (par.12)
13. How does a smart power system work? (par.13)
14. In what way could engineers boost power capacity in Florida? (par.14)
15. Why will it be possible to defer construction of new power plants? (par.15)

I. Answer the following question in English.

1. What two strategies do power plants use to prevent blackouts? (par. 2)
 - a. _____
 - b. _____

II. Check (✓) TRUE or FALSE. Correct if the statement is FALSE (use one word only).

Do not use NOT as a correction.

1. High-voltage silicon switches might offer an inefficient solution to the commercial issues faced by utilities. (par. 4)
☐ TRUE ☐ FALSE
 - a. Copy the incorrect word: _____
 - b. Write the correction: _____
2. Westinghouse was in favour of using alternating current so direct current did not adequately answer the needs of all consumers. (pars. 6 and 7)
☐ TRUE ☐ FALSE
 - a. Copy the incorrect word: _____
 - b. Write the correction: _____

III. Choose the best answer, according to the information in the text.

1. A failure in power supply for a fraction of a second can result in (par. 1)
 - a. malfunctions of various electronic appliances
 - b. the continuous operation of advanced electronic devices
 - c. an increased reliability of computer screens
 - d. a higher quality of electric power supply

2. Electromechanical switches do not assist engineers in dealing with failures or supply requirements since (par. 3)
 - a. they can be quickly turned on or off whenever convenient
 - b. they are devices which are traditionally relied upon
 - c. they are incapable of the adaptations necessary for optimal performance
 - d. they operate noiselessly and achieve a high level of stability

3. The main advantage of high-voltage silicon switches over the older devices lies in their allowing (par. 4)
 - a. to decrease the operational capacity of transmission lines
 - b. to regulate electricity supply in novel ways
 - c. to reduce the quality of the available electrical power
 - d. to increase the amount of power utilities must generate

4. The difference between direct and alternating currents is that
 - a. the latter flows only from producer to consumer while the former periodically changes its direction
 - b. the former periodically reverses itself whereas the latter always flows in the same direction
 - c. both the former and the latter can be produced by changes in the relative orientation of a coil and a magnetic field
 - d. as opposed to the latter, the former is the unidirectional flow of electric charge

5. The difficulty in controlling AC power is caused by (par. 9)
 - a. the regulated stability of voltage
 - b. the uncomplicated rise and fall of line voltages
 - c. the perfect sinusoidal rhythm of voltage and current
 - d. the unstable electric and magnetic fields generated by current

6. What is the topic of paragraph 11?
 - a. The dependence of impedance on the electric and magnetic fields produced by the power flow.
 - b. The causes for the complexity of operating a real power transmission system.
 - c. The difference between the impedance created by transformers and that created by capacitors.
 - d. The effects of various distortions on the magnitude of capacitive and inductive effects.
7. The impedance of transmission lines
 - a. is easily and speedily modified
 - b. can be altered only in small stages
 - c. is impossible to modify without investing much time
 - d. is not altered by the insertion of inductors
8. The low power consumption of high-power silicon devices (par. 13)
 - a. is the sole reason for their profitability
 - b. will probably lead to significant financial advantages
 - c. prevents their being cost-effective
 - d. will make them the means of ensuring adequate power delivery
9. An investment in solid-state controllers would
 - a. quickly allow to save as much money as would be spent
 - b. result in insufficient financial yields
 - c. be less profitable than establishing new lines
 - d. considerably decrease power transfer capacity
10. Large sums of money will be saved if (par. 15)
 - a. decreasing generating reserve margins advance construction of new power plants
 - b. transfer of power is impeded by utilities lowering their generating reserve margins
 - c. facilitation of adequate power supply enables utilities to increase their reserve margins
 - d. construction of new power plants is postponed due to lower generating reserve margins

IV. Choose the best answer, according to the information in the text.

1. The word "operating" (par. 1, line 7) is used as
 - a. a verb
 - b. an adverb
 - c. a noun
 - d. an adjective
2. The word "its" (par. 2, line 2) refers to
 - a. maximum capacity
 - b. power grid
 - c. engineers
 - d. electric power
3. The word "capacity" (par. 2, line 4) means
 - a. volume
 - b. capability
 - c. position
 - d. competence
4. The word "power" (par. 3, line 7) is used as
 - a. a noun
 - b. a verb
 - c. an adjective
 - d. an adverb
5. The word "them" (par. 4, line 2) refers to
 - a. utilities
 - b. silicon switches
 - c. technology
 - d. economic constraints
6. The phase angle between the waves of voltage and current is zero _____ they rise and fall synchronically. (par. 10)
 - a. whereas
 - b. unless
 - c. until
 - d. once
7. The word "As" (par. 11, line 5) means
 - a. when
 - b. because
 - c. like
 - d. although

Materials Too Hot To Resist

Paragraph 1 - Carrying energy along superconducting wire is the ultimate dream of electrical engineers. Yet it remains tantalisingly just out of reach. A number of specialised applications exist, but technological problems still stand in the way of wiring the world with resistance-free circuitry. What makes things even more frustrating for researchers is that no one knows how superconductors work.

Paragraph 2 - Physicists have been toiling away at theories of superconductivity for years. How is it, they ask, that in certain conditions some metals and alloys at very low temperatures offer no resistance to an electrical current? From a common starting point their theories have now diverged into a range of possibilities, from the suggestion of new particles to special magnetic states. But do these theories stand up to experimental tests? And do any of them bring any closer the dream of a superconducting world where energy is transported without waste?

Paragraph 3 - Late last year there was widespread anticipation that the dream might come true. Researchers at laboratories in Paris and Grenoble reported observations of superconductivity at temperatures as high as 250 Kelvin (-23°C) in both bismuth-based and mercury-based materials. The previous high had been 150 Kelvin for a copper oxide-based material. But there was a certain amount of scepticism too.

Paragraph 4 - The continuing debate over the French results has once more highlighted the lack of a theory against which to test observations. The French groups reported that the phenomenon had recognisable hallmarks of superconductivity—conduction of electricity without resistance below a certain temperature and the expulsion of magnetic fields. But the signals were too weak for many critics to take seriously. Physicists have seen too many USOs (unidentified superconducting objects) to celebrate before seeing these results reproduced. Discovering a new superconductor will always be a hit-and-miss affair until a comprehensive theory is developed for high temperature superconductors.

Paragraph 5 - Few doubt that such a theory will eventually emerge from the ferment of current ideas. When it does, many believe it will be exotic: "There is going to come a time when some physicist walks away with a Nobel Prize mumbling about Spinons or something," says Robert Snyder, director of the Institute for Ceramic Superconductivity in Alfred, New York, referring to one of the new particle candidates. But he also stresses that the answer is not yet in sight: "I know of not a single theory that allows any prediction of where we can find other superconducting materials," he adds.

Breaking up pairs

Paragraph 6 - The theoretical starting point is the BCS theory, announced in 1957 by John Bardeen, Leon Cooper and Robert Schrieffer, all then at the University of Illinois, Urbana-

Champaign. This is accepted as the explanation for basic low temperature superconductivity, which has long been observed in metals at temperatures near to absolute zero. The key phenomenon, electrons moving without resistance, according to the theory, can be put down to them travelling in pairs, called Cooper pairs.

Paragraph 7 - So how do the electrons perform their binding trick? After all, two electrons, both of which are negatively charged, should repel each other. It is at this point that the theories begin to diverge. Bardeen, Cooper and Schrieffer had an answer that seems to work for standard superconductors which work below 4 Kelvin but doesn't seem to fit superconductors such as the copper oxides which work at higher temperatures.

Paragraph 8 - The BCS model says that a travelling electron can induce vibrations, called phonons, of atoms or of the lattice that makes up the crystal. As the electron moves it attracts positive ions to its side, creating a positively charged region in the crystal. But since the electron continues to travel faster than the ions, it leaves a trail of displaced positive ions that in turn attracts a following electron. As a result, the second electron accelerates through the positive ion "pass", as if it were attracted to the first electron. This is called phonon-mediated electron attraction.

Paragraph 9 - A proper description of electrons involves thinking of them in terms of their energy states and momentum at a quantum mechanical level. In this description, energy and angular momentum are directly related. So an exact calculation of angular momentum is crucial to finding out how much energy is needed to break up a Cooper pair. Describing the angular momentum of a spinning globe is easy; you can describe it by saying it spins this fast and in that direction. But for a quantum object, such as an electron, the most convenient way is in terms of waves.

Paragraph 10 - The most likely angular momentum for a phonon-mediated Cooper pair is zero. Other momentum values are possible but have so far not been observed. The zero angular momentum state is called an s-wave. In an s-wave, the Cooper pair is so stable that a comparatively large amount of energy is needed to break it up.

Paragraph 11 - Moreover, according to BCS theory, this effect is symmetrical, or isotropic. No matter in which direction the Cooper pair travels through a superconductor, the energy needed to break up the pair—the energy gap—remains the same. Cooper pairs in materials at low temperatures do not have enough thermal energy to cause them to break up. This explains why standard superconductors don't superconduct unless very cold. As the temperature rises, however, the Cooper pairs do begin to break up, as they gain enough energy to breach the energy gap.

Paragraph 12 - Any theory on how a Cooper pair breaks up—or, more usefully, how they form—predicts a certain wave state, or angular momentum, for that pair. And from that angular momentum, certain observable characteristics must follow, such as how many free electrons, as opposed to Cooper pairs, should exist in a superconductor at a particular temperature. A few years ago researchers found that whereas standard superconductors have Cooper pairs but no free electrons at low temperatures, high temperature cuprate-based superconductors have both. This puzzling finding led some scientists to postulate a different pairing wave, called a d-wave.

Paragraph 13 - Unlike the symmetrical s-wave pair, the energy needed to break up a d-wave pair varies depending on which direction the pair is travelling through the material. So the energy gap is said to be anisotropic. If the pair travels with the grain of a superconducting lattice or perpendicular to it, much energy will be needed to break the pair up, just as in an s-wave. But if the pair begins to travel off these axes the energy gap becomes smaller, reaching zero along the diagonals. As a result, Cooper pairs simply cannot exist along these diagonals. Rather, the diagonals would be the sole province of single electrons. If you picture the s-wave energy gap as a circle, then the d-wave energy gap looks more like a four-leaf clover. Moreover, the Cooper pairs travelling with the grain are 180 degrees out of phase with those travelling perpendicular to it, so that the d-wave has a phase shift.

Paragraph 14 - The difference between the s and d-waves is subtle, but crucial. Whether or not the copper oxide Cooper pairs are s or d-wave has become one of the main dividing lines between most of superconductivity theories. "I think that the d-wave crowd is a real Mafia," says Richard Klemm, a theorist at Argonne National Laboratory. "They are just trying to control it by screaming loudly." Other scientists, meanwhile, suggest that they don't respect anyone who still thinks that a cuprate Cooper pair bonds in an s-wave.

Paragraph 15 - One of the best known d-wave theories was formulated by David Pines at the University of Illinois, Urbana-Champaign and Doug Scalapino at the University of California, Santa Barbara. It includes the idea of a Cooper pair, but leaves behind the phonon-electron interaction. The researchers suggest that it is the spin of the copper atoms, rather than the atomic lattices that is fluctuating. They assume that the spins are highly ordered into a pattern like a chessboard, where white and black squares correspond to up and down spins. (Whether the superconductor is indeed this highly ordered remains another unanswered question). Where in the BCS theory a passing electron is thought to change the surrounding charge concentration, here it is thought to change the spins of the surrounding atoms. These altered spins then affect a second electron, pulling it through sharply so that it slams into the first.

Wave debate

Paragraph 16 - On the s-wave side of the fence, a well-known theory has been backed by Phil Anderson, a physicist at Princeton University. Anderson strongly believes that his s-wave theory will prove correct. He agrees that the wave might be anisotropic, but believes it to be an anisotropic s-wave. He says his theory fits results from every type of experiment. "Everyone else is kind of guessing and prodding at his theory to see if he can get everyone to agree with it. But my theory fits everything," he states.

Paragraph 17 - Anderson's theory is more easily understood by describing the Cooper pairs in terms of particles. Anderson believes that the energy thought to belong to electrons moving in each copper oxide layer creates two new types of energy state or particle, known as the holon and spinon. The holon carries charge but no spin and the spinon carries spin but no charge. In his theory, holons bind to holons and spinons to

spinons to create the analogue of the Cooper pair. If waves from holons are superimposed on waves from spinons, they create an asymmetrical, or anisotropic, s-wave. It may look a lot like a d-wave, but it is a strangely shaped s-wave because it has no phase shift.

Paragraph 18 - Whereas most physicists agree that the copper oxide layers are important to the superconducting behaviour of cuprates, Anderson goes a step further. He says that holons and spinons exist only in the copper oxide plane. In order to explain how they move vertically from plane to plane, he theorizes that the spinons or holons must reform into an electron and cross the vacuum. This is a phenomenon common to all superconductors, whereby a current jumps from superconductor to superconductor at the quantum level.

Paragraph 19 - While the idea of electrons separating into different particles may sound odd, many scientists accept this aspect of Anderson's theory. Patrick Lee, a physicist at the Massachusetts Institute of Technology, accepts the spinon/holon concept but suspects the Cooper pair is a d-wave.

Paragraph 20 - Making use of holons and spinons, Lee has developed something called Gauge Field theory. Instead of the perfectly ordered spin chessboard of Pines and Scalapino, with only two spin directions— up and down—Lee proposes that an inherent property of the copper oxide layer is that the spins point in different directions, at an angle of around 50 degrees to each other. Lee says that this forest of diagonal spins, known as the Gauge Field, would change the phase of a moving holon to produce a d-wave.

Paragraph 21 - Now researchers are trying to shed light on the debate by investigating the shape of the wave that describes the angular momentum. Is it the circular s-wave or the four-leafed clover shape produced by a d-wave? And is there a phase shift? If they could eliminate either the s or d-wave this would narrow the range of possible theories considerably.

Paragraph 22 - During the past year two groups have carried out experiments that point to an anisotropic energy gap, providing evidence for either a conventional d-wave or an anisotropic s-wave. One of these involved nuclear magnetic resonance tests which measured how long it took a nucleus that has been excited to high energy to "relax" to a lower energy state. When Charles Slichter at the University of Illinois lowered the temperature in his cuprate samples to near zero, where BCS theory says they should superconduct and therefore show Cooper pairs, the nuclei in the sample continued to relax. He reasoned as follows. An excited nucleus cannot release its energy unless there is something to release it to. Cooper pairs cannot accept this energy because there is not enough to break them up. So, there must be free electrons even at these low temperatures to receive the released energy—evidence for an anisotropic gap.

Paragraph 23 - A second set of experiments, by Walter Hardy's group at the University of British Columbia in Vancouver, was based on another observable characteristic of superconductors. It involved measuring the size of magnetic fields they expelled. Single electrons hinder this expulsion, so the size of the field indirectly measures how many such electrons there are. If the energy gap is anisotropic, theory says that the amount of

magnetic field expelled should increase linearly with temperature, as the number of free electrons increases. This is what they observed. This linear variation, though it does not prove that the energy gap is necessarily a d-wave, does point to a highly anisotropic wave. For this reason it is often cited in support of the d-wave school.

Paragraph 24 - Hardy himself is quick to say that he remains outside anyone's theoretical camp. "That's dangerous for an experimentalist," he says. "You really want to make sure you've got your measurement right and you can't allow something you're expecting to influence your judgment." Klemm believes that experiments such as Hardy's and Slichter's do not necessarily prove the d-wave camp's point. "Some of these experiments that have been cited as evidence for the d-wave are probably not," he says. He points out that Hardy's experiments only examine the surface of the crystal and what happens there is not always what happens on the inside.

Paragraph 25 - Neither set of experiments, however, can distinguish between a d-wave and an anisotropic s-wave—the difference being that the d-wave has a phase shift. To detect the phase shift, researchers have relied on the Josephson tunneling effect, in which a current can jump from superconductor to superconductor, even if it has to change its wave state to do so. By joining traditional low temperature superconductors which are known to carry s-waves to YBCO, or even YBCO to YBCO, researchers have been able to measure whether the Cooper pairs change from s to d-wave as they cross these junctions.

Paragraph 26 - A number of experiments have made use of this effect by setting up different configurations of closed loops. The first was proposed by Tony Leggett and performed last year by Dale Van Harlingen and his student David Wollman, all at the University of Illinois. They clipped a C-shaped piece of lead, which behaves as a conventional superconductor at low temperatures, to the top and side edges of a square crystal of YBCO. This circuit was designed so that a current would run down both arms of the C-shaped piece of lead and reunite in the crystal on the far side of the circuit. In order to reunite the two halves would have to be in phase when they meet.

Paragraph 27 - If the YBCO supported an s-wave, like the lead, the two halves could recombine and the current would travel easily, since s-waves do not change phase. But if the YBCO supported a d-wave, an overall current would not flow since the currents from each arm would be out of phase and cancel each other out. This is indeed what the researchers found. Since then, Hans Ott of the Swiss Federal Institute of Technology in Zurich has had similar results from the same experiment. In March the d-wave theory was given a further boost when a team from IBM, in Yorktown Heights, New York, announced a different setup which several scientists regard as the most conclusive proof for the d-wave yet.

Narrowing the field

Paragraph 28 - The team, led by John Kirtley and Chang Tsuei, made use of an effect from quantum theory. Quantum mechanics predicts that electrons travelling in a closed loop must have the same phase when they return to where they started. They can switch

phase along the way, like flipping a playing card, but when they reach the start again, the same side must be face up. This effect is so consistent that a closed loop will spontaneously create current and magnetic fields to ensure that the phase is conserved. The existence of these currents and magnetic fields does not point unequivocally to a phase shift. Kirtley and Tsuei showed not only that they exist, but also that they are the right size. Their experiment makes a strong case for d-waves. "If it's correct it's one of the most important things that's happened in this field," says Kirtley. "And if it's not correct I'm going to have to leave town."

Paragraph 29 - But there has also been strong experimental support for the s-wave theory. Over the past year, Robert Dynes at the University of California in San Diego has created a circuit of lead and YBCO, a little like the one made by Leggett and Van Harlingen. They laid a square of lead on top of a square of YBCO and then ran a current from one to the other. The theory was that if the YBCO supports an s-wave, then supercurrent should flow normally. But if it is a d-wave, Cooper pairs of all phases should flood into the crystal and cancel each other out to produce zero current, which supports the s-wave theory.

Paragraph 30 - The jury is still out on the final theory. But experimental evidence has begun to narrow the field. Deciding which experiments are valid remains the most daunting task for theorists. "But," as Klemm says, "This is not philosophy and this is not politics. Sooner or later there's going to be a right and a wrong."

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Materials Too Hot to Resist – Questions

Questions to be answered orally

Each student in the class is to be allocated a paragraph to read, translate and understand. The student will then be required to answer the questions set for that paragraph.

1. What is the main idea of paragraph 1?
2. What is the main idea of paragraph 2?
3. What was unique about the experiment? (par. 3)
4. Why were the results not accepted? (par.4)
5. What is the main idea of paragraph 5?
6. What are Cooper pairs? (par.6)
7. What is the main idea of paragraph 7?
8. What is phonon-mediated electron attraction? (par.8)
9. Explain the content of paragraph 9.
10. What is unique about an s-wave? (par.10)
11. What does the word “isotropic” mean? (par.11)
12. Translate paragraph 12 into Hebrew.
13. What does the word “anisotropic” mean? (par.13)
14. What is the main idea of paragraph 15?

I. Answer in English:

- 1) What is the main target of electrical engineers, according to the first paragraph?

II. TRUE or FALSE. Correct the false statements (use one word only)

- 1) Physicists are attempting to reach superconductivity at lower temperatures.

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

- 2) According to the text, the French physicists were successful in obtaining superconductivity.

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

III. Circle the best answer:

- 1) Knowledge about the operation of superconductors
 - a) is irrelevant to the engineers' purpose
 - b) would hinder the work of engineers
 - c) could facilitate the achievement of the engineers' goal
 - d) is a minor factor preventing engineers from reaching a breakthrough

- 2) Theories of superconductivity developed by physicists
 - a) have led to important discoveries
 - b) have now become limited to new particles and special magnetic states
 - c) have been successfully proven through experiments
 - d) have developed in diverse directions and are inconclusive

- 3) In order to be certain that a theory of superconductivity has been achieved
 - a) it must be tested on one or two materials
 - b) it must apply to a great number of materials
 - c) it must be based on unidentified superconducting objects
 - d) it must be compared to existing known standards of superconductivity

- 4) The key to achieving superconductivity as envisioned by the BCS theory lies in
 - a) maintaining low temperatures of materials used
 - b) the tendency of resistance free electrons to travel in pairs
 - c) electrons moving without resistance
 - d) accepting the validity of the BCS theory

- 5) After studying the movement of electrons, Bardeen, Cooper and Schrieffer
 - a) cannot explain why electrons are attracted to each other
 - b) are at disagreement with one another regarding the movement of electrons
 - c) suggest that the attraction of electrons to each other is performed by means of ions
 - d) theorize that resistance free electrons repel each other

- 6) The most critical element in breaking up a Cooper pair is
 - a) the speed and direction of a spinning globe
 - b) the angular momentum
 - c) the energy of the wavelength
 - d) the quantum mechanical level

- 7) The closer a Cooper pair is to an s-wave,
- a) the less energy is need to break it up
 - b) the angular momentum is zero
 - c) the more energy will be needed to break it up
 - d) the easier it is to break it up
- 8) We can infer from the text that the breaking up of Cooper pairs
- a) is conducive to superconductivity
 - b) only occurs at very low temperatures
 - c) leads to standard superconductivity
 - d) hinders superconductivity
- 9) A d-wave is characterized by
- a) a combination of free electrons and Cooper pairs
 - b) no free electrons at low temperatures
 - c) Cooper pairs at high temperatures
 - d) standard superconductors
- 10) According to paragraph 14,
- a) the d-wave theory has gained unanimous supremacy
 - b) the s-wave theory is incorrect
 - c) the d-wave theory is more accurate than the s-wave theory
 - d) scientists' opinions are divided between supporters of each theory

IV. Circle the correct answer:

- 1) Paragraph 2 _____ the information given in paragraph 1.
- a) contradicts
 - b) summarizes
 - c) defines
 - d) elaborates on
- 2) In paragraph 9, an electron is described as
- a) a quantum object
 - b) a spinning globe
 - c) an angular momentum
 - d) a Cooper pair

- 3) The word "both" in paragraph 12, line 7 refers to
- a) Cooper pairs and free electrons
 - b) standard superconductors and Cooper pairs
 - c) free electrons and low temperatures
 - d) high temperatures and cuprate-based superconductors
- 4) The expression "with the grain" in paragraph 13 means
- a) through a grain-sized hole in the lattice
 - b) in the same direction
 - c) little by little
 - d) following a pattern
- 5) The word "those" in paragraph 13, line before last, refers to
- a) 180 degrees
 - b) Cooper pairs travelling with the grain
 - c) d-waves
 - d) Cooper pairs
- 6) The word "s-wave" in paragraph 13, line 8 is used as
- a) an adjective
 - b) a noun
 - c) an adverb
 - d) a verb
- 7) The word "Where" in paragraph 15, line 8 could be replaced by the word
- a) Because
 - b) Whereas
 - c) Therefore
 - d) In addition

Batteries of the Future

Par. 1 – One of the most challenging issues regarding mobile devices from smartphones to electric cars is improving rechargeable batteries' performance. A good rechargeable car battery will set you back around \$10,000, for a product that is toxic, degrades substantially after a few years and must be carefully designed to avoid catastrophic overheating. Samsung's recent woes with exploding batteries in its Galaxy Note 7 smartphone caused it to recall all the devices and cease production. Better, cheaper batteries are top of the wish list for almost any technology that's not powered by fossil fuels.

Par. 2 – Might there be a better way? That's the claim of researchers championing a long-overlooked device to store and supply energy. They think it could actually stand more of a chance of delivering the power we need, how we need it, and so revolutionize the way we use energy. Is it time to look beyond batteries?

Par. 3 – Rechargeable batteries store energy by performing a reversible chemical reaction in which ions are stored in and flow between positive and negative electrodes. The right materials, such as the lithium compounds common to both Tesla and Samsung's batteries, can store lots of energy but are slow to charge and discharge, and heat up when they do. What exactly caused the Note 7 fault is not yet clear, but lithium ion batteries need tiny separators to keep components apart. If these are poorly designed or damaged, they can fail, creating a short circuit that heats and damages other parts of the battery causing a runaway reaction. Such safety concerns, plus the sheer cost of lithium ion batteries, have long had chemists casting around for something better.

Par. 4 – But chemistry isn't the only way to store electric charge. In devices known as capacitors, energy is physically stored in an electric field between metal electrodes. Capacitors are sprinters to the battery's long-distance runner, charging and discharging in a blink, and doing this over and over again without their performance suffering. They are already used to power the flash on a camera.

Par. 5 – But you can't run a car on a camera flash. A kilogram of petrol contains about 4000 Watt hours of useful energy, 30 times as much as the batteries in Tesla's current crop of vehicles. Traditional capacitors hold 1000 times less – just 0.1 Watt hours per kg. If your car could drive 500 km on a tank of petrol, it would run little more than 6 meters using the same weight of capacitors.

Par. 6 – It's unthinkable, then, that a traditional capacitor could ever compete with a battery. But many have had that exact thought – even Elon Musk, CEO of Tesla and a pioneer in electric car manufacture. "If I were to make a prediction, I'd think there's a good chance that it

is not batteries but capacitors that will deliver a breakthrough," he said in 2011. According to his opinion, it's just a case of guiding the continuing evolution of the capacitor.

Par. 7 – That evolution stretches back to 1966, when Robert Rightmire at Standard Oil of Ohio was part of a team considering the future of fuel storage. He knew that the key to replacing batteries lay in the fact that the charge a capacitor could store depended on the surface area of its electrodes. So why not make these surfaces spongier in order to make it possible to store more energy? He produced a capacitor where the electrodes were coated with thin layers of carbon chemically punctured with millions of tiny holes. This so-called activated carbon is typically used for jobs like decaffeinating coffee and has an internal area about 100,000 times larger than its outside surface. And it worked. Rightmire's "super-capacitors" stored 10 times as much energy as traditional capacitors.

Par. 8 – By the 1990s, small super-capacitors (or ultra-capacitors) had become a commercial reality. They provided instant, short-lived back-up power to computers if the main supply failed, so they could shut down safely. Some wind turbine companies even use them as an emergency alternative to batteries. Turbine blades need to be constantly adjusted to face the wind. If their electricity supply fails, the blades must quickly return to a neutral position to avoid strong gusts damaging or even destroying the turbine. That calls for a speedy short-term power supply – at which ultra-capacitors excel. Moreover, conventional batteries are heavy and eventually need replacing. "Once you've put something into a turbine, you're not going to want to go up and service it. You just want to forget it," says Kim McGrath from Maxwell Technologies, an ultra-capacitor manufacturer.

Par. 9 – Using batteries as the sole storage medium isn't ideal for two reasons. First, constant charging and discharging shortens their life. Second, batteries can't release all their energy quickly, so power grids need excess battery capacity to cope with short surges in demand over and above normal fluctuations. Therefore, adding ultra-capacitors instead of supersizing the battery is a vast improvement. "The net effect is a reduction in the upfront expenditure and lower operating costs," says McGrath. "And the technology has reached the stage where it blows the market open for us."

Par. 10 – Ultra capacitors have come so far from their humble beginnings that it is tempting to wonder if they might replace batteries entirely. We're far from that day, but perhaps betting against it ever arriving is unwise. We know that Elon Musk started a PhD studying ultra-capacitors before quitting for his first Silicon Valley start-up. And Tesla Motors' patents still make references to ultra-capacitors. The man once so dedicated to ultra-capacitors hasn't entirely lost faith, then. Maybe they are still evolving behind the doors of his huge factory in Nevada.

Adapted from Mark Harris, "Beyond batteries: This technology could revolutionise energy," NewScientist, November 14, 2016

Batteries of the Future - Questions

I. Answer the following question in English.

1. What other reason besides lithium ion batteries being slow and unsafe makes chemists look for a more valid solution? (par. 3)
-

II. Check [✓] TRUE or FALSE for each statement. Only if a statement is FALSE then:

- a. Copy the incorrect word and correct the FALSE statement (use one word only).
- b. Do not use the word 'NOT' as a correcting word.
- c. If both statements are marked as TRUE, none will be accepted.

1. One of the main advantages of lithium ion batteries is that they are incapable of charging and discharging with enough speed. (paragraph 3)

TRUE FALSE

Copy the incorrect word: _____

Write the correction: _____

2. Turbine companies use ultra-capacitors as an emergency alternative due to their ability to provide power quickly. (paragraph 8)

TRUE FALSE

Copy the incorrect word: _____

Write the correction: _____

III. Check [✓] the best answer, according to the information in the text.

1. The example of Galaxy Note 7 in paragraph 1 illustrates the idea that
 - a. a quality battery is quite expensive as well as fallible
 - b. battery technology is essentially based on fossil fuels
 - c. poor design of rechargeable batteries can lead to disastrous consequences
 - d. Samsung's devices are on the top list of many customers
2. According to paragraph 3, the rechargeable battery technology is based on a chemical reaction
 - a. of short circuiting and heating the electrodes to produce electricity
 - b. in which a current of ions between the positive and negative poles is maintained
 - c. which both Tesla and Samsung have developed
 - d. in which lithium flows between positive and negative electrodes
3. According to paragraph 4, capacitors
 - a. work more slowly than batteries
 - b. can charge and discharge in an efficient manner
 - c. degrade over time and their performance suffers
 - d. have to be charged over a long period of time

4. The purpose of paragraph 5 is to show that
 - a. petrol has much higher potential energy than batteries
 - b. various Tesla vehicles are powered by capacitors
 - c. cars could run as much as 500 km using capacitors instead of petrol
 - d. while capacitors may be promising, they are still inefficient
5. We can understand from paragraph 6 that Elon Mask thinks that capacitors will solve the problems of energy supply if
 - a. the engineers will optimize their future development
 - b. they will be incapable of competing with batteries
 - c. they will be used primarily in electric cars
 - d. the breakthrough in the energy industry will be postponed
6. Rightmire's "super-capacitors" (par. 7) were more powerful than ordinary capacitors because
 - a. their storage capacity was not affected by the surface area of the electrodes
 - b. the surface of the electrodes was 10 times smoother
 - c. the size of their internal area was reduced by the use of activated carbon
 - d. the electrodes' surface underwent treatment which made it porous
7. We can infer from paragraph 8 that one significant advantage of ultra-capacitors in wind turbines is that they
 - a. do not need servicing or replacing
 - b. prevent turbine blades from returning to the neutral position
 - c. power up the turbine all the time
 - d. weigh more than conventional batteries
8. What is the main idea of paragraph 8?
 - a. Ultra-capacitors are still not ready for commercial use
 - b. The advantages of conventional batteries outweigh the ones of ultra-capacitors
 - c. Ultra-capacitors have several important practical applications
 - d. Servicing wind turbines is a complex problem
9. According to paragraph 9, a good solution to cope with batteries' disadvantages is
 - a. supersizing conventional rechargeable batteries
 - b. using ultra-capacitors in addition to batteries
 - c. compensating for the slow release of their energy by excess capacity
 - d. constant recharging and discharging of the batteries

10. It might be possible that ultra-capacitors will take the place of batteries since (par. 10)
- a. no improvement of their humble capacities would be wise
 - b. it would be tempting to use this solution
 - c. Elon Mask no longer believes in them
 - d. they may continue to undergo major evolutionary changes

IV. Check [✓] the best answer.

1. When a product “degrades substantially after a few years” (line 3), its condition
 - a. doesn’t undergo any considerable change over a long period of time
 - b. significantly improves after several years of use
 - c. suffers major deterioration after a relatively short period of time
 - d. worsens at a negligible rate after prolonged use
2. The word "do" (par. 3, line 4) refers to
 - a. heat up
 - b. store
 - c. discharge
 - d. charge and discharge
3. The word “these” (par. 3, line 6) refers to
 - a. tiny separators
 - b. lithium Ion batteries
 - c. fault
 - d. components
4. The word “its” (par. 7, line 3) refers to
 - a. key
 - b. batteries
 - c. charge
 - d. capacitor
5. The word “spongier” (par. 7, line 4) is used as
 - a. a noun
 - b. an adjective
 - c. a verb
 - d. an adverb

6. The word “short-lived” (par. 8, line 2) is used as
- a. a noun
 - b. an adjective
 - c. a verb
 - d. an adverb
7. The main structure of paragraph 9 is that of
- a. exemplification
 - b. comparison
 - c. cause and effect
 - d. time sequence

Creating Human-Level Artificial Intelligence

Paragraph 1 - However you look at it, the future appears bleak. The world is under immense stress environmentally, economically and politically. It's hard to know what to fear the most. Even our own existence is no longer certain. Threats loom from many possible directions: a giant asteroid strike, global warming, or a new plague.

Paragraph 2 - Another threat is artificial intelligence. In December 2014, Stephen Hawking told the BBC that "the development of full artificial intelligence could spell the end of the human race. It would take off on its own and redesign itself at an ever increasing rate. Humans, who are limited by slow biological evolution, couldn't compete, and would be superseded." Last year, he followed that up by saying that AI is likely "either the best or the worst thing ever to happen to humanity." Other prominent people, including Bill Gates and Steve Wozniak, have made similar predictions about the risk AI poses to humanity. Nevertheless, billions of dollars continue to be funnelled into AI research and stunning advances are being made.

Paragraph 3 - Hawking's fears revolve around the idea of the technological singularity. This is the point in time at which machine intelligence starts to accelerate. The idea is roughly the same age as research into AI itself. More recently, the idea of a technological singularity has been popularized by Ray Kurzweil, who predicts it will happen around 2045, and Nick Bostrom, who has written a bestseller on the consequences.

Paragraph 4 - There are several reasons to be fearful of machines overtaking us in intelligence. Humans have become the dominant species on the planet largely because we are so intelligent. Many animals are bigger, faster or stronger than us. But we used our intelligence to invent tools, agriculture and amazing technologies. These allowed us to dominate the planet. It is therefore not surprising that machines that think threaten to usurp us.

Paragraph 5 - The idea of an intelligence explosion, when machines continuously improve their intelligence and thus quickly exceed human intelligence, is not a particularly wild idea. The field of computing has profited considerably from many similar exponential

trends. Moore's law predicted that the number of transistors on an integrated circuit would double every two years and it has done so for decades. So it is not unreasonable to suppose AI will also experience exponential growth.

Paragraph 6 - Scientists working in AI predict that we are just 30 or 40 years away from AI achieving superhuman intelligence. But there are several strong reasons why a technological singularity is improbable. Silicon, for example, has a significant speed advantage over our brain's wetware. This advantage doubles every two years according to Moore's law. But speed alone does not bring increased intelligence. Even if we can make a dog think faster, it is still unlikely to play chess. It doesn't have the necessary mental constructs, the language and the abstractions. Intelligence is much more than thinking faster or longer about a problem than someone else. Of course, Moore's law has helped AI. We now learn faster and off bigger data sets. Speedier computers will certainly help us to build artificial intelligence. But at least for humans, intelligence depends on many other things, including years of experience and training. It is not at all clear that we can short circuit this in silicon simply by increasing the clock speed or adding more memory.

Paragraph 7 - In addition, the technological singularity supposes human intelligence is some special point to pass. Bostrom writes: "Human-level artificial intelligence leads quickly to greater-than-human-level artificial intelligence. The interval during which the machines and humans are roughly matched will likely be brief. Shortly thereafter, humans will be unable to compete intellectually with artificial minds." _____ if there is one thing we should have learned from the history of science, it is that we are not as special as we would like to believe. Copernicus taught us that the universe does not revolve around Earth. Darwin showed us that we are not so different from other apes. Watson, Crick and Franklin revealed that the same DNA code of life powers us and the simplest amoeba. And AI will no doubt teach us that human intelligence is itself nothing special. There is no reason to suppose that human intelligence is a special point, that once passed allows for rapid increase in intelligence. Even if we have enough intelligence to design super-human AI, the result may not be adequate to cause a technological singularity. Improving intelligence is far harder than just being intelligent.

Paragraph 8 – The idea of a technological singularity supposes that each AI generation is getting some fraction better than the last. However, the performance of most of our AI systems has so far been that of diminishing returns. There is often great promise at the

start, but we then run into difficulties when looking for improvements. This helps explain the overly optimistic claims made by many of the early AI researchers. An AI system may be able to improve itself an infinite number of times, but the extent to which its intelligence changes overall could be limited. Hence, the impact of AI on our economy and on our society may occur less quickly than people like Hawking fear.

Adapted from Toby Walsh “What if ...We create human-level artificial intelligence?”, NewScientist, November 19, 2016)

Creating Human-Level Artificial Intelligence - Questions

I. Answer the following question in English.

1. What fact do the scientific discoveries of Copernicus, Darwin, and Watson, Crick and Franklin exemplify? (par. 7)
-

II. Check [✓] TRUE or FALSE for each statement. Only if a statement is FALSE then:

a. Copy the incorrect word and correct the FALSE statement (use one word only).

b. Do not use the word 'NOT' as a correcting word.

c. If both statements are marked as TRUE, none will be accepted.

1. Vast sums of money are being invested into AI research, due to predictions regarding the risks it poses. (par. 2)

TRUE FALSE

Copy the incorrect word: _____

Write the correction: _____

2. The advantage of our brain's wetware compared to silicon is its incapacity to think faster. (par. 6)

TRUE FALSE

Copy the incorrect word: _____

Write the correction: _____

III. Check ☒ the best answer, according to the information in the text.

1. What is the purpose of the first paragraph in relation to the second paragraph?
 - a. To provide the background for introducing the main topic
 - b. To contrast between artificial intelligence and other threats
 - c. To warn that artificial intelligence is the worst threat of all
 - d. To dismiss the threat posed by artificial intelligence
2. Paragraph 2 introduces several contrasts. One of them is between
 - a. Hawking's view and those of Gates and Wozniak
 - b. the amounts of money invested in AI and the stunning advances made
 - c. the development of full artificial intelligence and the end of the human race
 - d. slow human biological evolution and the increasing rate of AI self-redesign
3. Paragraph 3 defines "technological singularity". What is the consequence of the technological singularity discussed in paragraph 4?
 - a. Humans designing machines that think
 - b. Humans having no fear of machines
 - c. Machines overtaking humans in intelligence
 - d. Humans becoming the dominant species on the planet

4. In paragraph 4, the author mentions the intelligence of animals in order to highlight the fact
 - a. that human intelligence is unsurpassable
 - b. that human intelligence could be surpassed by AI
 - c. that humans and animals are very similar
 - d. that human intelligence allows the invention of novel technologies
5. Moore's law is invoked in paragraph 5 in order to
 - a. predict that transistors used in AI will double in number every two years
 - b. show that Moore has correctly predicted the intelligence explosion of AI
 - c. define the concept of exponential trends
 - d. draw an analogy between the increase in sophistication of integrated circuits and AI
6. We can infer from paragraph 6 that
 - a. artificial intelligence lacks the mental constructs, language and abstractions of a human
 - b. the technological singularity can be achieved by using silicon components
 - c. faster clock speed and more memory will be necessary to create superhuman intelligence
 - d. the technological singularity depends entirely on the implications of Moore's law
7. The quote from Bostrom's writings _____ the idea that human intelligence is special.
 - a. contradicts
 - b. elaborates on
 - c. undermines
 - d. is irrelevant to
8. We can infer from paragraph 7 that greater-than-human-level intelligence
 - a. can never be achieved
 - b. has been discovered by scientists
 - c. is something that we don't have to fear
 - d. will be made possible by humans being special
9. In paragraph 8, the author bases his opinion regarding the improvement of AI systems on
 - a. the optimism of AI researchers
 - b. the idea that each generation improves by a fraction
 - c. the way they are likely to impact economy and society
 - d. the past performance of such systems

10. What is the connection between paragraphs 7 and 8?
- a. Their common topic is the achievements of artificial intelligence
 - b. Paragraph 8 exemplifies the idea presented in paragraph 7
 - c. Paragraph 8 contradicts the idea presented in paragraph 7
 - d. They both focus on reasons for the improbability of a technological singularity

IV. Check ☒ the best answer, according to the text.

1. The word "However" (par. 1, line 1) means
 - a. But
 - b. Despite
 - c. No matter how
 - d. Nevertheless
2. The word "its" (par. 2, line 3) refers to
 - a. development
 - b. artificial intelligence
 - c. the human race
 - d. an increasing rate
3. The word "increasing" (par. 2, line 3) is used as
 - a. an adjective
 - b. an adverb
 - c. a verb
 - d. a noun
4. The word "computing" (par 5, line 3) is used as
 - a. a verb
 - b. an adjective
 - c. an adverb
 - d. a noun
5. Which connector can best fill in the blank in par. 7?
 - a. Because
 - b. But
 - c. Therefore
 - d. In addition

6. The words “the last” (par. 8, line 2) refer to
 - a. fraction
 - b. idea
 - c. generation
 - d. singularity

7. The expression “diminishing returns” (par. 8, line 3) means
 - a. great promises
 - b. insufficient benefits
 - c. excellent results
 - d. technological singularities

Three Reasons Why the Periodic Table Needs a Redesign

By Joshua Howgego

Chemists can't agree on the best way to arrange the elements, prompting proposals of everything from spiral-shaped alternatives to radically elongated versions.

The illusion of order

The periodic table may seem a paragon of orderliness but some elements may be out of place

Group ①

1

H

Hydrogen

②

WHERE DOES HYDROGEN GO?

Key

Atomic number

Symbol

Name

③	④	⑤	⑥	⑦	⑧	⑨	⑩	⑪	⑫	⑬	⑭	⑮	⑯	⑰	⑱
B	C	N	O	F	Ne	B	C	N	O	F	Ne	B	C	N	O
Boron	Carbon	Nitrogen	Oxygen	Fluorine	Neon	Boron	Carbon	Nitrogen	Oxygen	Fluorine	Neon	Boron	Carbon	Nitrogen	Oxygen
13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
Al	Si	P	S	Cl	Ar	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni
Aluminium	Silicon	Phosphorus	Sulphur	Chlorine	Argon	Potassium	Calcium	Scandium	Titanium	Vanadium	Chromium	Manganese	Iron	Cobalt	Nickel
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46
Ga	Ge	As	Se	Br	Kr	Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd
Gallium	Germanium	Arsenic	Selenium	Bromine	Krypton	Rubidium	Strontium	Yttrium	Zirconium	Niobium	Molybdenum	Technetium	Ruthenium	Rhodium	Palladium
49	50	51	52	53	54	55	56	57-71	72	73	74	75	76	77	78
In	Sn	Sb	Te	I	Xe	Cs	Ba	Lanthanides	Hf	Ta	W	Re	Os	Ir	Pt
Indium	Tin	Antimony	Tellurium	Iodine	Xenon	Cesium	Barium		Hafnium	Tantalum	Tungsten	Rhenium	Osmium	Iridium	Platinum
81	82	83	84	85	86	87	88	89-103	104	105	106	107	108	109	110
Tl	Pb	Bi	Po	At	Rn	Fr	Ra	Actinides	Rf	Db	Sg	Bh	Hs	Mt	Ds
Thallium	Lead	Bismuth	Polonium	Astatine	Radon	Francium	Radium		Rutherfordium	Dubnium	Seaborgium	Bohrium	Hassium	Meitnerium	Darmstadtium
111	112	113	114	115	116	117	118		119	120	121	122	123	124	125
Rg	Cn	Nh	Fl	Mc	Lv	Ts	Og		Uut	Uuq	Uub	Uut	Uuq	Uub	Uut
Roentgenium	Copernicium	Nihonium	Flerovium	Moscovium	Livermorium	Tennesine	Oganesson		Ununtrium	Unquadium	Unbium	Untrium	Unquadium	Unbium	Untrium

THE F-BLOCK CONUNDRUM

WHY ARE MERCURY AND GOLD SO WEIRD?

57	La	58	Ce	59	Pr	60	Nd	61	Pm	62	Sm	63	Eu	64	Gd	65	Tb	66	Dy	67	Ho	68	Er	69	Tm	70	Yb	71	Lu
Lanthanum		Cerium		Praseodymium		Neodymium		Promethium		Samarium		Europium		Gadolinium		Terbium		Dysprosium		Holmium		Erbium		Thulium		Ytterbium		Lutetium	
89	Ac	Th	90	Pa	91	U	92	Np	93	Pu	94	Am	95	Cm	96	Bk	97	Cf	98	Es	99	Fm	100	Md	101	No	102	103	Lr
Actinium		Thorium		Protactinium		Uranium		Neptunium		Plutonium		Americium		Curium		Berkelium		Californium		Einsteinium		Fermium		Mendelevium		Nobelium		Lavenderium	

Par 1 - Run your fingers over the white keys of a piano. The notes get higher and higher as your hand moves to the right. On the eighth key, something beautiful happens: a note hangs in the air that embodies something of the first, only with a different pitch.

Par 2 - We began noticed that something similar was going on with the chemical elements more than 150 years ago. Scientists even called it the law of octaves. And it is this repetition in the properties of the elements that the periodic table captures so beautifully. Similar elements end up stacked in columns or groups. One group comprises noble gases like argon and neon that barely react with anything, another contains reactive metals, some of which, like francium, explode on contact with water.

Par 3 - But there are doubts over whether the periodic table is in the best possible configuration. Just as notes can be arranged in various ways to produce music, so the essence of the relationships between the elements could be depicted differently. There is no easy way to judge which is better, or more “true”. So arguments over perceived flaws in the current arrangement rumble on, with some chemists arguing that certain elements should be relocated – and others working on more radical ways to recompose the table.

Par 4 = At first, the elements were organised by atomic weight. Now we order them by the number of protons in their nucleus. We also know that their properties are largely determined by the arrangement of the negatively charged electrons that orbit in successive shells around the nucleus.

Par 5 - The lightest elements have just one shell, which can hold two of these particles. Heavier elements have more shells that can hold larger numbers of electrons. What really matters for each element's behaviour, however, is how many electrons it has in its outer shell.

Par 6 - That number tends to fit nicely with the way the table is arranged, namely to place elements with similar properties in the same group. For instance, group 1 elements have one electron in their outer shell and those in group 2 have two. But it doesn't always fit together quite as neatly as all that.

Where does hydrogen go?

Par 7 - Take the first element. Hydrogen has one electron in its outermost shell so you might assume it belongs exactly where it is, in group 1 above lithium and sodium, which also have one electron in their outermost shell. Yet hydrogen is a gas, not a metal, so its properties don't fit.

Par 8 - The complication arises because, with an outer shell that can only hold two electrons, hydrogen is one electron away from being full. Given that elements yearn for full outer shells that make it very reactive. In this sense, hydrogen resembles the elements in group 17, namely the halogens like chlorine. Their outer shells need only gain one electron to achieve a full shell of eight, which makes them similarly reactive. In terms of its properties, then, hydrogen is closer to chlorine than lithium.

Why are mercury and gold so weird?

Par 9 - Lower down the table there are no available spaces for misplaced elements. Even so, a couple of the incumbents look like outliers. Take mercury, also known as quicksilver because it is a liquid at room temperature. In that sense, it is quite different to the other members of group 12, including zinc and cadmium, which are all solid metals. What is happening here?

Par 10 - The further down the table you go, the more of the positively charged protons an element's nucleus contains. This creates a stronger pull on the orbiting electrons, meaning they must travel faster and faster. By the time you reach mercury, the electrons are travelling at 58 per cent of the speed of light. According to Einstein's special theory of relativity, this means their effective mass is significantly higher than an electron's normal mass, which exacerbates the inward pull they feel.

Par 11 - The upshot is that mercury's electrons orbit so tightly that they can't be shared to form bonds with other atoms, as is required to make a solid. The same thing explains why gold is gold, a unique colour among metals: relativistic effects change the way electrons absorb light.

The F-block confusion

Par 12 - Group 3 holds two elements that might belong elsewhere. As we move across the upper rows of the table, electrons fill up shells in a sequence of so-called orbitals, waiting until the innermost shell is full before entering the next. By element 57, lanthanum, the electrons begin to enter a new type of orbital, an f-orbital. To account for this, most periodic tables hive off the elements making up this f-block, putting it below the table, leaving a gap in group 3.

Par 13 - Fair enough. But there is debate over which of the elements in the f-block should come first. Some chemists maintain that the decision should come down to electron configuration, which would leave the table as it is, with lanthanum and actinium at the left-hand end of the f-block. Others point out that chemical properties such as atomic radius and melting point make lutetium and lawrencium, currently at the right end, a better bet. In 2016, the International Union of Pure and Applied Chemistry assembled a task group to settle the argument. But no one expects a decision soon.

Starting over

Par 14 - This confusion has persuaded some chemists that we need to redraw the periodic table – and there is no shortage of ideas. Mark Leach at Manchester Metropolitan University, UK, keeps the internet database of periodic tables, which contains hundreds of versions.

Par 15 - In an attempt to better represent the continuity where one row currently ends, retired Canadian chemist Fernando Dufour developed a 3D periodic system that looks like a Christmas tree, with the elements radiating from a trunk in circles that get larger closer to the bottom. An alternative is the spiral developed by Theodor Benfey, which allows the f-block to bulge outwards (see main image, above).

Par 16 - Eric Scerri at the University of California, Los Angeles, is among those who has argued for more fundamental changes. He previously proposed that the table could be arranged to maximise the number of “triads”, sets of three elements that share similar properties and are related by their atomic weights. These days, he is backing an even more drastic approach: make the table not 18 but 32 columns by slotting all 30 f-block elements between the current groups 2 and 3 (see “Going long”). This allows the atomic number to run in an uninterrupted sequence.

Par 17 - But Guillermo Restrepo at the Max Planck Institute for Mathematics in the Sciences, Germany, favours an alternative. He has explored whether chemical similarity of elements in the same columns still holds as well as it did 150 years ago, given our increased knowledge of chemical reactivity. His conclusion is that lanthanum belongs in group 3 – that is, out of sequence.

Par 18 - Redesigning the periodic table might seem a quixotic quest, but it could soon take on a new urgency. We are already on the trail of element 119. Where it will go, and how the table will morph to make space for it, remains to be seen.

Three Reasons Why the Periodic Table Needs a Redesign - Questions

Pre-reading

1. Read the title and the subheadings
 - a. What field of science does it refer to?

 - b. What do you know about the periodic table?

2. Look at the illustration. What is it? Do you see any problem?
 - a. _____
 - b. Match the sections' headings to the illustration. Can you guess what problem they suggest?

3. Read the first sentence of each paragraph. Predict the writer's purpose in this text.

Close reading

4. According to par 3, the elements can be arranged in various ways.
 - a. yes/ no
 - b. evidence: _____
5. How were the elements organized in the past compared to nowadays (par. 4)?

6. What is the difference between the lightest and the heavier elements (par. 5-6)?

7. According to par. 7-8, it is easy to place hydrogen in the periodic table.
 - a. yes / no
 - b. evidence: _____
8. Mercury is liquid at room temperature due to the fact that its electrons are travelling at _____ than half the speed of light (par 10).
9. Why does gold have its distinctive color (par. 11)?

10. The argument in par. 13 regarding the F-block was not resolved because
- no one expects a decision soon
 - there is a gap in group 3
 - a task group was assembled
 - Actinium and Lanthanum were placed at the left end of the f-block
11. The author of this article is _____ that a new design of the periodic table is at hand shortly (par. 18).
- unconfident/confident
 - evidence: _____

12. Reference questions

- Noble gasses (par. 2) refer to _____
 - Others (par. 3) refers to _____
 - It (par. 7) _____
 - He (par. 16) refers to _____
13. **Vocabulary**. The following words are taken from the text. Look up their definitions in the dictionary of and write them.
- Elements (par. 4) _____
 - Nucleus (par. 4) _____
 - Orbit (par. 4) _____
 - Properties (par. 4) _____
 - Shell (par. 5) _____
 - Incumbent (par. 9) _____
 - Outlier (par. 9) _____

Post-reading

14. What is the writer's main idea in this text?

15. What are the three reasons for changing the periodic table?

YouTube's Recommendation Algorithm Has a Dark Side

It leads users down rabbit holes By [Zeynep Tufekci](#) * |

Par. 1. It was 3 A.M., and the smoke alarm wouldn't stop beeping. There was no fire, so I didn't need to panic. I just had to figure out a way to quiet the darn thing and tamp down my ire. I had taken out the battery and pushed and twisted all the buttons to no avail.

Par. 2. Luckily for me, the possible solutions were all laid out in the YouTube tutorial I found. The video helpfully walked me through my options, demonstrating each step. And the fact that it had hundreds of thousands of views reassured me that this might work.

Par. 3. YouTube has become the place to learn how to do anything, from assembling an Ikea cabinet to making a Bluetooth connection with your earbuds. It is a font of tutorials, some very good, some meandering, some made by individuals who have become professionals at it and rake in serious sums through advertising. But many are uploaded by people who have solved something that frustrated them and want to share the answer with the world.

Par. 4. The native language of the digital world is probably video, not text—a trend missed by the literate classes that dominated the public dialogue in the pre-digital era. I've noticed that many young people *start* their Web searches on YouTube. Besides, Google, which owns YouTube, highlights videos in its search results.

Par. 5. “How do I” assemble that table, improve my stroke, decide if I'm a feminist, choose vaccinations, highlight my cheeks, tie my shoelaces, research whether climate change is real...? Someone on YouTube has an answer. But the site has also been targeted by extremists, conspiracy theorists and reactionaries who understand its role as a gateway to information, especially for younger generations.

Par. 6. And therein lies the dark side: YouTube makes money by keeping users on the site and showing them targeted ads. To keep them watching, it utilizes a recommendation system powered by top-of-the-line artificial intelligence (it's Google, after all). Indeed, after Google Brain, the company's AI division, took over YouTube's recommendations in 2015, there were laudatory articles on how it had significantly increased “engagement”: Silicon Valley—speak for enticing you to stay on the site longer.

Par. 7. These “recommended” videos play one after the other. Maybe you finished a tutorial on how to sharpen knives, but the next one may well be about why feminists are ruining manhood, how vaccinations are poisonous or why climate change is a hoax—or a nifty explainer “proving” the *Titanic* never hit an iceberg.

Par. 8 YouTube's algorithms will push whatever they deem engaging, and it appears they have figured out that wild claims, as well as hate speech and outrage peddling, can be particularly so.

Par. 9. Receiving recommendations for noxious material has become such a common experience that there has been some loud pushback. Google did ban a few of the indefensibly offensive high-profile “creators” (though not before helping them expose their views to millions of people), and recently the company announced an initiative to reduce recommending “borderline content and content that could misinform users in harmful ways.” According to Google, this content might be things like “a phony miracle cure for a serious illness” or claims that “the earth is flat.” The change, they say, will affect fewer than 1 percent of all videos.

Par. 10. While it's good to see some response from Google, the problem is deep and structural. The business model incentivizes whatever gets watched most. YouTube's reach is vast. Google's cheap and nifty Chromebooks make up more than half the computers in the K–12 market in the U.S., and they usually come preloaded with YouTube. Many parents and educators probably don't realize how much their children and students use it.

Par. 11. We can't scream at kids to get off our lawn or ignore the fact that children use YouTube for a *reason*: there's stuff there they want to watch, just like I really needed to figure out how to unplug that beeping catastrophe at 3 A.M. We need to adjust to this reality with regulation, self-regulation and education. People can't see how recommendations work—or how they're designed to keep eyes hooked to the screen. We could ask for no YouTube or “no recommendations” for Chromebooks in schools.

Par. 12. This is just tip of the iceberg of the dangerous nexus of profit, global scale and AI. It's a new era, with challenges as real as that iceberg the *Titanic* did hit—no matter what the video claims.

*[Scientific American April 2019](#)

YouTube's Recommendation Algorithm - Questions

Pre-reading

Read the title and the subtitle.

1.
 - a. What do you know about YouTube?

 - b. What is an algorithm?

2. What do you think is a "dark side"? Is it good or bad?

3. Read the topic sentences. Can you predict what the text is about?

Close reading

4.
 - a. What does the anecdote in par. 1 and 2 add to what you already know about YouTube?

5.
 - a. The information that the writer found was useful / useless (circle one).
 - b. Evidence from the text: _____
6. The word "meandering" can be replaced with
 - a. Rambling
 - b. Coherent
 - c. Focused
7. The phrase "rake in" in par. 3 is used as
 - a. an adjective
 - b. a noun
 - c. a verb
 - d. an adverb

8. Par. 5 presents a problem. What connector indicates it?

a. _____

b. What is the problem?

9. What is the dark side of YouTube, according to par. 6?

10. What point do the Titanic and the knives' sharpening tutorial come to illustrate? (par. 7)

11. In par. 9 we learn that Google has done something to deal with problematic materials. Was it serious? Yes/No (choose one). Provide evidence from the text to support your answer.

12. The word "noxious" (par. 9) can be replaced with

a. innocent

b. safe

c. repulsive

13. True or False. Is the following sentence correct according to par. 10? If not, **correct** it.

"Chromebooks make less than 50% of the computers in the school market in the U.S."

14. What does the *Titanic* in par. 12 come to illustrate?

15. References: What do the following words refer to?

a. "them" par. 6 _____

b. "they" par. 8 _____

c. "it" par. 10 _____

16. The word "nexus" (par. 12) can be replaced with

- a. split
- b. chain
- c. break
- d. disruption

17. According to par. 11-12, can YouTube be limited? If yes, how? Provide evidence from the text:

Post reading

18. In your own words, what is the main problem that this article raises?

19. Provide your own solution to the problem.

The Origin of Mathematics

Anil Ananthaswamy

New Scientist, 2 September 2017

Par. 1 - To the Iranian mathematician Maryam Mirzakhani, the first woman to win the Fields medal, mathematics often felt like “being lost in a jungle and trying to use all the knowledge that you can gather to come up with some new tricks”. “With some luck,” she added, “you might find a way out.” Mirzakhani ventured deeper into the mathematical jungle than most. _____, most of us have spent enough time on its periphery to have a sense of what the terrain looks like. Increasingly, it seems as if humans are the only animals with the cognitive ability to hack their way through the undergrowth. But where does this ability come from? Why did we develop it? And what is it for? Answering these questions involves diving into one of the hottest debates in neuroscience, and reimagining what mathematics really is. The natural world is a complex and unpredictable place. Habitats change, predators strike, food runs out. An organism’s survival depends on its ability to make sense of its surroundings, whether by counting down to nightfall, figuring out the quickest way to escape danger or weighing up the spots most likely to have food. And that, says Karl Friston, a computational neuroscientist and physicist at University College London, means doing mathematics. “There is a simplicity and symmetry to mathematics,” says Friston, “which, if you were treating it as a language, wins hands down over all other ways of describing the world.” From dolphins to slimemoulds, organisms throughout the evolutionary tree seem to make sense of the world mathematically, deciphering its patterns and regularities in order to survive.

Par. 2 - But how did we develop the tools to consciously understand what our bodies do instinctively? One long-standing idea says we are born with a conscious sense of numbers in the same way we are conscious of colours. In his 1997 book *The Number Sense*, Stanislas Dehaene of the INSERM-CEA unit for cognitive neuroimaging in Gif-sur-Yvette, France, hypothesised that evolution endowed humans and other animals with numerosity, an ability to immediately perceive the number of objects in a pile. In other words, three red marbles would produce a sense of the number 3 just as they would produce a sense of the colour red. Dehaene proposed that this numerosity was exact for numbers below 4 and fuzzier thereafter, but nonetheless represented a hardwired ability. Armed with such an instinct, our paths through the mathematical jungle would quickly start to clear.

Par. 3 - Evidence to support this “nativist” view soon started to accumulate. Elizabeth Spelke at the Massachusetts Institute of Technology and her colleagues showed that 6-month-old children could distinguish between an array of eight dots and one with 16 dots.

Then Dehaene and his colleagues reported that the Mundurucu Indians in the Brazilian Amazon, who don't have words for numbers larger than 5, could approximately discriminate between much larger quantities, suggesting that this ability was independent of culture. Other studies indicated that humans instinctively represent numbers spatially on an imaginary "number line", their values increasing from left to right. There was even evidence of numerosity in animals. This all pointed to an innate number sense that millennia of culture had helped expand. But before long, some researchers grew uncomfortable with the conclusions of these studies. Might subjects, for example, be distinguishing two arrays of dots based not on the number of dots, but on other attributes such as their spatial distribution or area of coverage? "These are cues that are usually correlated with number, so it would be unwise not to use them," says Tali Leibovich at the University of Haifa in Israel. "If you are an animal in nature and you need to hunt something and need to do it very quickly, you want to use all available cues." Indeed, on further examination, it seems that people also rely on these non-numerical cues. Soon, a different hypothesis emerged. Perhaps, instead of having an innate sense of number, we are born with a sense for quantities – such as size and density – that correlate with the numbers of things. "It takes time and experience to develop and understand this correlation," says Leibovich.

Par. 4 - More refined cognitive tests in children tend to support this view. For example, children younger than about 4 years of age cannot understand that five oranges and five watermelons have something in common: the number 5. To them, a bunch of watermelons simply represents more "stuff" than the same number of oranges. Even teaching young children to identify the order of numbers – going through the motions of counting – doesn't immediately impart their meaning, says developmental psychologist Daniel Ansari at the University of Western Ontario in Canada. That occurs informally through long-term exposure to parents and siblings. "This points to the strong influence of cultural practices on the learning of exact representations of number," he says. Study of the cultural aspects of numerical cognition has suffered from bias, says Ansari, in that not enough attention has been paid to data collected from non-industrialized cultures. These findings, he believes, cast serious doubts on the nativist hypothesis. Take the Yupno people of Papua New Guinea. Rafael Núñez at the University of California at San Diego has learned, for example, that they don't use the supposedly universal mental number line. Also, they have no comparatives in their language to say that one thing is bigger or smaller than another.

Par. 5 - This is not to say that the Yupno language is primitive. Far from it. Take demonstratives. In English, there are only four: this, that, these and those, to specify the proximal or distal nature of things. The Yupno, on the other hand, have words to indicate whether something is higher or lower than them in elevation (in keeping with their mountainous terrain), and they have nuanced words to capture not only how proximal or distal something is, but also by how much. The Yupno are not alone in having a language

that doesn't emphasize numbers. Núñez points to a study of 189 Aboriginal Australian languages, of which three-quarters were found to have no words for numbers above 3 or 4, while a further 21 went no further than 5. To Núñez, this suggests that exact numerosity is a cultural trait that emerges when circumstances, such as agriculture and trading, demand it. "Hundreds of thousands of humans who have language, sometimes very complicated and sophisticated language, don't have exact quantification," he says. Even languages that do, such as English or French, can only take you so far. Last year, Dehaene and his student Marie Amalric reported the results of scanning the brains of 15 professional mathematicians and 15 non-mathematicians of the same academic standing. They found a network of brain regions involved in mathematical thought that was activated when mathematicians reflected on problems in algebra, geometry and topology, but not when they were thinking about non-mathsy things. No such distinction was visible in the other academics. Crucially, this "maths network" doesn't overlap with brain regions involved in language. This suggests that once mathematicians have learned their symbolic language, they start thinking in ways that don't involve normal language.

Par. 6 - The exact moment when culture transformed whatever instincts we may have had into a recognizable mathematical ability is unclear. One of the earliest pieces of evidence of humans dealing with numbers comes from the Border cave in the Lebombo Mountains in South Africa. There archaeologists found 44,000-year-old bones with notches, including the fibula of a baboon etched with 29 such marks. Anthropologists think that such "tally sticks" were an aid to counting and represent evidence for an emerging symbolic understanding. Counting and measuring hit new heights sometime around the 4th millennium BC, in the sophisticated Mesopotamian culture of the Tigris-Euphrates valley, a region in modern-day Iraq. Eleanor Robson at the University of Oxford has argued that mathematics in Mesopotamia was a cultural invention needed to keep track of days, months and years, to measure areas of land and amounts of grain, and maybe even to record weights. And as humans took to the seas, or studied the skies, we began developing the mathematics required for navigation and for tracking celestial objects. But it was always, in the beginning, a product of cultural necessity (and if you think trading-driven mathematics is a thing of the past, think again: some of the most sophisticated mathematics is being developed for trading stocks and bonds on Wall Street).

Par. 7 - With the help of fundamental mathematical tools, humans have built an immense pyramid of mathematical knowledge. Over the past 5000 years or so, mathematics has expanded into ever more abstract domains, seemingly further removed from the processes that govern the world around us. And yet, the more we learn about the universe's hidden workings, the more such mathematical innovation seems to describe the things we see. When David Hilbert developed a highly abstract algebra that worked in an infinite number of dimensions rather than the familiar three dimensions of space, for example, nobody ____ have foreseen its use in the emerging field of quantum mechanics. But soon after, it turned out that the state of a quantum system could best be described

using such a Hilbert space – with the underlying mathematics remaining key to our attempts to make sense of the quantum world.

Par. 8 - The ubiquity of such connections between mathematics and physics led the physicist Eugene Wigner to comment on the “unreasonable effectiveness of mathematics” at describing the natural world. To many physicists today, the success of mathematics as a language speaks to its primacy in the organization of the universe. Max Tegmark of MIT is one of these. He thinks the universe is a mathematical structure in that it has only mathematical properties – and we are slowly uncovering this structure, brushing away the dust to reveal the theorems and proofs that underpin reality. “It used to be that it was very easy to list the small number of things in nature that you could describe with maths. Now it’s very easy to list the small number of things you cannot,” says Tegmark. Even biology, which long resisted mathematical rigour, is slowly succumbing: witness the proliferation of mathematics in genomics or computational neuroscience. From this perspective, mathematics is a discovery rather than an invention. For researchers like Núñez, however, that is an overly simplistic distinction. “When the question is asked – is mathematics invented or discovered?” he says, “there is a supposition that it’s exclusive. If you invented it, you don’t discover it, and so on.” But it’s not an either-or situation.”

The Origin of Mathematics - Questions

Answer in English.

1) How does the author define the following terms:

a) "numerosity"(paragraph 2) _____

b) "a Hilbert space" (paragraph 7) _____

Check [✓] TRUE or FALSE for each statement. Only if a statement is FALSE then:

- a. Copy the incorrect word and correct the FALSE statement (use one word only).
- b. Do not use the word 'NOT' as a correcting word.
- c. If both statements are marked as TRUE, none will be accepted.

1. Mundurucu Indians can distinguish between quantities larger than five can although their language includes words for such numbers. (par. 3)

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

2. Nuñez is in agreement with scientists like Tegmark who postulate that the universe is a mathematical structure waiting to be discovered. (par. 8).

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

Circle the correct answer.

- 1) The author starts out by quoting mathematician Maryam Mirzakhani in paragraph 1
- a) in order to demonstrate that mathematics is a difficult subject
 - b) by way of introducing the subject of mathematics as something that should be deciphered
 - c) to emphasize that she is among the few who could decipher mathematics
 - d) to suggest that mathematics is comparable to a jungle

- 2) The statement that "most of us have spent enough time in its periphery" (paragraph 1) means that the majority of people
- a) have studied relatively advanced mathematics
 - b) try to stay away from mathematics
 - c) have had encounters with mathematics, albeit not as professionals
 - d) consider mathematics to be a subject of little interest to them
- 3) Mirzakhani and Friston present _____ views on mathematics.
- a) similar
 - b) unclear
 - c) complex
 - d) opposite
- 4) Stanislas Dehaene uses the example of three red marbles to demonstrate (par. 2)
- a) the perception of numbers and colours
 - b) numerosity
 - c) accuracy in perceiving numbers below 4
 - d) fuzziness in perceiving numbers above 3
- 5) The "nativist" view implies that
- a) we are born with mathematical abilities
 - b) cognitive neuroimaging is innate
 - c) only humans have native mathematical ability
 - d) native Indians have mathematical cognition
- 6) How does the example of Mundurucu Indians demonstrate that numerosity does not depend on culture?
- a) They do not have words for numbers larger than 5
 - b) Despite their limited vocabulary, they possess rudimentary mathematical ability
 - c) Their culture places a great value on mathematical symbols
 - d) Their culture is deficient in all respects that are related to numerosity
- 7) Leibovich's conclusion is in agreement / disagreement with the conclusions drawn by Spelke and Dehaene .

Support your answer by quoting from the text.

- 8) The example of the young children capable of distinguishing between different arrays of dots can be interpreted in two ways. What are they?
- a. _____
- b. _____
- 9) Why do five watermelons seem more than five oranges to children and which of the two above-mentioned views does this perception demonstrate?
- _____
- _____
- 10) According to Ansari, counting is innate / culturally biased. (circle one). Support your answer with two examples from paragraph 4.
- _____
- _____
- 11) We can understand from the examples given in paragraph 5 that
- a) English is far from being a highly developed language
 - b) the Yupno language does not have demonstratives
 - c) numerosity is a cultural phenomenon
 - d) the Yupno language is not comparable to the Australian Aboriginal languages
- 12) The study conducted by Dehaene and Amalric demonstrates that
- a) while mathematicians have the capacity for mathematical thought, other academics do not
 - b) only mathematicians seem to have developed a "maths network"
 - c) mathematical thought is an inherent characteristic of languages that have quantification
 - d) mathematicians stop using normal language once they have learnt symbolic mathematical language.
- 13) What is the main idea of paragraph 6?
- a) Mathematics in Mesopotamia was a cultural invention
 - b) Mathematics developed out of cultural necessity
 - c) There are numerous examples of the use of mathematics in many cultures
 - d) The exact moment of transition from mathematics as an instinct to mathematics as a cultural product is unknown

Circle the correct answer.

- 1) What connector can best fill in the blank in par. 1, line 5?
 - a) Furthermore
 - b) Therefore
 - c) Nonetheless
 - d) Subsequently

- 2) The word "its" (par. 1, line 19) refers
 - a) patterns and regularities
 - b) world
 - c) organisms
 - d) evolutionary tree

- 3) The word "long-standing" (par. 2, line 2) is used as
 - a) a noun
 - b) a verb
 - c) an adjective
 - d) an adverb

- 4) The word "hardwired" (par. 2, line 9) is used as
 - a) a noun
 - b) a verb
 - c) an adjective
 - d) an adverb

- 5) The word "their" (par. 3, line 8) refers to
 - a) numbers
 - b) values
 - c) humans
 - d) studies

- 6) The word "These" (par. 3, line 13) refers to
 - a) attributes
 - b) the number of dots
 - c) clues
 - d) spatial distribution and area of coverage

- 7) The word "further" (par. 5, line 9) is used as
 - a) a noun
 - b) a verb
 - c) an adjective
 - d) an adverb

- 8) The word "as" (par. 6, line 12) means
- a) when
 - b) like
 - c) since
 - d) so
- 9) What modal can best fill in the blank in par. 7, line 8?
- a) would
 - b) should
 - c) might
 - d) could

Hello, Cool World

Fred Pearce

We can limit global warming to 1.5 °C, finds **Fred Pearce**. But it's going to take a radical approach

Par. 1 - ECSTATIC celebrations broke out in Paris just before Christmas. World leaders acclaimed their freshly forged deal to fight climate change. The Paris agreement says governments will find a way to limit global warming to “well below 2 °C” and do their best to have no more than 1.5 °C. But is that even possible? Were the celebrations justified?

Par. 2 - _____ aiming for 1.5 °C is definitely a good idea, since two degrees of warming will almost certainly wipe low-lying islands off the map, it is also an impossible goal, according to some climate scientists. Others are more optimistic, and think 1.5 °C might just be within reach. One thing is clear: if we are going to do it, we will have to create a whole new industry to suck vast quantities of carbon dioxide directly from the air. Without these “negative emissions” we can bid farewell to 1.5 °C.

Par. 3 - The essential task is to control the build-up of CO₂ in the atmosphere. Humans have pushed levels from 280 parts per million(ppm) before the Industrial Revolution to 400 ppm at present by burning fossil fuels and trashing nature. As a result, in 2015, meteorological stations around the world recorded a 1 °C rise in global temperatures above pre-industrial times. The key number here is 430 ppm – that's the level that the UN Intergovernmental Panel on Climate Change (IPCC) says corresponds to 1.5 °C of warming. At this stage, it is effectively impossible to stop us overshooting that number. But we could, theoretically, get back down to 430 ppm by the end of the century – assuming we are willing to create a whole new mega-industry of negative emissions.

Par. 4 - Joeri Rogelj of International Institute for Applied Systems Analysis (IIASA) is one of the first to analyse how we could do this. From his calculations, it's possible to draw up a hypothetical road map to 1.5 °C. The first step is to bring net global emissions to zero by 2050, emitting no more than 800 gigatonnes cumulatively between now and then . Keeping emissions at zero after 2050 should be enough to limit warming to below 2 °C. To

bring warming back down to 1.5 °C, we would then have to remove some 500 Gt from the air.

Par. 5 - Net zero emissions by 2050 is no small feat. Some fossil fuels could be burned, but only as much as can be sucked out of the atmosphere. Still, there is a glimmer of hope. According to a preliminary assessment by the Global Carbon Project, last year CO₂ emissions from power stations and industrial sources fell back to 2013 levels, despite a global economy that was 6 per cent larger – the first time this has happened outside a global recession.

Par. 6 - Corinne Le Quéré, who led the analysis and is head of the Tyndall Centre for Climate Change Research at the University of East Anglia, UK, cites three causes. First, we are burning less coal. China is getting out of coal fast, motivated by its smog problem as much as anything else. _____, coal's contribution to US electricity generation has shrunk from 53 per cent to 35 per cent in five years. Second, investment in renewable energy now exceeds that going into fossil-fuel power stations globally. Optimists hope the Paris deal will boost this trend. Finally, energy efficiency continues to improve in most places. Le Quéré says emissions will probably still rise again, but we may be surprisingly close to their peak. China, the US and the European Union – the world's biggest emitters – are on the right track.

Par. 7 - There are several ways to chemically absorb CO₂ directly from the air – you may have heard of them as “artificial trees”. All are prohibitively expensive, both in cash and energy demand, says Pete Smith of the University of Aberdeen. **A program big enough** to meet Rogelj's 500 Gt target would cost about \$270 trillion and consume a quarter of the world's energy supply, he says. Equally unlikely on cost grounds is the idea of augmenting the natural erosion of carbonate rocks, a process that removes CO₂ from the atmosphere.

Par. 8 - If chemicals and rocks can't do the job, what about actual trees? Deforestation is currently responsible for more than 10 per cent of our CO₂ emissions. If we are to stop temperatures rising, the target agreed in New York in 2014 for zero global deforestation by 2030 is a must. After that, replanting could make forests into negative emitters. Where land is available, this is a bargain option. Smith estimates that the cost is lower than that of the chemical or geological options. That said, it still adds up to a total bill of \$14 trillion over 50 years.

Par. 9 - Another problem is that trees take a lot of land. Planting enough of them to soak up 500 Gt of CO₂ over 50 years would call for 10 million square kilometres – an expanse as big as the United States, which isn't available. _____, trees could still be a part of the solution. Why not, asks Smith, plant some trees, cut them when they reach maturity – at which point they stop soaking up as much CO₂ anyway – and burn them in power stations? If we then planted the cleared land with new trees or fast-growing energy crops, they could eventually soak up the same amount of CO₂ again, resulting in carbon-neutral energy over the long term. Then we could capture the concentrated CO₂ before it leaves the power station and bury it. The net effect would be negative emissions that also generate energy. Bioenergy with carbon capture and storage (BECCS), as this is known, effectively reverses the way we generate energy today. Instead of digging up concentrated carbon in the shape of coal, oil and gas, burning it and releasing the waste into the atmosphere, we could grab diffuse CO₂ from the air with plants, burn them, and bury the carbon in concentrated form underground. The more electricity we generate, the bigger the negative emissions.

Par. 10 - Smith's study found that BECCS could capture 500 Gt of CO₂ over the second half of the century, while contributing a large fraction of the world's electricity. "Of all the ways of achieving negative emissions, BECCS seems to be the most promising," says Florian Kraxner of IIASA. But the big issue with BECCS is land. It requires less than simply planting trees, but at its height, delivering 500 Gt of negative emissions would still take anywhere between 3.8 and 7 million square kilometres, says Smith. And ideally most of that would be in the tropics, where plants grow faster – but where the need for land to grow food is also greatest. "We have to feed a population of 9 to 10 billion people in 2050," Smith says, "which is a challenge of similar proportions to that posed by limiting climate change to 1.5 °C."

Par. 11 - The danger is that people will find new land to grow their food by clearing forests or grasslands. An analysis last year by Andrew Wiltshire of the UK Met Office and Taraka Davies-Barnard of the University of Exeter argued that this could reduce the effectiveness of BECCS by up to 80 per cent. What is really needed is a solution that sucks CO₂ out of the atmosphere while boosting food production. A good solution would be to bury biochar: agricultural waste like straw, manure and unused food that has been pyrolysed to charcoal. Burying biochar is an industrialised version of traditional methods practised by tribes in the Amazon and elsewhere to enrich their soils. Because biochar contains tightly

bound carbon atoms, it is stable for hundreds of years. Smith calculates that biochar could provide up to 125 Gt of negative CO₂ emissions between 2050 and 2100.

Par. 12 - Finally, there is another option that could help. Consider the fact that most of the planet is ocean – Earth's largest carbon sink. Tricking the oceans into taking in more CO₂ by fertilising plankton with iron or nitrogen, something that has been widely discussed in recent years, is stymied by our ignorance of how carbon cycles work below the waves, says Smith, who dismisses this idea as unrealistic. But farming the oceans might work. Last year, Brian Walsh at the IIASA made the case for cultivating microalgae on a massive scale. The microalgae could be used as fuel for BECCS power stations or as animal feed, replacing soy and other land-hungry crops. Prototype microalgae farms are at work in Australia. By mid-century, there could be up to 50 million hectares of them growing billions of tonnes of biomass.

Par. 13 - Walsh estimates that such an expanse of microalgae could absorb up to 25 Gt of CO₂ a year. It might take a while to get going at scale. But even at a quarter of Walsh's rate, just 20 years could deliver the remaining 125 Gt needed to reach Rogelj's 500 Gt target, while feeding up to 10 per cent of the world's livestock. Forests could be saved for nature; fields for growing food; and global warming could be capped at 1.5 °C.

Par. 14 - Whether this will work depends on how well we know the climate. If we are unlucky, and greenhouse gases have a greater impact on temperatures than we think, then the chance of hitting 1.5 °C is zero. If current sensitivity estimates are right, then we stand a chance. We can dream. Or perhaps the world should get on with it and see if it works.

Adapted from New Scientist, 20 February 2016, vol. 229 (3061), pp. 30-33

Hello, Cool World - Questions

Answer in English

1. What are negative emissions, according to paragraph 2?

2. What is the definition of biochar? (paragraph 11)

Check [✓] TRUE or FALSE for each statement. Only if a statement is FALSE then:

- a. Copy the incorrect word and correct the FALSE statement (use one word only).
b. Do not use the word 'NOT' as a correcting word.
c. If both statements are marked as TRUE, none will be accepted.

1. The drop in emissions from power stations and industrial sources was in direct correlation with the growth in the global economy. (paragraph 5)

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

2. According to Corinne Le Quéré, emissions will continue rising in the more distant future. (paragraph 6)

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

3. We can infer that when atoms of a certain material are loose, that material is unstable. (paragraph 11)

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

4. Augmenting the oceans' carbon storage capacity is currently possible.
(paragraph 12)

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

Circle the correct answer.

- 1) The value of 430 ppm is important because (paragraph 3)
 - a) decreasing emissions level to this number will guarantee a 1 degree Celsius rise in temperature
 - b) by lowering emissions to 430 ppm we could actually limit global warming to a reasonable extent
 - c) this is what is needed to stop temperatures from rising
 - d) it will enable a return to pre-industrial global temperatures

- 2) What steps are necessary in order to limit global warming to 1.5 degrees?
(paragraph 4)
 - a) Emit up to 800 Gt every year until 2050
 - b) Keep emissions at zero before and after 2050
 - c) Bring emissions down to zero, avoid emissions altogether and remove 500 Gt from the air
 - d) Bring emissions to zero, emit no more than 800 Gt cumulatively and finally remove 500 Gt from the air

- 3) We can infer from paragraphs 5 and 6 that sucking emissions out of the air
 - a) must be complemented by a reduction in the burning of fossil fuels
 - b) is sufficient in itself to maintain net zero emissions
 - c) depends entirely on developments in the global economy
 - d) causes nations to become more efficient in their use of energy

- 4) What is the main idea of paragraph 7?
 - a) Artificial trees and carbonate rocks are viable solutions for absorbing CO₂
 - b) Augmenting the natural erosion of carbonate rocks is a better solution than artificial trees
 - c) Using artificial trees is the only solution for removing CO₂ out of the air which is not prohibitively expensive
 - d) The two methods suggested for removing CO₂ from the air are unviable due to high costs

- 5) Smith suggests cutting trees when they reach maturity because (paragraph 9)
- a) trees become useless once they mature and they take up too much land
 - b) at this point they could be more effectively used to produce energy rather than to absorb CO₂
 - c) burning trees produces cleaner energy than burning coal, oil and gas
 - d) there is a growing demand for electricity generated by burning trees
- 6) The main idea of paragraph 10 is that
- a) BECCS is not the optimal solution for limiting gas emissions
 - b) the advantages of BECCS outweigh its disadvantages
 - c) BECCS may solve the problem of emissions, but threatens food sources
 - d) using BECCS to reduce emissions will at the same time free up lands for food production
- 7) The ideal viable option for limiting greenhouse gas emissions might be (paragraph 12)
- a) using the BECCS method
 - b) cultivating microalgae
 - c) fertilizing plankton with carbon and oxygen
 - d) learning how carbon cycles work below the waves
- 8) The advantage of microalgae is that they
- a) can be used to fertilize the oceans
 - b) provide a way of farming the oceans
 - c) eliminate the land problem
 - d) can be used to fuel BECCS power stations
- 9) The numbers in paragraph 13 indicate that
- a) Walsh's estimate regarding the absorption capacity of microalgae is realistic
 - b) microalgae can help achieve the target global warming limitation in a timely manner
 - c) producing sufficient crops of microalgae make take up to 20 years
 - d) cultivating microalgae has advantages as well as disadvantages
- 10) In the last paragraph, the author _____ with regard to limiting global temperature.
- a) expresses a large degree of skepticism
 - b) is relatively pessimistic
 - c) brings up two alternative scenarios
 - d) expects positive developments

Circle the correct answer.

1. What connector can be used in the blank space in paragraph 2?
 - a. Because
 - b. While
 - c. For example
 - d. Thus

2. The phrase "these negative emissions" (paragraph 2) refers to
 - a. aiming for 1.5 degrees Celsius
 - b. avoiding two degrees of warming
 - c. taking a pessimistic view with regard to preventing warming
 - d. eliminating the greenhouse gases already found in the atmosphere

3. The word "build-up" (paragraph 3) is used as
 - a. a verb
 - b. an adjective
 - c. a noun
 - d. an adverb

4. The phrase "it is effectively impossible to stop us overshooting that number (paragraph 3) means that
 - a. We will exceed 1.5 degrees Celsius of warming
 - b. We will overcome the difficulty of preventing 1 a 1.5 degrees Celsius increase in temperature
 - c. We can get back to 430 ppm by the end of the century
 - d. We will fail in lowering temperatures by 1.5 degrees Celsius

5. The word "this" (paragraph 5) refers to
 - a. a larger global economy
 - b. 6 percent
 - c. a drop in emissions
 - d. a global recession

6. Which connector could be used in the blank in paragraph 6?
 - a. As a result
 - b. However
 - c. By contrast
 - d. In addition

7. The word "prohibitively" (paragraph 7) is
- a. an adjective
 - b. an adverb
 - c. a verb
 - d. a noun
8. What connector can be used in the blank in paragraph 9?
- a. Consequently
 - b. However
 - c. For example
 - d. For this reason
9. The word "remaining" (paragraph 13) is used as
- a. a noun
 - b. a verb
 - c. an adverb
 - d. an adjective
10. The word "sensitivity" (paragraph 14) is used as
- a. an adverb
 - b. a noun
 - c. an adjective
 - d. a verb

Welcome to Moon City!

Did you ever wonder what could have happened if NASA's Apollo space program hadn't been canceled? Henry Spencer thinks about the challenges faced by "Moon architects", design possibilities and overall potential.

Par. 1 – In the late 1960s, the Apollo space program which succeeded in preparing and landing the first humans on the Moon, was faced with difficulties. Just three years after Neil Armstrong's famous historic first steps on the Moon, human exploration of the Moon was coming to an end. At that time, the US was spending too much of its finances on the Vietnam war. There were also various incidents on the Apollo launch pad in 1967, which did not look good for NASA. And on the other side of the globe, there was no apparent competition from the Soviet Union – its moon effort seemed to be going nowhere.

Par. 2 – The congress therefore decided to refuse the further funding of moon exploration, in the summer of 1967. But what if things had turned out differently? Henry Spencer ponders an alternate future where the Apollo program continued and how it would have affected human exploration of space.

Par. 3 – Imagine - it's July 2009, and on the Moon there is a city - Johnson City. America's permanent colony on the Moon, named after president Lyndon B. Johnson, the one who authorized it. Citizens of the Moon, already living there for three generations by now, are celebrating a special occasion for "Moon Americans": the first child born to parents themselves born on the Moon. With just 5000 inhabitants, "city" is perhaps too grandiose a term, but it is more than enough.

Par. 4 – From an architect's point of view, it is imperative to fit the design to the harsh reality. Building a city on the Moon requires a lot of planning and realistically grounded thinking. _____, those who had anticipated a city covered with a space dome, of the kind they once read about in science fiction books and comics, would be disappointed. A dome would never survive the Moon environment, in which cosmic rays and meteorites continuously bombard the surface.

Par. 5 – Fortunately for Moon City planners and builders, there was a ready-made alternative: subsurface lava tubes carved by extinct volcanic flows. Lava tubes were formed on the Moon due to eruption of basaltic lava. As the surface of the lava cooled and hardened, it created a tube-like pathway for the remaining lava to flow through beneath the surface. The early Apollo missions had seen evidence for partly collapsed tubes in the form of intermittent surface trenches. The extreme dryness of the rock and the Moon's weak gravity mean that tubes can be a kilometer or more in diameter and remain stable.

Par. 6 – Lunar lava tubes that run underneath the Moon's surface are ideal for building a city. The arched shape of the tubes provides an effective barrier against micrometeorites, radiation, and extreme temperature swings. When constructing a structure on the Moon in such a manner, the living quarters will need to be either buried under the surface or further protected with a protective covering. Furthermore, it is possible to house large structures within lava tubes. Moon settlers could even seal and fill specific sections with air, providing ample space for them. Another benefit of lava tubes is their widespread

presence across different areas of the Moon, which can play a significant role in the initiation of commercial mineral mining.

Par. 7 – As for the latter, many scientists believe that the most valuable resource on the Moon is helium-3, which can serve as fuel for thermonuclear reactors. While the largest deposits of this element are located near the lunar equator, where there are no craters containing water ice, there are lava tubes in this area that can house all the necessary infrastructure.

Par. 8 – In the beginning of this project, the government had run almost everything in Johnson City. These days, though, lunar exploration increasingly depends on the colony's independent businesses. Shipping costs of material from Earth are still astronomical, and local extraction of water and oxygen is also a prosperous business. There are thriving schools for the colony's children, but more advanced education _____ be beamed up from Earth.

Par. 9 – Forty years on from Apollo, passenger flights into space are routine, but traveling to Earth is still expensive, and lunar inhabitants can't easily adapt to its higher gravity. Even if, like 90 per cent of the population, you were born on Earth, you can't easily go back. These problems are not shared by space tourists, who pay good money for about a week's stay in orbit – and soon in the first hotel in Johnson City. There's a Russian lunar colony too, a legacy of Nikita Khrushchev's desire to establish a permanent moon presence for the Soviet Union. In fact, the Moon in 2009 is more multinational than ever: European and Japanese voices fill Johnson City, and China and India are considering starting their own colonies.

Par. 10 – On the other hand, exploring beyond the Moon has been less of a success. In 1980, three US astronauts landed on Mars for a stay of a week, but that was a hastily conceived "flags and footprints" effort to keep ahead of the Soviet competition. A Soviet landing and a couple more US missions followed, but by the late 1980s crewed missions to Mars had stopped. NASA's fixed budget was strained by cost overruns in establishing Johnson City, and the tottering Soviet economy was unable to match even that.

Par. 11 – So apart from a visit to two near-Earth asteroids using leftover Mars hardware in the early 1990s, exploration beyond the Moon stopped until the turn of the millennium. Technological improvements then inspired a cautious revival. Not least of these was the ability to supply rocket fuel extracted from the Moon's unexpectedly rich polar ice deposits, meaning long-distance missions could be launched far more cheaply from the Moon than from Earth. In January 2004, a team of four Americans and two Europeans made it back to Mars, spending 14 months on the first extended surface exploration. They found spectacular scenery, and the odd hint of past riches: traces of organic compounds, tiny rock features that might be micro fossils. But life on Mars was clearly never common. If it is still around, it probably only inhabits niches such as the neighborhood of geothermal vents – if they exist.

Par. 12 – Coming back to reality, progress in space has been slower than in the dreams of the pioneers of the early 1960s. But with a permanent presence on the Moon, and plans for bases further afield, the idea is firmly entrenched that, for humanity, the sky is no limit.

Adapted from Henry Spencer, "Welcome to Lunarville", *New Scientist*, 11 July 2009

Welcome to Moon City – Questions

I. Answer the following questions in English.

1. What were the challenges faced by the Apollo space program in the late 1960s? (par. 1)

2. What were the advantages of using subsurface lava tubes for building a city on the moon? (par. 6)

II. Check (v) TRUE or FALSE. Correct if the statement is FALSE (use one word only).

1. The US government decided to discontinue the successfully started Apollo space program regardless of a number of obstacles. (par. 1)

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

2. The construction of a protective space dome on the Moon would become a highly realistic project. (par. 4)

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

3. Lunar lava tubes could host humans and provide comfortable living conditions. (par. 6)

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

III. Choose the best answer, according to the information in the text.

1. What is NOT one of the reasons for Apollo's Moon exploration coming to an end?

(par. 1)

- a. The US money expenditure on the Vietnam War
- b. Neil Armstrong's first steps on the Moon
- c. Soviet Union's state moon exploration efforts
- d. The Apollo launch pad problems

2. What was the main reason of naming the permanent colony on the moon "Johnson City" after President Lyndon B. Johnson? (par. 3)

- a. He was the first president to land on the Moon
- b. He had a personal connection with the Moon
- c. He approved the building of a permanent colony on the Moon
- d. He was a scientist who was interested in exploring the Moon

3. What was the biggest challenge faced by architects while building a city on the moon?

(par. 4)

- a. Building a city with a space dome
- b. Building a city without the help of NASA
- c. Building a city based on science fiction books
- d. Building a city that fits the lunar environment

4. How were lunar lava tubes created? (par. 5)

- a. By extinct volcanic flows that carved a way over the surface
- b. By the constant fluctuations in the Moon's gravity
- c. By the changes in molten lava beneath the surface
- d. By the presence of moisture in the rocks on the Moon

5. What is the main idea of paragraph 6?
 - a. Mineral mining can be efficiently carried out in many different areas of the Moon
 - b. Lava tubes can be filled with air so as to provide habitable space
 - c. People on the Moon are exposed to micrometeorites and radiation
 - d. Lava tubes can be used to make building a city on the Moon possible

6. Why were lunar lava tubes ideal for building a city on the moon? (par. 6)
 - a. They were able to safely accommodate living areas within them
 - b. They provided a defective barrier against micrometeorites and radiation
 - c. They provided enough space only for compact buildings with protective covering
 - d. They were widespread in a specific area of the Moon

7. We can infer from paragraph 8 that the reason for the government running almost everything in Johnson City at the beginning of the lunar colony project was
 - a. the astronomical shipping costs of material
 - b. the gradual increase of successful lunar exploration
 - c. the lack of independent lunar businesses
 - d. the extraction of oxygen being economically advantageous

8. What is the topic of paragraph 9?
 - a. The routine of space flights forty years on from Apollo
 - b. The problems of space tourists forty years on from Apollo
 - c. The legacy of Nikita Khrushchev forty years on from Apollo
 - d. The state of life on the Moon forty years on from Apollo

9. We can infer from paragraph 9 that Moon inhabitants cannot easily adapt to Earth's gravity since
 - a. The Earth's weaker gravity makes them more susceptible to disease
 - b. They have lived in a low-gravity environment for too long
 - c. Their bodies are much stronger and more flexible than those of Earth people
 - d. Passenger flights to Earth are too expensive for them to often leave the lunar environment

10. What has caused the renewed interest in space exploration beyond the Moon?

(par. 11)

- a. Scientific innovations
- b. The discovery of rocket fuel
- c. Cost overruns in establishing Johnson City
- d. An American-European mission to Mars

IV. Choose the best answer, according to the text.

1. The word “which” (par. 1, line 6) refers to

- a. the Vietnam war
- b. the Apollo launch pad
- c. various incidents
- d. finances

2. The word “further” (par. 2, line 1) is used as

- a. an adjective
- b. a noun
- c. a verb
- d. an adverb

3. The word “grounded” (par. 4, line 2) is used as

- a. a noun
- b. a verb
- c. an adjective
- d. an adverb

4. What connector can best fill in the blank in paragraph 4, line 3?

- a. Moreover
- b. However
- c. For example
- d. Therefore

5. The word "while" (par. 7, line 2) means
- a. since
 - b. although
 - c. during
 - d. as
6. The word "house" (par. 7, line 4) is used as
- a. a noun
 - b. an adjective
 - c. a verb
 - d. an adverb
7. What modal can best fill in the blank in paragraph 8, line 6?
- a. is able to
 - b. has to
 - c. should
 - d. would
8. The word "its" (par. 9, line 2) refers to
- a. space
 - b. inhabitants
 - c. gravity
 - d. Earth
9. The relationship between paragraphs 9 and 10 is that of
- a. comparison
 - b. contrast
 - c. illustration
 - d. addition

10. The word "hardware" (par. 11, line 2) means
- a. mechanical part of computers
 - b. major items of equipment
 - c. educational programs for a particular purpose
 - d. set of instructions

More Evidence, Less Poverty

Money and good intentions are not enough to fight poverty effectively. We also need data about what works and what doesn't

Dean Karlan, *Scientific American*, 1 October 2015

Par. 1 - You can't make money without money. That was the exciting and intuitively obvious idea behind microloans, which took off in the 1990s as a way of helping poor people out of poverty. Banks wouldn't give them traditional loans, but small amounts would carry less risk and allow entrepreneurs to jump-start small businesses. Economist Muhammad Yunus and Bangladesh's Grameen Bank figured out how to scale this innovation and won the 2006 Nobel Peace Prize for their work.

Par. 2 - The trouble is that _____ microloans do have some benefits, recent evidence suggests that on average they increase neither income nor household and food expenditures—key indicators of financial well-being.

Par. 3 - That a program could be celebrated for more than 20 years and lavished with money and still fail to help people out of poverty underscores the paucity of evidence in antipoverty programs. Individual Americans, for instance, spend \$335 billion a year on charity, yet most people give on impulse or a friend's recommendation—not because they have evidence that their giving will do any good. Philanthropies also often give money to projects without really knowing if they are successful.

Par. 4 - Fortunately, we are living in the age of big data: decisions that used to be made on instinct can now be based on solid evidence. In recent years social scientists have begun to marshal the tools of big data to ask the hard questions about what works and what doesn't. The goal is to turn philanthropy into a science, where money gets directed to programs for which there is strong evidence of their effectiveness.

Par. 5 - I learned about microloans in 1992, on what was supposed to be a short detour from a career in hedge funds. As a 22-year-old intern in El Salvador for one of the largest microlenders, I was struck by how little the organization knew about their effect on clients—usually women—and the local economy. They knew that many customers were coming back for more loans and saw “client retention” as proof of their success. Why else would customers keep borrowing if it was not helping? But the microlenders did not have any serious evidence that the loans were helping women get their families out of poverty.

When I asked about evidence on impacts, I was directed to a perfunctory questionnaire. I wondered: maybe repeat borrowing is not good if the client's business does not continue to grow. Perhaps true success would be to provide *one* loan to help someone in need and then down the road to discover the borrower to be stable enough not to need another.

Par. 6 - Here was a huge nongovernmental organization pulling in large grants to help the poor, with no real measurement of whether their efforts were working. For-profit businesses have benchmarks to know how they are performing, but most donors are not accustomed to asking charities about their results. Sometimes they ask what proportion of money goes to overhead, but that number is mostly meaningless. The question that needs to be asked—and that needs to be asked every time someone writes a check to a charity or a government commits to a multimillion-dollar aid project—is, Will this actually work to alleviate poverty? In other words, how will people's lives change, compared with how their lives would have changed without the program?

Par. 7 - This question knocked me off my Wall Street track and into graduate school for economics. One of my professors, Michael Kremer, had just started conducting randomized controlled trials to learn what programs work to help kids stay in school and improve the education they receive. He was borrowing this method from health and other sciences—randomly assigning schools to either receive a particular resource (the treatment group) or remain as they would have been otherwise (the control group) and then comparing school performance across these two groups.

Par. 8 - His approach gave me an idea about how to return to the microlending questions that had brought me to academia in the first place. So off I went in my fourth year of graduate school to South Africa to set up my first experiment on the question of whether microlending is effective. But it turns out to be difficult for academics at universities to carry out studies far away with the level of detail that good scientific trials require. You need reliable staff on the ground who understand the science but who also have the social skills to work with partners and manage field operations.

Par. 9 - By 2002, as I was starting out as a professor, I founded a nonprofit called Innovations for Poverty Action (IPA) to help fill these knowledge gaps in finance, health, education, food, and peace and postconflict recovery. IPA connects academics at the Massachusetts Institute of Technology, Yale University, and the like, with a trained staff of more than 500 people working in 18 countries on randomized controlled trials. We have now conducted upward of 500 trials. A chief insight has been that simple interventions that take human behavior into account can have outsized effects. Putting chlorine dispensers right next to water sources, to make it easy to remember and publicly observable, increases use of clean water sixfold. Adding a simple bag of lentils to a

convenient monthly immunization camp for families in India roughly sextuples rates of full immunization for kids (while making the entire process cheaper because more families show up). And cheap and simple text message reminders can be effective in helping people accomplish their goals, from saving money to completing their medication regimens. Naturally not everything works. We must figure out what works and what doesn't.

Par. 10 - We have also learned that information is only part of the solution. Having strong relationships with local governments, nonprofits, businesses and banks keeps the academic experts working on questions that matter and gets answers into hands of the people who can use them.

Par. 11 - Over the years microloans kept nagging at my colleagues and me. Fifteen years after my first study attempt in South Africa, we now have seven randomized trials completed on traditional microloans and one on consumer lending back in South Africa. The seven projects are spread out around the world and have been conducted by different researchers with similar research designs: in Bosnia and Herzegovina, Ethiopia, India, Mexico, Mongolia, Morocco and the Philippines. These studies found some benefits of microloans, such as helping families weather hard times, pay off goods over time and even make small investments in businesses. But there was no average impact on the main financial well-being indicators—income and household and food expenditures. To the chagrin of microloan critics, there also were no big negative effects. So what *does* work to increase income for the world's poorest?

Par. 12 - We just recently studied another program that addresses some of the shortcomings of microloans. One sad failure of many programs (including microloans) has been in reaching the poorest of the poor—known in the field as the ultrapoor. They live on less than what \$1.25 would buy in the U.S. a day, and they account for more than a billion people, or one seventh of the world's population. The things keeping them poor are usually complicated enough that no one individual fix is going to help, but one program being run in Bangladesh by BRAC, the world's largest nonprofit organization, and a few other places stands out. It saw extreme poverty as a complex problem deserving of a complex solution. Its “graduation” approach, designed to move the extreme poor out of their current conditions, offers a package of six items:

1. A “productive asset,” that is, a way to make a living (livestock, beehives to make honey or supplies to start a simple store).
2. Technical training on how to use the asset.

3. A small, short-term regular stipend, to meet immediate needs for daily living so the individual does not have to sell the asset while learning how to use it.
4. Access to health support, to stay healthy enough to work.
5. A way to save money for the future.
6. Regular (usually weekly) visits from a coach, to reinforce skills, build confidence and help participants handle any challenges they encounter.

Par. 13 - The Ford Foundation and Consultative Group to Assist the Poor in Washington, D.C., came to me with an ambitious idea: test an identical program, implemented by different organizations in multiple places. We ended up conducting similar studies in six places: Ethiopia, Ghana, Honduras, India, Pakistan and Peru. What we found was unprecedented—everywhere the program worked, it worked well. When we came back a year after the program had ended, we found the impact had lasted: people had more money to spend and food to eat. When we calculated the costs (labor, asset costs, transportation and overhead) as compared with the benefits, the overall returns were positive in five out of six countries—ranging from 133 percent in Ghana to 433 percent in India. In other words, every dollar invested in India yielded \$4.33 more food and spending for ultrapoor households.

Par. 14 - The one exception was Honduras, where the productive asset most used by the local organization—chickens—was an outside breed that was not resistant to local disease and so became sick and died. This was a humanitarian failure, but it demonstrated that the asset is an essential component of the program. Remove that component, and the other five components did not generate positive impacts on their own. As the programs are expanding in Ethiopia, India and Pakistan, we hope to learn more about how to make this program work better, either by reducing costs or by improving the services.

Par. 15 - There is no panacea in the fight against poverty. Even a graduation program for the ultrapoor, which is ready to scale and yields an excellent return for a charitable buck, is not going to transform the ultrapoor into car-buying middle-class households. The vision statement for Innovations for Poverty Action is appropriately modest: more evidence, less poverty. We are not going to end poverty, but with proper evidence we can make important strides.

More Evidence, Less Poverty - Questions

1) Find definitions for the following terms in the article.

- a) microloans _____
- b) big data _____
- c) client retention _____
- d) the ultrapoor _____
- e) productive asset _____

2) Match the words from the text with their synonyms.

Word	Paragraph	Synonym
underscores	3	ease
paucity	3	solution
impacts	5	standards
benchmarks	6	spending
alleviate	6	cure-all
insight	9	influences
expenditures	2	emphasizes
fix	12	scarcity
shortcomings	12	realization
panacea	15	disadvantages

Check [✓] TRUE or FALSE for each statement. Only if a statement is FALSE then:

- a. **Copy the incorrect word and correct the FALSE statement (use one word only).**
- b. **Do not use the word 'NOT' as a correcting word.**
- c. **If both statements are marked as TRUE, none will be accepted.**

1. The article mainly praises microloans.

TRUE FALSE

Copy the incorrect word: _____

Write the correction: _____

2. A decrease in food expenditure is one of the main indicators of financial prosperity.

TRUE FALSE

Copy the incorrect word: _____

Write the correction: _____

3. According to the author, client retention indicated that microlending is a successful tool.

TRUE FALSE

Copy the incorrect word: _____

Write the correction: _____

Comprehension and Critical Reading

4. Consider the title of the article. What kind of evidence is required? Find what connector is missing in the gap in paragraph 2?
- _____
5. Why does the author compare and contrast between aid projects / philanthropies and for-profit businesses? (paragraph 6)
- _____
6. How are Michael Kremer's randomized controlled trials, mentioned in paragraph 7, relevant to Karlan's type of research?
- _____
7. Putting chlorine dispensers next to water sources, or adding a bag of lentils to an immunization camp are examples of _____ (paragraph 9)
8. What does the author mean when he says that "microloans kept nagging at my colleagues and me" (paragraph 11)?
- _____
9. The six items listed in paragraph 12 constitute an example of _____
10. Were the programs supported by the Ford Foundation and CGAP successful in the long term? YES / NO (paragraph 13). Support your answer with a quote from the text.
- _____
11. "The one exception was Honduras" (paragraph 14) means that the study conducted in Honduras was successful / unsuccessful.
- _____

Tone of the article

- 1) The tone in paragraph 6 is mostly
- a. appreciative
 - b. critical
 - c. optimistic
 - d. sarcastic
- 2) The tone in the last paragraph of the text is
- a. pessimistic
 - b. ironic
 - c. realistic
 - d. indifferent

Parts of Speech

Group the following words from the text according to their function in the sentence. Check (V) the suitable box.

Word	Paragraph	Noun	Verb	Adjective	Adverb
scale	1				
household	2				
key	2				
well-being	2				
lavished	3				
antipoverty	3				
giving	3				
works	4				
proof	5				
helping	5				
impacts	5				
repeat	5				
borrowing	5				
For-profit	6				
overhead	6				
check	6				
just	7				
approach	8				
microlending	8				
connects	9				
Putting	9				
monthly	9				
nonprofits	10				
weather	11				
ultrapoor	12				
car-buying	15				

How to Be Curious

Paragraph 1 -- Curiosity is the driving force behind science, exploration and discovery. It can also guide us into passionate, purpose-filled lives. _____, it can lead us to waste time on clickbait and fake news, doomscroll through social media feeds or chase dangerously extreme experiences. In a modern world awash with many diversions, it would be good to know how to make the most of our curiosity while avoiding its pitfalls. Recent research on the double-edged nature of curiosity has not only shed light on its many benefits for learning and creativity, but also on the reasons that it can lead us astray – and why we should sometimes seek to curb our curiosity.

Paragraph 2 -- While curiosity is undoubtedly a complex psychological state, most researchers broadly define it as a drive to know things and gather information about the world – something that all organisms need to do. “Information is as fundamental for life as energy”, says cognitive neuroscientist Jacqueline Gottlieb of Columbia University. “A worm or amoeba gathering information about its environment, like where it can find food, shows curiosity, even if it’s a very limited and immediate kind.” Curiosity in humans _____ be more expansive, open-ended and powerful. But at its most basic level, our hunt for information is sparked by a desire to deal with uncertainty.

Paragraph 3 -- Recent research suggests that this is particularly important for babies and children. Celeste Kidd, a psychologist at the University of California, has devised a series of visual scenes of varying predictability. These videos showed objects like toy fire engines being momentarily hidden by a screen that was repeatedly dropped and raised. Each time the screen lifted, the object would be present with a certain probability. In some cases, the object was almost always present, making the sequence very predictable, while in other cases the chance of it reappearing was much more random, with low predictability. They also saw scenarios in between. Kidd then measured participants’ attention with eye-tracking tools to find out exactly where they were looking. In children as young as 7 months old, she found that sequences of intermediate predictability elicited more visual exploration than either tediously predictable or confusingly random sequences.

Paragraph 4 -- Kidd describes this gap between predictability and uncertainty as the Goldilocks effect. It is the relatively unpredictable – but not totally regular or completely

erratic - situations that offer the most useful opportunities to learn. We are particularly curious when we face an “information gap” – some unsolved mystery or unanswered question. In much the same way that those visual scenes for children needed to have the Goldilocks effect of predictability and uncertainty to grab their attention, the size of the information gap matters. If it is too big, the question feels unmanageable and daunting; too small, and it feels like irrelevant minutiae best ignored. We are most curious about stuff that falls somewhere between the two - something that is surprising and useful, but not wholly unfamiliar.

Paragraph 5 -- Finding that balance will be crucial for learners and their teachers. Numerous studies have shown that the more curious people feel about getting the answer to a trivia question, the better they remember the fact. Matthias Gruber at Cardiff University has discovered the reasons why this is so. Gruber asked people to report their own curiosity while they viewed different facts and scanned their brain activity. In Gruber’s model, curiosity begins with some kind of uncertainty or information gap that activates the hippocampus (a brain area responsible for memory as well as for detecting new stimuli) and the anterior cingulate cortex, which monitors informational mismatches in the brain. The brain then appraises how rewarding it will be to fill in the hole in its knowledge. If that appraisal is positive, we enter a state of curiosity that activates circuits in the brain related to reward processing and memory. “Curiosity warms up these circuits to prepare the brain to learn and create long-term memories,” says Gruber.

Paragraph 6 -- But curiosity also has some downsides. The most obvious are distraction and time-wasting. Clickbait headlines like the phrase “You won’t believe what happens next!” exploit information gaps. For many people, the desire to close those gaps can be so strong that they can’t resist clicking, even if the topic isn’t of particular importance and even if it may lead to untrustworthy information. To explore this idea, psychologists have developed various tools to measure our desire to fill information gaps.

Paragraph 7 – For instance, one such tool, which is common, is a questionnaire that asks you to rate sentences such as “I work intensively at problems that I feel must be solved” – a tendency that is assumed to be driven by curiosity about the potential answer. Such tendencies can be a red flag. A study published by Claire Zedelius at the University of California and her colleagues found that people who scored highly on these scales were prone to score worse on measures of general knowledge. They were also more receptive

to fake news. They were at a higher risk of believing a purported news story about a plane that had “disappeared and landed 37 years later”, for example, or a claim that former US president Barack Obama had been arrested for espionage. Overall, the participants that showed a great desire to resolve unanswered questions just weren’t very discriminating in the information they were willing to consume.

Paragraph 8 -- As long as you are mindful of the potential for distraction and ensure that you feed your curiosity with intellectually nutritious sources of stimulation, rather than clickbait and fake news, this should lead you to greater happiness and fulfillment. As psychologist Todd Kashdan puts it, curiosity opens up pathways that can lead you to understand your purposes in life.

Adapted from Dan Jones, “How to be curious”, *New Scientist* 15 October 2022, 38-43.

How to Be Curious -- Questions

I. Answer the following question in English.

1. What is the definition of hippocampus given in paragraph 5?

II. Check (v) TRUE or FALSE. Correct the FALSE statements.

1. In Kidd's experiment, the more the toy was absent when lifting the screen, the higher the predictability of the sequence. (par. 3)

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

2. People who have a tendency to try intensively to solve a problem are usually more knowledgeable. (paragraph 7)

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

III. Choose the best answer, according to the text.

1. What is the main idea of the first paragraph?
 - a. While curiosity has some disadvantages, its benefits are significant
 - b. Curiosity is the driving force behind science despite its disadvantages
 - c. Despite its benefits, curiosity can have significant negative consequences
 - d. The benefits of curiosity significantly outweigh its possible pitfalls

2. Curiosity is described as “double-edged” since it (par. 1)
 - a. leads to worthwhile knowledge as well as to time waste and danger
 - b. can lead us astray while at the same time causing us to believe in false news
 - c. can be distractive as well as lead us into unavoidable and dangerous pitfalls
 - d. leads to purpose-filled lives as well as to science, exploration and discovery

3. The example of the worm or amoeba in paragraph 2 illustrates the fact that curiosity
 - a. takes more complex forms in humans
 - b. is ultimately linked to survival
 - c. is equally complex for all species
 - d. receives only limited expressions in simple organisms

4. In the experiment conducted by Celeste Kidd, (par. 3)
 - a. children found the highly predictable sequences confusing
 - b. the random dropping and raising of the screen provided the best stimuli
 - c. children expressed more interest in sequences they could easily foresee
 - d. the moderately predictable cues were of most interest to participants

5. What method was used to determine the level of interest in the Kidd experiment? (par. 3)
 - a. Assessing the predictability of sequences
 - b. Asking participants to report on their feelings
 - c. Using eye-tracking tools
 - d. Making a number of educated guesses

6. We can infer that the Goldilocks effect (par. 4)
 - a. confronts both children and adults with unpredictable scenarios
 - b. describes a sequence of activities in which children tend to lose interest
 - c. refers to an unmanageable information gap
 - d. strikes the right balance between the predictable and the uncertain

7. Gruber found that curiosity prepares the brain for learning by using (par. 5)
 - a. a number of trivia questions
 - b. the learners' personal reports
 - c. surveys and statistics
 - d. empirical lab procedures

8. According to Gruber's model, curiosity and memory correlate due to the fact that (par. 5)
 - a. as we retain information in our memory, our curiosity increases accordingly
 - b. if the brain finds the filling of the information gap worthwhile, circuits responsible for memory are activated
 - c. the hippocampus counteracts mismatching information produced by the anterior cingulate cortex
 - d. the brain circuits responsible for reward processing are activated prior to the positive evaluation of information

9. What is the topic of paragraph 7?
 - a. The correlation between working intensively at problems which trigger curiosity and intellectual inferiority
 - b. The tendency of people who possess above average intelligence to concentrate on an intriguing issue
 - c. The positive effects of the indiscriminate consumption of information on the acquisition of vast general knowledge
 - d. The importance of questionnaires using sentences which allow to rate curiosity in psychological research

10. The last paragraph emphasizes the importance of
 - a. staying away from sources of knowledge that enrich our intellect
 - b. limiting our curiosity to clickbait rather than fake news
 - c. maintaining a critical viewpoint when consuming information
 - d. being cautious when absorbing information even from trustworthy sources

IV. Choose the best answer, according to the text.

1. The word “driving” (par. 1, line 1) is used as
 - a. a noun
 - b. an adjective
 - c. a verb
 - d. an adverb

2. What connector can best fill in the blank in paragraph 1, line 2?
 - a. Therefore
 - b. On the other hand
 - c. Similarly
 - d. Subsequently

3. The word "its" (par. 1, line 7) refers to
 - a. curiosity
 - b. nature
 - c. research
 - d. pitfalls

4. What modal can best fill in the blank in paragraph 2, line 7?
 - a. must
 - b. ought to
 - c. can
 - d. is able to

5. The word “face” (par. 4, line 3) is used as
 - a. a noun
 - b. a verb
 - c. an adverb
 - d. an adjective

6. The word "like" (par. 6, line 2) refers to
- a. the phrase
 - b. distraction and time-wasting
 - c. downsides
 - d. clickbait headlines
7. The word "information" (par. 6, line 7) is used as
- a. a noun
 - b. a verb
 - c. an adjective
 - d. an adverb
8. The relationship between paragraphs 6 and 7 is that of
- a. addition
 - b. comparison
 - c. elaboration
 - d. cause and effect
9. The phrase "such tendencies" (par. 7, line 4) refers to
- a. working relentlessly at a problem
 - b. being driven by curiosity
 - c. raising a red flag
 - d. making an assumption about potential answers
10. The word "consume" (par. 7, line 11) means
- a. eat or drink
 - b. expend
 - c. waste
 - d. watch or read

Cyberchondriacs Just Know They Must Be Sick

Researchers are unraveling the psychological reasons why some people relentlessly self-diagnose themselves online for hours a day

- By [Charles Schmidt](#) : *Scientific American* April 5, 2019

Par 1. One day after nursing her six-month-old baby, Colleen Abel developed an itchy red rash on her right breast. The cause was a mystery. Abel speculated that scratches left by her son while he fed might have gotten infected, or that bedbugs might have bitten her. The 36-year-old writer from Illinois opened her laptop and investigated her symptoms on Google. What she found shocked her. The first result blared inflammatory breast cancer, “and that scared me out of my mind,” Abel says. Other diagnoses such as dermatitis were far more likely, but Abel was convinced that a fast-growing malignancy was in her body. Before long, she was spending three to four hours every day reading about breast cancer on the internet.

Par 2. Abel was in the throes of cyberchondria—a technology-enabled malady that drives people to repeatedly query their symptoms on search engines. Nearly one in three people among the millions who seek health information online report feeling more anxious afterwards. Yet individuals with cyberchondria paradoxically keep searching, even as their distress continues to grow. The term cyberchondria was coined by the media nearly 20 years ago as a play on the much older medical term hypochondria, whose sufferers are convinced they are sick with an illness they do not really have.

Par 3. Because cyberchondria has never been added to the American Psychiatric Association’s list of diagnosable mental disorders, reliable estimates of the number of people who have it are not available. But what is known, says Eoin McElroy, a psychologist at University College London, is that “cyberchondria has the potential to interrupt many aspects of your life.” Studies have linked it to depression, and its sufferers are prone to either see their doctors too often, or not enough for fear of what they might hear.

Par 4. Now scientists are starting to define cyberchondria and untangle its psychological roots. These are early days, but researchers are revealing fresh insights into what drives people to engage in a behavior that leads mostly to unhappiness. A key finding is that the relentless searching, more than anything else, is for assurances of personal safety—which the internet by its very nature is unable to provide.

THINKING ABOUT THINKING

Par 5. Thomas Fergus, a professor of psychology at Baylor University, is a pioneering investigator in assurance-seeking behavior. He links cyberchondria to a dysfunctional web of metacognitive beliefs, which are really just thoughts about thinking. Everyone has these sorts of belief systems. For instance, it is normal to believe that ruminating over a challenging problem will lead to a well-considered outcome. But in cyberchondria, metacognitive beliefs morph into a mental trap that chains people to online health content.

Par 6. Together with his collaborator, Marcantonio Spada, an academic psychologist at London South Bank University, Fergus has shown that metacognitive beliefs in cyberchondria overlap to some degree with those of other anxiety disorders that have longer diagnostic histories. People with health anxiety, for instance, worry irrationally about having a serious medical problem, and hold maladjusted views about the role worry plays in maintaining their emotional and physical well-being. These cycling beliefs can either be positive (such as: worrying about a medical problem will help me prepare before it is too late) or negative (my worry is getting out of control and it is making me sick). Even after their worries subside, people with health anxiety will eventually begin worrying that they have not worried enough about their medical state, thus triggering the cycle anew. It is these same sorts of dysfunctional belief systems, Fergus says, “that send people with cyberchondria back for long sessions at the computer.”

Par 7. Fergus and Spada published research in 2018 that further links cyberchondria with features of obsessive compulsive disorder (OCD). People with OCD believe that performing a ritualistic behavior will ease their anxiety, and they engage in that ritual until they hit a predetermined threshold or stopping point. Fergus and Spada discovered that

similar metacognitive beliefs propel online search habits in cyberchondria. Individuals search ritualistically for health information to dispel their anxiety, and will only stop when they feel the search has sufficiently reduced the uncertainties they have about the state of their health. “What they’re really looking for is reassurance that nothing bad is going to happen,” Fergus says.

Par 8. Unfortunately, online health content is too diverse and conflicting to deliver the certainty. So, sufferers keep looking. Knowing who to trust online amounts to little more than a guessing game. Abel says the websites sparking her initial fears of inflammatory breast cancer appeared authoritative. “The more I learned, the more I found reasons to believe my worst-case scenario was correct,” she explains. “I ignored the information that should have reassured me—such as the rarity of the disease—and focused instead on what confirmed my suspicions.”

ESCAPING THE TRAP

Par 9. Current treatments for cyberchondria range from antidepressants to talk therapy and mindfulness training, although the relative benefits of one over another are unclear. According to Robin Bailey, a psychologist at Liverpool John Moores University, in the U.K., the condition has to be more fully defined before clinical studies can get underway. But appropriate therapies, Bailey says, could draw from the same principles used in treating other anxiety disorders. Health anxiety, for instance, has traditionally been treated with cognitive behavioral therapies that encourage people to question the evidence leading them to believe they have a dangerous medical problem. Cognitive behavioral therapy for health anxiety can last 30 weekly sessions or more, according to Bailey.

Par 10. Alternatively, therapists can try metacognitive techniques that shift the focus away from a perceived health threat to the worry that surrounds it. Individuals are encouraged to “observe” thoughts about illness or injury without letting them intrude emotionally. Or they might postpone their health worries until a scheduled time of day. Applied in a cyberchondria context, Bailey says, metacognitive approaches would encourage people to question the psychological value of going online to relieve their anxiety, “when evidence from their own experience shows it has the opposite effect.”

Par 11. Abel found a way out. After two months spent in chatrooms devoted to inflammatory breast cancer, too scared to go to an urgent care clinic, Abel finally went to a primary care doctor who told her she did not have cancer. Instead, she had thrush—a benign condition that cleared up fast with antifungal treatment. Remarkably, thrush had never appeared on any of her Google searches, an omission that Abel says reflects the internet’s deficiency as a tool for self-diagnosis.

Par 12. Abel admits to a lifelong battle with anxiety, and says she is still prone to bouts of cyberchondria, which she knows can wreak more havoc on her life than any of the health ailments that prompt it. “You shouldn’t be embarrassed to show up at the doctor once in a while if you’re worried,” she says. “That’s normal. And if you are in the throes of anxiety or panic, you need a doctor for that too. Don’t be afraid to reach out for mental or physical health. Never be ashamed of that.”

Cyberchondriacs - Questions

Answer the following questions according to the instructions.

Pre-reading

1. What is the meaning of *cyber*?

2. Read the title and the sub-heading. Can there be a connection between cyber and self-diagnosis? If yes / no, explain why

3. What idea does the anecdote in par. 1 illustrate?

4. Try to predict what this article discusses.

Close reading

5. What is *cyberchondria*, according to par. 2?

6. There is a common denominator between cyberchondria and hypochondria. What is it (par. 2)?

7. What does the word "it" refer to (par. 3)?

- a. number
- b. people
- c. cyberchondria
- d. hypochondria

8. Why, according to par. 4, do people who suffer from cyberchondria engage in relentless searching on the internet?

9. The word "ruminating" (par. 5) can be replaced by

- a. chewing
- b. contemplating
- c. forgetting
- d. behaving

10. Complete the table according to pars. 6 and 7.

Reseacher	Research conclusion	Example
Fergus and Spada		People with health anxiety, for instance, worry irrationally about having a serious medical problem, and hold maladjusted views about the role worry plays in maintain their emotional and phusical well-being
	Cyberchondria is linked with features of OCD	

11. Decide if the following statement is true or false (par. 8). If false, find the incorrect word and correct it with ONE word only

Online health content can deliver the certainty that sufferers of cyberchondria need.

- True/False
- incorrect word _____
- correct word _____

12. How can cyberchondria be treated (par. 9)?

a. _____

b. _____

c. _____

13. Decide if the following statement is true or false (par. 8). If false, find the incorrect word and correct it with ONE word only

According to Bailey, Cognitive behavioral therapy for health anxiety can last 30 daily sessions.

- True/False
- incorrect word _____
- correct word _____

14. Did Abel find a solution to her problem (par. 11)?

a. Yes/No

b. Evidence _____

15. The word "diagnoses" (par. 1, line 6) is used as

- a. a noun
- b. an adjective
- c. a verb
- d. an adverb

16. The word "they" (par. 6, line 10) refers to _____

17. The words "such as" (par. 8, line 6) refer to _____

18. The word "wreak" (par. 12, line 2) is used as

- a. a noun
- b. an adjective
- c. a verb
- d. an adverb

Post reading

19. Based on this article, is it possible to say that the Internet is a good source of information for self-diagnosis? Express your opinion in a few sentences.

AI Gets Emotional

Paragraph 1 -- Rana el Kaliouby was alone in her flat, messaging her husband. “How are you doing?” he typed. “I’m fine,” she typed back. Except that wasn’t true. The couple had been apart for weeks and she was feeling miserable. Had he been in the room, he could have read the emotions on her face at a glance. But he was miles away. It is a scene that could easily have played out during a coronavirus pandemic lockdown, when colleagues, friends and even families were cut off from one another. But it actually took place 20 years ago, soon after el Kaliouby had moved from Egypt to the UK to study, leaving her husband behind. It was in that moment, she says, that she realized how technology was blind to human emotions. Ever since, el Kaliouby has dreamed of building an emotionally intelligent computer – or as she puts it “a mind-reading machine”. With so many relationships mediated by text or video call these days, it is a technology that couldn’t be more relevant. These days, the company el Kaliouby co-founded, Affectiva, and others like it, claim to have systems capable of detecting human emotions. The promises they make about the potential of this emotion artificial intelligence (AI) are staggering. Computers, they say, will know if we are distracted while driving, angrily typing an email that we may regret or when our mental health is beginning to slump. In fact, systems like this already exist. But do they perform as expected? And do we really want machines that know how we feel?

Paragraph 2 -- To make her dream a reality, el Kaliouby is relying on AI systems that learn from data how to do certain tasks. Take an AI that recognizes cat pictures. It is trained by being shown lots of photos of cats and, in the process, learns what shapes, colors and other features to look out for. To begin with, the system is told whether the pictures contain cats or not. Eventually, given enough examples, it can learn to discern this itself. Like spotting a cat, recognizing a human emotion is a skill that, in principle, could be mastered by AI. Far more is at stake, though, at least for el Kaliouby. “These technologies that interface with us on a day-to-day basis need to know human, they need to understand human,” she says. We all know that video calls and messaging apps can’t fully replicate the experience of a face-to-face chat and part of that is because it is harder to

remotely read other people's emotions. Yet computer-mediated conversations are set to become more common, especially as more of us work from home. One possible application of emotion AI is to help us judge whether an audience is bored, excited or somewhere in between.

Paragraph 3 -- The field of affective computing – making machines that can recognize, interpret and simulate human emotions – has been going for years yet is in some ways still in its infancy. We are a long way from machines that can realistically simulate our emotions. But when it comes to machines that can recognize our feelings, we appear to be getting there.

Paragraph 4 -- Since Affectiva was founded in 2009, it has mostly worked with marketing firms to help them understand how audiences react to videos and other marketing content. This is often done by recording videos of people's faces as they encounter a poster, say, or watch an advert. By tracking how facial expressions change when they encounter these things, Affectiva says it can predict how successful an ad campaign will be. It isn't alone. Among others, Realeyes, a company headquartered in London, has also worked with advertisers to measure how attentive viewers are.

Paragraph 5 -- Other companies reckon they can glean insights from the sound of our voices alone. Behavioral Signals, a tech firm in Los Angeles, has developed software to classify emotion based on a speaker's tone. Its algorithm tracks pitch, volume, rhythm, intonation and other features of speech. "We focus on how something is being said," says chief executive Rana Gujral. "Oftentimes we don't even convert the audio into text." This kind of intelligence could be handy for businesses, not least firms that field lots of phone calls. Behavioral Signals claims to be able to tell within 30 seconds of a call whether a customer will take a particular action, such as commit to paying off a debt. The company says it has sold its tech to a major European bank, which used tone-of-voice analysis to match callers with staff in call centers.irate customers were automatically triaged to especially calm and collected handlers. There was a 20 per cent improvement in call outcomes, says Gujral, which translated into an expected \$300 million, had the bank relied on the system across its business for a whole year.

Paragraph 6 -- But Behavioral Signals is going even further. Gujral says he is working with another company, which he won't name, that hopes to use AI in order to pick up vocal signals associated with depressive symptoms. The idea is to predict the likelihood of someone going on to attempt suicide based on their tone of voice during conversations with a psychologist or caregiver. This is experimental, but "they're actually working on live patients right now", says Gujral. Clearly, emotional AI could be useful. Can algorithms really grasp human emotions, though? It isn't easy to evaluate the various companies' claims directly because their algorithms typically aren't made public. Even so, the science of emotion recognition can offer some insights.

Paragraph 7 -- We know it is possible to classify facial expressions. In the mid-20th century, psychologist Paul Ekman pinpointed facial movements that he argued could be associated with broad emotional states. There are now considered to be seven such emotional states under Ekman's Facial Action Coding System (FACS): anger, fear, disgust, happiness, sadness, surprise and contempt. These are often referred to as the "universal emotions" and well-practiced humans can get good at identifying them from facial signals alone, with one study suggesting they get it right 77 per cent of the time.

Paragraph 8 -- There is a rather large catch, nevertheless: accurately identifying the "anger" state like this doesn't necessarily mean the person displaying it is angry. One researcher who doubts the usefulness of this approach is Lisa Feldman Barrett, a psychologist at Northeastern University in Massachusetts. She and her colleagues recently completed a large review of existing research to find out whether there was any relationship between specific facial expressions and internal emotional states. The gist of their findings is that the evidence is scant – a scowl can be associated with a wide range of emotions, from anger to confusion to concentration. What's more, the use of various expressions varies noticeably between cultures. Fully decoding expressions isn't easy in fact. If humans aren't perfect, can AI do better? Well, FACS is still an important ingredient in the algorithms designed by AI companies, including Affectiva and Realeyes. This makes Barrett highly skeptical of them. She argues that emotions are more like "episodes": there is no one brain signal for "anger" and there isn't one recognizable way of expressing that emotion on the face either.

Paragraph 9 -- Similar quibbles plague voice-based emotion recognition. Margaret Lech at RMIT University in Melbourne, Australia, and her colleagues reviewed a series of studies on the accuracy of such systems and found that they _____ correctly identify emotions around 60 per cent of the time on average – hardly a glowing result. Some proponents of emotion AI accept that this is valid argument and admit that the systems only recognize expressions or intonations, not necessarily the true emotions beneath. Rosalind Picard, who co-founded Affectiva and is now director of the Affective Computing group at the Massachusetts Institute of Technology, is one of them. “I am not proposing one could measure affective state directly, but rather measure observable functions of such states,” she wrote in a 1995 paper. _____, others see the distinction as providing a very weak argument. Barrett’s critique might hold if a person or AI observes a face with no contextual information, they would argue, but that isn’t what happens in real life. Emotion AI is typically used to see how a person reacts to a specific thing, like a funny video clip. In that context, a grin is probably just a grin: an uncomplicated indication of amusement.

Paragraph 10 -- What happens as this technology begins to be used more widely, though? Some companies working in affective computing have made their tech available to study and this has led to the discovery of some examples of bias. Lauren Rhue at the University of Maryland catalogued the results of emotion AIs developed by Microsoft and Chinese tech firm Megvii when the systems were fed pictures of white and black basketball players. “Both services interpret black players as having more negative emotions than white players,” she wrote in a 2018 paper. New Scientist asked both firms for a response to these claims. Megvii says that it puts great importance on fairness and doesn’t use its algorithms in scenarios that don’t meet its standards. Microsoft declined to comment.

Paragraph 11 -- Such ethical considerations aren’t stopping some police forces from trialing emotion AI in the wild. Lincolnshire Police in the UK recently hit the news because it has received government funding to deploy a system intended to detect the emotions of people captured on CCTV footage. This will, for now, be limited to a trial, and footage will be deleted after 31 days, according to the force. Researchers at New York University’s AI Now Institute recently questioned how fair it is to use emotion-detecting AI on members of the public without explicit consent. Among the systems mentioned in the

institute's 2019 annual report is one offered by Oxygen Forensics, a US-based, Russian-owned firm that sells software to the FBI, London's Metropolitan Police and Interpol. Among its products, the company offers facial recognition technology that it claims can detect emotion. The institute's report argued that there was little to no scientific basis for this technology in general and said its use in areas including criminal justice was "concerning". Lee Reiber at Oxygen Forensics has previously said that being able to detect anger, stress or anxiety can help investigators.

Paragraph 12 -- "I don't think we should be using emotion AI broadly in public life, particularly to make decisions that have legal consequence for people," says Jevan Hutson at the University of Washington School of Law in Seattle. Hutson likens emotion AI to phrenology, the discredited 19th-century idea that bumps on people's heads were correlated with personality traits. He says countries should adopt policies and laws to restrict the use of emotion AI in certain areas. Law enforcement, job recruitment, surveillance in public places – all of these applications and more could be deeply problematic if we get this stuff wrong, he says. Meanwhile, there are plenty of people working to soothe such issues by making emotion AI more accurate. The most obvious way to do this is to provide the AI with more context. Imagine an AI trying to work out whether a grimace signifies laughter or despair. If it had the tools to recognize whether a person is hanging out with friends or strangers – perhaps by recognizing the other faces nearby – then it could more confidently settle on laughter. This isn't perfect, of course, but in general the more contextual information an AI has, the more likely it is to draw accurate inferences.

Paragraph 13 -- Take this to its logical conclusion and we might end up with emotion AI systems that Hoover up information on our voices, our body movements, faces and data about our environment, who we hang out with and what the people around us are doing. This might sound a long way off, but there is one place where we are already seeing the first steps in this direction: cars. El Kaliouby and her colleagues at Affectiva are among the engineers now developing AI to monitor the behavior of vehicle occupants. "We've expanded to things like activity detection," she says. "Are you holding a cellphone to your ears? Are the kids fighting in the back seat?" Ultimately, such a system could form the basis of an autonomous car that even knows when to take control of the wheel. Vehicle

maker Toyota is already building a prototype car equipped with an emotion AI system developed by US-based SRI International. Other car-makers are working with Affectiva, el Kaliouby says, and she expects to launch the first commercial versions of the technology in two to three years. All this leaves computers that can detect our emotions at a pivotal juncture. They still aren't entirely trusted, and yet they are on the cusp of becoming far more widespread. Society hasn't fully grappled with the consequences. But one thing is for sure: we have come a long way since el Kaliouby sent those lonely messages two decades ago.

Adapted from Chris Baraniuk, "AI gets emotional," *NewScientist* 21 November 2020

AI Gets Emotional – Questions

I. Answer the following question in English.

What is the definition of affective computing given in paragraph 3?

II. Check (✓) TRUE or FALSE. Correct the FALSE statements.

1. On the whole, it is infeasible to develop AI capable of identifying various kinds of human feelings. (par. 2)

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

2. We can infer from paragraph 3 that designing AI that can express human emotions is a much easier task than creating machines that can detect them.

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

III. Choose the best answer, according to the text.

1. One of the reasons communicating through various technologies is different from face-to-face interaction is that (par. 2)

- a. other people's feelings can be easily identified from afar
- b. people will increasingly depend on computers for having conversations
- c. distance makes it harder to determine what other people feel
- d. modern ways of communication necessitate better understanding of human emotions by AI

2. Affectiva and Realeves work with video recordings of people's reactions in order to (par. 4)
 - a. efficiently achieve more successful marketing
 - b. classify potential customers according to their emotions
 - c. dispense with the need to assess the impact of advertising campaigns
 - d. divert the attention of potential clients from marketing content

3. The example of a major European bank is used to demonstrate that (par. 5)
 - a. classifying emotions based on different features of speech can provide significant commercial benefits
 - b. all customers prefer to interact with calm and collected employees, regardless of their own emotional state
 - c. businesses which handle large amounts of phone calls do not profit from paying attention to their customer's emotions
 - d. it is possible to rely just on the sound of our voices to accurately evaluate our emotions

4. It is difficult to be sure whether AI can understand human emotions since (par. 6)
 - a. the claims of various companies are based mainly on the insights from the science of emotion
 - b. the proofs deduced from the merely experimental use of the companies' algorithms are inconclusive
 - c. there are companies which go too far in their use of algorithms to assess patients' symptoms
 - d. the companies keep their algorithms secret so that their efficiency cannot be assessed accurately

5. Doubts concerning the usefulness of identifying emotions from facial expressions are based on (par. 8)
 - a. the absence of cultural differences in the ways various emotions are conveyed
 - b. the solid evidence showing that specific facial signals correlate to what people truly feel
 - c. the insufficient amount of evidence that human feelings are reflected by specific facial signals
 - d. the fact that it is usually not difficult to interpret the ways in which humans show their emotions

6. What is the main idea of paragraph 9?
 - a. Some specialists acknowledge that emotion AI can only identify various exterior signs of emotions rather than real feelings
 - b. It is possible that in a specific context, emotional AI may be used as a reliable tool for recognizing human feelings
 - c. Affective states can be accurately observed both by people and AI without any reference to their measurable functions
 - d. Specialists disagree over whether the efficiency of emotional AI is affected by its ability to identify only the external indications of feelings

7. We can understand from paragraph 10 that the problem making the emotion AIs developed by Microsoft and Megvii unreliable is
 - a. the restricted use of the algorithms in sorted out scenarios
 - b. the lack of objective evaluation by independent specialists
 - c. the underlying presence of racial prejudice
 - d. the refusal of the companies to comment on the results of the study

8. Various police forces are interested in exploiting emotional AI despite (par. 11)
 - a. solid evidence that its use in criminal justice is unfair
 - b. ethical concerns surrounding its use on the members of the public
 - c. their opinion that it cannot accurately detect emotion
 - d. scientific reports establishing the efficiency of its widespread use

9. According to paragraph 12, the most effective solution to the problem of emotional AI's lack of accuracy is
- a. limiting its use to areas where mistakes are less damaging to people
 - b. alleviating public concern by providing strict and impartial control
 - c. teaching it to distinguish between the different signs of the same emotion
 - d. providing as much data about the circumstances as possible
10. Although the implications of the extensive practical use of emotional AI are still not fully evaluated, (par. 13)
- a. society has full faith in its reliability
 - b. it will probably soon become very common
 - c. it is already successfully incorporated into self-driving vehicles
 - d. we should not be concerned about losing our privacy

IV. Choose the best answer, according to the text.

1. The word "marketing" (par. 4, line 2) is used as
- a. a noun
 - b. an adjective
 - c. a verb
 - d. an adverb
2. The word "field" (par. 5, line 6) is used as
- a. a noun
 - b. an adjective
 - c. a verb
 - d. an adverb
3. The word "which" (par. 5, line 12) refers to
- a. call outcomes
 - b. call centers
 - c. a 20 per cent improvement
 - d. \$300 million

4. The relationship between paragraphs 7 and 8 is that of
 - a. elaboration
 - b. addition
 - c. illustration
 - d. contrast

5. The word "them" (par. 8, line 12) refers to
 - a. algorithms
 - b. AI companies
 - c. FACS
 - d. humans

6. Which modal can best fill in the blank in paragraph 9, line 3?
 - a. had to
 - b. were able to
 - c. should
 - d. must

7. What connector can best fill in the blank in paragraph 9, line 10?
 - a. Moreover
 - b. Consequently
 - c. Finally
 - d. However

Artificial Imagination: How Machines Could Become More Humanlike

Paragraph 1 -- If you ever feel cynical about human beings, a good antidote is to talk to artificial- intelligence researchers. You might expect them to be triumphalist now that AI systems match or beat humans at recognizing faces, translating languages, playing board and arcade games, and remembering to use the turn signal. To the contrary, they're always talking about how marvelous the human brain is, how adaptable, how efficient, how infinite in faculty. Machines still lack these qualities. They're inflexible, they're opaque and they're slow learners, requiring extensive training. Even their well-publicized successes are very narrow.

Paragraph 2 -- Many AI researchers got into the field because they want to understand, reproduce and ultimately surpass human intelligence. Yet even those with more practical interests think that machine systems should be more like us. A social media company training its image recognizers, for example, will have no trouble finding cat or celebrity pictures. But other categories of data are harder to come by, and machines could solve a wider range of problems if they were quicker-witted. Data are especially limited if they involve the physical world. If a robot has to learn to manipulate blocks on a table, it can't realistically be shown every single arrangement it might encounter. Like a human, it needs to acquire general skills rather than memorizing by rote.

Paragraph 3 -- In getting by with less input, machines also need to be more forthcoming with output. Just the answer isn't enough; people also want to know the reasoning, especially when algorithms pass judgement on bank loans or jail sentences. You can interrogate human bureaucrats about their biases and conflicts of interest; good luck doing that with today's AI systems. In 2018 the European Union gave its citizens a limited right to an explanation for any judgement made by automated processing. In the U.S., the Defense Advanced Research Projects Agency funds an "Explainable AI" research program because military commanders would rather not send soldiers into battle without knowing why.

Paragraph 4 -- A huge research community tackles these problems. Ideas abound, and people debate whether a more humanlike intelligence will require radical retooling. Yet it's remarkable how far researchers have gone with fairly incremental improvements. Self-improvement, imagination, common sense: these seemingly quintessential human qualities are being incorporated into machines, at least in a limited way. The key is clever coaching. Guided by human trainers, the machines take the biggest steps themselves. More than most fields of science and engineering, AI is highly cyclical. It goes through waves of infatuation and neglect, and methods come in and out of fashion. Neural networks are the ascendant technology. Such a network is a web of basic computing units: "neurons." Each can be as simple as a switch that toggles on or off depending on the state of the neurons it is connected to. The neurons typically are arrayed in layers. An initial layer accepts the input (such as image pixels), a final layer produces the output (such as a high-level description of image content), and the intermediate, or "hidden," layers create arithmetic combinations of the input. Some networks, especially those used for problems that unfold over time, such as language recognition, have loops that reconnect the output or the hidden layers to the input.

Paragraph 5-- A so-called deep network has tens or hundreds of hidden layers. They might represent midlevel structures such as edges and geometric shapes, although it is not always obvious what they are doing. With thousands of neurons and millions of interconnections, there is no simple logical path through the system. And that is by design. Neural networks are masters at problems not amenable to explicit logical rules, such as pattern recognition.

Paragraph 6 -- Crucially, the neuronal connections are not fixed in advance but adapt in a process of trial and error. You feed the network images labeled "dog" or "cat." For each image, it guesses a label. If it is wrong, you adjust the strength of the connections that contributed to the erroneous result, which is a straightforward exercise in calculus. Starting from complete scratch, without knowing what an image is, let alone an animal, the network does no better than a coin toss. But after perhaps 10,000 examples, it does as well as a human presented with the same images. In other training methods, the network responds to vaguer cues or even discerns the categories entirely on its own.

Remarkably, a network can sort images it has never seen before. Theorists are still not entirely sure how it does that, but one factor is that the humans using the network must tolerate errors or even deliberately introduce them. A network that classifies its initial batch of cats and dogs perfectly might be fudging: basing its judgment on unreliable cues and variations rather than on essential features. This ability of networks to sculp themselves means they can solve problems that their human designers have no idea how to solve. And that includes the problem of making the networks even better at what they do.

Paragraph 7 -- Teachers often complain that students forget everything over the summer. In lieu of making vacations shorter, they have taken to loading them up with summer homework. But psychologists such as Robert Bjork of the University of California, Los Angeles, have found that forgetting is not inimical to learning but essential to it. That principle applies to machine learning, too. _____ a machine learns a task, then forgets it, then learns another task and forgets it, and so on, it can be coached to grasp the common features of those tasks, and it will pick up new variants faster. It won't have learned anything specific, but it will have learned *how* to learn – what researchers call meta-learning. When you do want it to retain information, it'll be ready. "After you've learned to do 1,000 tasks, the 1,001st is much easier," says Sanjeev Arora, a machine-learning theorist at Princeton University. Forgetting is what puts the meta into meta-learning. Without it, the tasks all blur together, and the machine can't see their overall structure. Meta-learning gives machines some of our mental agility. "It will probably be key to achieving AI that can perform with human-level intelligence," says Jane Wang, a computational neuroscientist at Google's DeepMind in London. Conversely, she thinks that computer meta-learning will help scientists figure out what happens inside our own head.

Paragraph 8 -- In nature, the ultimate meta-learning algorithm is Darwinian evolution. In a protean environment, species are driven to develop the ability to learn rather than rely solely on fixed instincts. In the 1980s AI researchers used simulated evolution to optimize software agents for learning. But evolution is a random search that goes down any number of dead ends, and in the early 2000s researchers found ways to be more systematic and therefore faster. In fact, with the right training regimen, any neural

network can learn to learn. As with much else in machine learning, the trick is to be very specific about what you want. If you want a network to learn faces, you should present it with a series of faces. By analogy, if you want a network to learn how to learn, you _____ present it with a series of learning exercises. In 2017 Chelsea Finn of the University of California, Berkley, and her colleagues developed a method they called model-agnostic meta-learning. Suppose you want to teach your neural network to classify images into one of five categories, be it dog breeds, cat breeds, car makers, hat colors, or what have you. In normal learning, without the "meta," you feed in thousands of dog images and tweak the network to sort them. Then you feed in thousands of cats. That has the unfortunate side effect of overriding the dogs; taught this way, the machine can perform only one classification task at a time.

Paragraph 9 -- In model-agnostic meta-learning, you interleave the categories. You show the network just five dog images, one of each breed. Then you give it a test image and see how well it classifies that dog – probably not very well after five examples. You reset the network to its starting point, wiping out whatever modest knowledge of dogs it may have gained. But – this is the key step – you tweak this starting point to do better next time. You switch to cats – again, just one example of each breed. You continue for cars, hats, and so on, randomly cycling among them. Rotate tasks and quiz often. The network does not master dogs, cats, cars or hats but gradually learns the initial state that gives it the best head start on classifying anything that comes in fives. By the end, it is a quick study. You might show it five bird species: it gets them right away.

Paragraph 10 -- Finn says the network achieves this acuity by developing a bias, which, in this context, is a good thing. It expects its input data to take the form of an image and prepares accordingly. "If you have a representation that's able to pick out the shapes of objects, the colors of objects and the textures and is able to represent that in a very concise way, then when you see a new object, you should be able to very quickly recognize that," she says. Finn and her colleagues also applied their technique to robots, both real and virtual. In one experiment, they gave a four-legged robot a series of tasks to run in various directions. Going through meta-learning, the robot surmised that the common feature of those tasks was to run, and the only question was: Which way? So the machine prepared by running in place. "If you're running in place, it's going to be easier to

very quickly be adapted to running forward or running backward because you're already running," Finn says.

Paragraph 11 -- This technique, like related approaches by Wang and others, does have its limitations. Although it reduces the amount of sample data needed for a given task, it still requires a lot of data overall. "Current meta-learning methods require a very large amount of background training," says Brenden Lake, a cognitive scientist at New York University, who has become a leading advocate for more humanlike AI. Meta-learning is also computationally demanding because it leverages what can be very subtle differences among tasks. If the problems are not sufficiently well defined mathematically, researchers must go back to slower evolutionary algorithms. "Neural networks have made progress but are still far from achieving humanlike concept learning," Lake says.

The most fun part of an AI conference is when a researcher shows the silly errors that neural networks make, such as mistaking random static for an armadillo or a school bus for an ostrich. Their knowledge is clearly very shallow. The patterns they discern may have nothing to do with the physical objects that compose a scene. "They lack grounded compositional object understanding that even animals like rats possess," says Irina Higgins, an AI researcher at DeepMind.

Paragraph 12 -- In 2009 Yoshua Bengio of the University of Montreal suggested that neural networks would achieve some genuine understanding if their internal representations could be disentangled – that is, if each of their variable corresponded to some independent feature of the world. For instance, the network should have a position variable for each object. If an object moves, but everything else stays the same, just that one variable should change, even if hundreds or thousands of pixels are altered in its wake. In 2016 Higgins and her colleagues devised a method to do that. It works on the principle that the real set of variables – the set that aligns with the actual structure of the world – is also the most economical. The millions of pixels of an image are generated by a relatively few variables combined in multitudinous ways. "The world has redundancy – this is the sort of redundancy that the brain can compress and exploit," Higgins says. To reach a parsimonious description, her technique does the computational equivalent of squinting – deliberately constricting the network's capacity to represent the world, so it is

forced to select only the most important factors. She gradually loosens the constriction and allows it to include lesser factors.

Paragraph 13 -- In one demonstration, Higgins and her colleagues constructed a simple "world" for the network to dissect. It consisted of heart, square and oval shapes on a grid. Each could be one of six different sizes and oriented at one of 20 different angles. The researchers presented all these permutations to the network, whose goal was to isolate the five underlying factors: shape, position along the two axes, orientation and size. At first, they allowed the network just a single factor. It chose position as most important, the one variable without which none of the others would make much sense. In succession, the network added the other factors. To be sure, in this demonstration the researchers knew the rules of this world because they had made it themselves. In real life, it may not be so obvious whether disentanglement is working or not. For now that assessment still takes a human's subjective judgment.

Paragraph 14 -- Like meta-learning, disentanglement has lots of applications. For starters, it makes neural networks more understandable. You can directly see their reasoning, and it is very similar to human reasoning. A robot can also use disentanglement to map its environment and plan its moves rather than bumbling around by trial and error. Combined with what researchers call intrinsic motivation – in essence, curiosity – disentanglement guides a robot to explore systematically. _____, disentanglement helps networks to learn new data sets without losing what they already know. For instance, suppose you show the network dogs. It will develop a disentangled representation specific to the canine species. If you switch to cats, the new images will fall outside the range of that representation – the type of whiskers will be a giveaway – and the network will notice the change. "We can actually look at how the neurons are responding, and if they start to act atypically, then we should probably start learning about a new data set," Higgins says. At that point, the network _____ adapt by, for example, adding extra neurons to store the new information, so it won't overwrite the old.

Paragraph 15 -- Many of the qualities that AI researchers are giving their machines are associated, in humans, with consciousness. No one is sure what consciousness is or why we have a vivid mental life, but it has something to do with our ability to construct models

of the world and of ourselves. AI systems need that ability, too. A conscious machine seems far off, but could today's technologies be the baby steps toward one?

Adapted from George Musser, "Artificial Imagination," *Scientific American* May 2019

Artificial Imagination: How Machines Could Become More Humanlike -- Questions

I. Choose the best answer.

1. The sentence "self-improvement, imagination, common sense: these seemingly quintessential human qualities are being incorporated into machines, at least in a limited way" (par. 4, lines 4-5) means that
 - a) although qualities such as self-improvement, imagination and common sense inherently belong to human beings, artificial intelligence will probably obtain them in a rudimentary form
 - b) self-improvement, imagination and common sense are intrinsically human characteristics, and as such they cannot be integrated into machines' skills kit
 - c) only a restricted incorporation of self-improvement, imagination and common sense into artificial intelligence is feasible, despite the fact that these characteristics merely seem to belong exclusively to humans
 - d) it is practical to make humans and machines share some essential properties, such as self-improvement, imagination and common sense, since their quintessential dissimilarity is an illusion

II. Check (v) TRUE or FALSE. Correct the FALSE statements.

1. In order to solve a wider range of problems, robots have to improve their broad mental capacity instead of honing their memory skills. (par. 2)
☐ TRUE ☐ FALSE
 - a. Copy the incorrect word: _____
 - b. Write the correction: _____
2. Although scientists lack precise understanding of the process by which artificial intelligence can identify unfamiliar images, it is crucial to reject its fallacies. (par. 6)
☐ TRUE ☐ FALSE
 - a. Copy the incorrect word: _____
 - b. Write the correction: _____

III. Choose the best answer, according to the text.

1. Artificial-intelligence researchers are very appreciative of the qualities of the human mind (par. 1)
 - a) despite the fact human intelligence cannot compare with the outstanding mental capabilities of machines
 - b) since when compared to the human brain, machines seem underdeveloped and inefficient
 - c) in spite of the advantages of human intelligence being easily outweighed by the marvelous benefits offered by AI systems
 - d) as they are the only scientists capable of accurate evaluations of the capabilities and limitations of the human brain

2. Most researchers agree that artificial intelligence should acquire the qualities which characterize human intelligence, although
 - a) they were not initially interested in machines outstripping the abilities of the human brain
 - b) the practical ramifications of this development process would be insignificant
 - c) they are aware that reproducing the qualities of the human brain is an infeasible task
 - d) their ultimate goal might be to create machines whose abilities will be capable of triumphing over those of the human mind

3. After receiving a large amount of images, the network (par. 6)
 - a) still cannot cope with the data in a way similar in its efficiency to the capacities of a human being
 - b) shows no improvement in its functioning in comparison to its initial performance
 - c) can only deal with the information in a random way, just as a human would do
 - d) can proceed from random inefficient performance to a human-like processing of the data

4. The topic of paragraph 7 is
 - a) the advantages of combining learning and forgetting for the development of human intelligence
 - b) meta-learning as the potential main condition of artificial intelligence becoming a match for the human mind
 - c) the definition of meta-learning and its practical applications in education and science
 - d) computer meta-learning and its potential for improving human intelligence

5. Paragraph 8 mentions Darwinian evolution in order to
 - a) underscore the analogy between the development of species in nature and the improvement of software networks
 - b) focus on machine meta-learning as opposed to the natural process of biological enhancement
 - c) demonstrate that learning is a natural phenomenon necessary for both the survival of biological species and for technological development
 - d) illustrate the dependance of continuous technological growth on its adaptation to the changes in the natural environment

6. According to Finn, the network acquires quick perception by
 - a) preparing itself for any unexpected task via careful differentiation between the information it must process
 - b) being able to completely ignore the common features of the data it receives
 - c) discovering the most important shared characteristic of the data it has to work with
 - d) disregarding the visual aspects of the information provided to it and analyzing its logical content

7. The large amounts of data needed for the efficient functioning of artificial intelligence
- a) facilitate its achieving optimal performance for any given task
 - b) are the most important hurdle in the way of its achieving human-like capabilities
 - c) prove the only obstacle to its gaining limitless abilities through meta-learning
 - d) are one of the factors lowering its efficiency and impeding its future development
8. The main idea of paragraph 12 is that
- a) new techniques could remove the obstacles in the way of artificial intelligence achieving a higher level of comprehension
 - b) artificial intelligence can be improved by broadening its perception of the external reality
 - c) the world is redundant, and so artificial intelligence can easily grasp its phenomena
 - d) Higgins method is based on the assumption that the most useful set of variables is the one closest to the structure of the external reality
9. The demonstration prepared by Higgins and her colleagues
- a) presented an objective picture of how disentanglement might work in the real world
 - b) offered a merely subjective evaluation of the efficiency of disentanglement in improving the performance of a given network
 - c) was useless since it cannot offer tools for future research of artificial intelligence
 - d) was based on the researchers' ignorance of the world in which the network had to operate
10. According to the last paragraph,
- a) artificial intelligence ever achieving a human-like conscience is an impossibility
 - b) achieving human-like consciousness may not be crucial to the further successful development of artificial intelligence
 - c) the current advances in the field of artificial intelligence might enable it to gain the capabilities of the human mind in distant future
 - d) discovering the nature of consciousness is essential for ensuring the future evolution of human-like artificial intelligence

IV. Choose the best answer.

1. The word "bank" (par. 3, line 3) is used as
 - a) a noun
 - b) an adjective
 - c) a verb
 - d) an adverb
2. The word "computing" (par. 4, line 9) is used as
 - a) a verb
 - b) an adverb
 - c) a noun
 - d) an adjective
3. The word "Each" (par. 4, line 10) refers to
 - a) neuron
 - b) web
 - c) network
 - d) technology
4. The word "those" (par. 4, line 14) refers to
 - a) combinations
 - b) layers
 - c) neurons
 - d) networks
5. The connection between paragraphs 4 and 5 is that of
 - a) elaboration
 - b) addition
 - c) comparison
 - d) exemplification

6. What connector can best fill in the blank in paragraph 7, line 5?
- a) Since
 - b) Unless
 - c) If
 - d) Although
7. The word "it" (par. 7, line 12) refers to
- a) machine
 - b) task
 - c) information
 - d) forgetting
8. The word "simulated" (par. 8, line 3) is used as
- a) a noun
 - b) an adjective
 - c) a verb
 - d) an adverb
9. What modal can best fill in the blank in paragraph 8, line 10?
- a) could
 - b) would
 - c) ought to
 - d) are able to
10. The word "They" (par. 11, line 13) refers to
- a) neural networks
 - b) objects
 - c) patterns
 - d) errors
11. What connector can best fill in the blank in paragraph 14, line 6?
- a) Consequently
 - b) Furthermore
 - c) Nevertheless
 - d) Similarly

12. What modal can best fill in the blank in paragraph 14, line 13?

- a) will
- b) will have to
- c) might
- d) is able to

True/False Statements – An Exercise

Check (✓) TRUE or FALSE. CORRECT the FALSE statements.

1. Bridge structures are too complex for complete computer analysis since simulations require a host of simplifying assumptions.

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

2. Engineers have learned at great expense to direct water away from structural elements and bearing surfaces, where the combination of salt and stress can stabilize a bridge in a few years.

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

3. Throughout the monument's history, architects and engineers have attempted to halt the lean, but since regular monitoring began in 1911, the offset at the top of the tower has decreased at a fairly consistent rate of about 1.2 millimeters each year.

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

4. Initial efforts focused on the exterior of the tower, but in the next few months, engineers plan to try more radical techniques which will be applied directly to the soil modifying the monument's footing.

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

5. Water molecules' ability to interact electrostatically with ions such as chloride results from water's capacity to dissolve sodium chloride, common salt, a substance made up exclusively of ions.

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

6. Utilities using high-voltage silicon switches will benefit from being able to deliver more power of better quality while reducing transmission losses and thus the amount of energy they must generate.

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

7. Increased flexibility in distribution will make it more difficult to transfer power from one region to another on demand, thus permitting individual utilities to reduce their generating reserve margins.

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

8. The batteries in Tesla's current crop of vehicles contain 30 times less Watt hours of useful energy than a kilogram of petrol, while traditional capacitors hold 1000 times less. It is clear that the former are noncompetitive.

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

9. People with health anxiety worry irrationally about having a serious medical problem and hold maladjusted views about the role worry plays in maintaining their emotional and physical well-being.

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

10. Although cyberchondria has never been added to the American Psychiatric Association's list of diagnosable mental disorders, reliable estimates of the number of people who have it are not available.

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

11. Last year CO2 emissions from power stations and industrial sources fell back to 2013 levels. Specialists are optimistic and claim that although emissions will probably still rise again, we may be surprisingly close to their lowest point.

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

12. While recent studies found some benefits of microloans, there was no average impact on the main financial well-being indicators — income and household and food expenditures.

☐ TRUE ☐ FALSE

a. Copy the incorrect word: _____

b. Write the correction: _____

Identifying the Main Idea of a Paragraph – An Exercise

1. Today's supersonic aircraft engineers believe they have come up with a cost-effective way not only to make jet engines more efficient, but also to solve the problem of noise during the supersonic flight. They have found all kinds of designs to achieve their aim. Back in the 1970s, Richard Seebass at the University of Colorado calculated that simply blunting the aircraft's nose would virtually eliminate the boom. A blunt nose would raise the air-pressure ahead of the aircraft, increasing air temperature and stretching the bow shock out in front of the aircraft. Then in 1998 they found a better way to achieve this effort. It involved a flat flap that extends forwards from the aircraft's nose like the drop nose on Concorde. When this flap tilts upwards a few degrees, it acts just like Seebass's blunt nose, stretching the bow shock wave and reducing its peak pressure.

What is the main idea of the paragraph?

- a. Blunting the aircraft nose can help solve the problem of noise during the supersonic flight.
 - b. The solution found in 1998 is more efficient than the design created by Richard Seebass in the 1970s.
 - c. It is highly probable that technical barriers to quiet supersonic flight have been overcome.
 - d. A blunt nose or a flat flap are utilized for the same purpose: to stretch the bow shock wave.
-
2. Simple problems, if allowed to progress unchecked, can lead to severe damage. Debris on a bridge deck, for example, may block drains, causing water to accumulate. In cold regions, water may freeze inside the deck, cracking it. Deicing compounds, however, form salt solutions that rapidly corrode reinforcing bars and other structural members. When the salt content reaches a critical level, the concrete must be replaced even though the cement and aggregate are still sound.

What is the main idea of the paragraph?

- a. Water which accumulates inside the deck due to drain blockage can seriously damage a bridge.
- b. Salt solutions resulting from deicing compounds pose a serious threat to bridge safety.
- c. Bridge maintenance can be made more costly by the need to replace the concrete which deteriorates more quickly than the other materials.
- d. Even seemingly uncomplicated issues can have an adverse and far-reaching impact on the condition of bridge structures.

3. Producing a plastic that can emit light would have seemed outlandish just two or three years ago. Even the idea that plastics can conduct electricity comes as a surprise to most people. After all, plastics are familiar as insulating sheaths round wires and as the casings around all manner of electrical appliances. But in the mid-1970s, a group of scientists discovered that plastics do not have to be insulators, when they stumbled across a polymer that behaved as a semiconductor. This discovery opened up the prospect of materials that might combine the electrical properties of metals with the ease of processing and versatility that have made plastics a runaway success over the past half-century. These pioneers predicted that a world of plastic wires and batteries was just around the corner.

What is the main idea of the paragraph?

- a. The idea that plastics can produce light is a very recent one and so is the realization that they have the capacity to conduct electricity.
- b. The discovery that plastics can conduct electricity was an important breakthrough which occurred in the mid-1970s.
- c. Despite the commonly accepted opinion that plastics are insulators, scientists found out that they can have electrical conductivity.
- d. Materials combining the advantages of metals with those of plastics will very soon become available for general use.

4. More recently, the ability of supercritical fluids to dissolve things that will not dissolve in everyday solvents has attracted the attention of Britain's brewing industry. In a bid for consistency that will make drinkers of real ale turn pale, some brewers have begun to use a similar technique to "clean up" hops for the brewing industry. Originally a German idea, it was taken up by a company called English Hops to remove some of the natural flavours and colours from hops and so give a more consistent taste to beer. The food industry uses SCFs in similar ways. Universal Flavours of Slough, which extracts oily essences for use in everything from celery soup to aerosol sprays, uses SCFs to dissolve out the chemicals responsible for flavour from vegetable matter. SCFs are used similarly by analytical chemists to separate tricky mixtures of chemicals. Now, thanks to recent developments in chemists' understanding of supercritical fluids, researchers are extending their repertoire even further.

What is the main idea of the paragraph?

- a. Britain's brewing industry has begun to exploit supercritical fluids to standardize the taste of beer.
 - b. Supercritical fluids are already used industrially, but they are currently being considered for wider utilization.
 - c. Chemists use supercritical fluids in ways parallel to their exploitation in the brewing and food industries.
 - d. The latest breakthrough research on supercritical fluids has enabled their widespread use in the food industry.
5. One of the best known d-wave theories was formulated by David Pines at the University of Illinois, Urbana-Champaign and Doug Scalapino at the University of California, Santa Barbara. It includes the idea of a Cooper pair, but leaves behind the phonon-electron interaction. The researchers suggest that it is the spin of the copper atoms, rather than the atomic lattices that is fluctuating. They assume that the spins are highly ordered into a pattern like a chessboard, where white and black squares correspond to up and down spins. (Whether the superconductor is indeed this highly ordered remains another unanswered question). Where in the

BCS theory a passing electron is thought to change the surrounding charge concentration, here it is thought to change the spins of the surrounding atoms. These altered spins then affect a second electron, pulling it through sharply so that it slams into the first.

What is the main idea of the paragraph?

- a. The distinctive feature of Pine's and Scalapino's d-wave theory is its focus on the changes in the spin of the atoms.
- b. The fluctuation of atomic lattices provides the basis for the explanation of superconductivity offered by Pine and Scalapino.
- c. According to the BCS theory, the alteration caused by a passing electron in the surrounding charge concentration results in superconductivity.
- d. The premise on which Pine's and Scalapino's theory is based is that superconductors are highly ordered.

Inferences – An Exercise

- 1. If recycled goods are so good for the environment, why aren't more people using and buying them?**

Which of the following can be inferred?

- a. Recycling takes up a lot of space.
- b. It takes time to change people's habits.
- c. Recycled goods are associated with used products.

- 2. Suicides among soldiers – Officers, beware!**

We can infer that the text is expected to be about

- a. officers who are too lenient with their soldiers
- b. desperate soldiers and officers' accountability
- c. soldiers who offend their officers

- 3. Hundreds of people went on strike when the local authorities raised taxes on water.**

Which of the following can be inferred?

- a. People are unhappy with the authorities' decision.
- b. The authorities can't change their decision once it is made.
- c. Going on strike is an instrument to influence the authorities' decision.
- d. The authorities raising taxes is an insufficient reason to go on strike.

- 4. Export today is a big and profitable business. While hundreds of years ago people used domestic products that were made within the geographic boundaries of their own countries, nowadays they are using both domestic and imported products.**

Which of the following can be inferred?

- a. Export is a profitable business.
- b. Homemade products are the best.
- c. Many products we buy today are brought from abroad.
- d. Export is a strategy which was used hundreds of years ago.
- e. The export marketing strategy will be adopted by many businesses.

- 5. The store was operating at a loss until a new marketing manager was appointed by the top management.**

Why was a new manager appointed?

- a. The former manager underperformed.
- b. The former manager was transferred to another department.
- c. Top management wished to recoup their losses.

6. **John is very good when it comes to making new friends. His outgoing nature works in his favor, as he makes the most of the opportunities he gets without worrying about questions like "What if?"**

Which of the following can be inferred?

- a. John is an extrovert.
- b. John seeks his own personal space and seldom likes anyone invading it.
- c. John is suitable for a job which requires interaction with people.
- d. John has excellent communication skills which help him in getting along with people quite well.

7. **Twenty thousand Inuit dogs were killed in the eastern Canadian Arctic in the 50s, 60s, and 70s by the Royal Canadian Police.**

Which inference would best serve the interests of the Inuit people?

- a. The killings were part of a government effort to force Canada's Inuit out of the traditional lifestyle.
- b. The RCP had killed the dogs because these animals were diseased, starving, or dangerous.

8. **Ivory Coast normally produces nearly 30% of the world's raw cocoa. There is currently a political instability in Ivory Coast which affects the production of cocoa.**

Which of the following can be inferred?

- a. Ivory Coast's production of cocoa is now well below normal.
- b. Chocolate lovers will need to pay more for a bar of chocolate.
- c. The market price of raw cocoa will go up.
- d. The economy of Ivory Coast is linked to its political situation.

9. **Ghana borders Ivory Coast. Unlike Ivorian cocoa, Ghana's cocoa is of premium quality.**

Which of the following can be inferred?

- a. Ivorian cocoa is of a lesser quality than that of Ghana.
- b. The price of Ivorian cocoa is lower than that of Ghana's.
- c. Ghana's cocoa is not of the best quality.

10. **According to the results of the December 2010 CFA Level 1 exam, a mere 34% of candidates passed globally. December 2009's pass rate was also dire, at 35%. However, the June 2010 pass rate was fairly good by comparison, at 46%.**

Which of the following are NOT inferences?

- a. The pass rate in June 2010 was undoubtedly higher than that in December of the same year.
- b. The candidates who took the test in June were better prepared.
- c. The questions in the June test were somewhat easier.
- d. The pass rate in December 2009 and 2010 was almost the same.
- e. One may have higher chances to pass the test in June.

Listening Comprehension 1: How to Spot a Misleading Graph

1. Graphs may help to persuade people who would otherwise remain unconvinced since they
 - a. are a very powerful way to present opinions
 - b. are used mainly by experienced politicians
 - c. offer them a possibility to form their own arguments
 - d. supposedly present objective statistical evidence

2. According to the bar graph presented in the advertisement,
 - a. Toyota trucks are much more reliable than Chevy trucks
 - b. the performance of Chevy trucks far outstrips that of Toyota trucks
 - c. the number of Toyota trucks still running after a decade of service is twice that of Chevy trucks
 - d. the performance of Chevy trucks after ten years of exploitation equals that of Toyota trucks

3. If the scale on the Y-axis of the graph were not distorted,
 - a. it would be easy to detect that the difference it shows is insignificant
 - b. it would present truthful information by going from 95 to 100 percent
 - c. the bars on it would still show the same significant disparity
 - d. the figure for Toyota would be higher than 96.5 percent

4. Inconsistencies along the X-axis are usually used
 - a. to misrepresent the differences between the things being compared in a bar graph
 - b. to manipulate the representation of alterations taking place during a certain period of time
 - c. to correct irregularities in the representation of various time-consuming processes
 - d. to prevent line graphs from giving a wrong impression about the presented information

5. Cherry picking means
 - a. representing the influence of a major event by including it in the time range of a graph
 - b. deliberately choosing a misleading time range and selecting convenient units of information
 - c. showing all the relevant data points so as to produce an objective graphical illustration of a process
 - d. providing a visual representation of every important change occurring during a certain period of time

6. The chart of how many people watch the Super Bowl each year is misleading
 - a. because of the distortions deliberately made on both of its axes
 - b. since irrelevant information included in it gives a wrong impression
 - c. due to the exclusion of some highly important data
 - d. as it focuses on the ratio of football fans to the general TV audience

7. In order not be misled by a graph, it is necessary to
 - a. concentrate on the lines and curves and pay careful attention to them
 - b. look closely at the labels and the numbers while disregarding the context
 - c. take into consideration the context in addition to scrutinizing the labels, the numbers and the scale
 - d. inspect the details of the story behind it without focusing too much on its visual elements

Listening Comprehension 2: How Statistics Can Be Misleading

1. The problem with making decisions based on statistics is that
 - a. sometimes statistics may not be persuasive enough
 - b. statistics may be less reliable than organized data
 - c. some sets of statistics are made invalid by obvious flaws
 - d. a collection of statistical data may hide a disruptive factor
2. In spite of appearances, hospital B is the better choice
 - a. for patients arriving with serious health issues
 - b. for patients arriving without serious diseases
 - c. for all patients, regardless of their health condition
 - d. for patients who are not in need of a complicated operation
3. Simpson's paradox is a phenomenon in which
 - a. the same statistical information points to different conclusions depending on how it is presented
 - b. a trend appears in several different groups of data and persists when these groups are combined
 - c. a trend is observed in a combination of several groups of data but reverses itself within each group
 - d. statistical information is unreliable when presented both in separate groups and in various combinations

4. A lurking variable is
 - a. a concealed factor which has no impact on the results of a statistical analysis
 - b. an easily detectable variable which may greatly influence the outcomes of a statistical experiment
 - c. an invisible element which has a major impact on the conclusions drawn from a statistical analysis
 - d. a feature which influences a statistical experiment in the same way whether it remains hidden or becomes visible

5. The study in the UK is an example of
 - a. a negligible influence of Simpson's paradox on serious real-life situations
 - b. a lurking variable drastically changing our understanding of an important issue
 - c. Simpson's paradox enabling researchers to present a hypothetical scenario
 - d. the vital importance of ignoring lurking variables for obtaining reliable results

6. In the example of Florida's death penalty cases, the lurking variable is
 - a. the race of the murderer
 - b. the racial disparity in sentencing
 - c. the sentencing rate
 - d. the race of the victim

7. In order to avoid being manipulated by statistics, we should
 - a. always carefully divide the data into arbitrary categories
 - b. look for lurking variables while thoroughly evaluating the context
 - c. disregard the context of each given statistical analysis
 - d. ignore the presence of lurking variables and focus on the given data

Listening Comprehension 3: The Turing Test -- Can a Computer Pass for a Human

1. Many specialists think that the prospects for the development of artificial intelligence depend on whether
 - a. neurons in the brain can be sparked artificially
 - b. a machine can function efficiently without thinking
 - c. we can find out what makes up the human mind
 - d. the complex issues related to the human mind can be disregarded

2. The idea behind the Turing test was
 - a. to add another criterion to already existing parameters
 - b. to replace uncertain criteria by a basic standard
 - c. to proceed without any established set of criteria
 - d. to use abstract ideas instead of specific standards

3. To pass the Turing test, a computer would have to
 - a. successfully replace the human judge and evaluate the conversation of the players
 - b. radically transform the responses of one of the players
 - c. talk in a way nearly identical to human conversation
 - d. produce responses easily identifiable as those of a machine

4. Despite Turing's predictions,
 - a. nowadays computers have the capacity to easily succeed in passing his test
 - b. to this day no machine has been able to overcome the difficulties presented by his test
 - c. computers still do not have enough memory to pass his test successfully
 - d. enhanced memory has not significantly improved computers' chances of passing his test

5. Eliza's and Parry's opposite approaches have led to success since
 - a. the Turing test is perfectly reliable although subjective
 - b. people can be easily misled into detecting intelligence
 - c. the Turing test is based on objective criteria
 - d. people are always able to correctly assess intelligence

6. Cleverbot's use of statistical analysis
 - a. is not enough to enable it to sustain human-like conversation
 - b. allows it to talk on a vast range of novel topics in a human-like manner
 - c. makes its responses sound very artificial and robotic
 - d. gives it the capacity to successfully enact a human persona

7. It might be necessary to understand what conscience is in order to
 - a. enhance computers' memory and processing capacity
 - b. set new goals different from that of Turing
 - c. make machines adept at performing a variety of complex tasks
 - d. create human-like artificial intelligence

Connectors – An Exercise

1. _____ there is a great deal of pollution in the Mediterranean, many tourists continue to swim in its seas.
 - a. Since
 - b. Although
 - c. However
 - d. As

2. _____ photovoltaic cells are still not widely used to generate power, their potential for the future is great.
 - a. Even though
 - b. As
 - c. Therefore
 - d. Due to

3. _____ the planet's heating up, man is continuing to destroy the ozone layer; _____, it is likely that future generations will suffer greatly from man's failure to respect nature.
 - a. Due to; so
 - b. Despite; therefore
 - c. Although; nevertheless
 - d. While; subsequently

4. _____ the fact that technology is continually advancing, man has little notion of how to save the planet.
 - a. Whereas
 - b. Owing to
 - c. In spite of
 - d. Even though

5. The Japanese railroad system is one of the most advanced in the world; _____, it is considered to be extremely safe.
- moreover
 - therefore
 - on the other hand
 - like
6. _____ attending their English classes at Sami Shamoon, the students have been able to read academic texts with greater ease; _____, there are still some texts which are difficult to comprehend.
- As; furthermore
 - Due to; thus
 - Since; however
 - Although; hence
7. _____ a company produces a new product, it should research the current needs of the market _____ decide whether there will be sufficient demand for it; _____, they might carry out a survey on a small sample of the population.
- Before; in order to; for instance
 - After; such as; for example
 - While; so as; that is
 - During; as well as; on the other hand
8. The Leaning Tower of Pisa is closed to the public _____ it is in danger of falling over. _____ the beginning of its construction in 1173, the tower has continued to tilt. Scientists have said that with such a tilt the tower should have fallen over long ago; _____, to the amazement of everybody, it continues to stand.
- owing to; Until; nevertheless
 - as; Once; therefore
 - because; Since; however
 - thus; During; while

9. Landfills are a common way of getting rid of much of the rubbish produced by the population; _____, this is becoming rather problematic, _____ the landfills are beginning to fill up and the country is running out of space. _____, other solutions to cope with the country's rubbish must be found.
- a. whereas; similarly; Subsequently
 - b. therefore; as; So
 - c. nevertheless; thus; Since
 - d. however; since; Hence
10. Computers in the world are forever growing in importance, _____ schools have decided to start teaching children how to use them from an early age. _____, not everyone finds using a computer easy; one must have the capacity to understand elementary logic _____ to be able to use it.
- a. due to; Likewise; as well as
 - b. thus; However; such as
 - c. since; Therefore; in order to
 - d. so; Nevertheless; so as
11. Many TV, radio, and print media advertisements for a particular product advance virtually the same message or appeal; _____, _____ the advertisements share a common theme, they are constructed differently to fit each media.
- a. likewise; while
 - b. nevertheless; although
 - c. therefore; as
 - d. however; because
12. Advertisements for expensive jewelry often picture the item on a plain-coloured velvet cloth with only a few words, _____ not to distract from the beauty of the item. _____, a busy advertisement with a jumble of words and pictures and very little whitespace may suggest low price and quality.
- a. in order; Although
 - b. although; Similarly
 - c. so as; On the other hand
 - d. such as; However

Connectors – Exercise 2

1. _____ engineers agree on the general mechanisms of bridge failure, the details are not well understood.
 - a) Despite
 - b) Since
 - c) Although
 - d) Once
2. Some scientists predicted that a world of plastic wires and batteries was just around the corner. _____, the predictions have not yet come true because of several major issues.
 - a) Consequently
 - b) Moreover
 - c) Finally
 - d) However
3. Certain gases enter the no-man's-land between a liquid and a gas _____ they are put under enough pressure and heated beyond a certain temperature.
 - a) whereas
 - b) so
 - c) once
 - d) unless
4. Water molecules are "polar"; _____, a water molecule will interact electrostatically with ions such as chloride and dissolve sodium chloride, common salt.
 - a) hence
 - b) subsequently
 - c) furthermore
 - d) nevertheless
5. At the same time that users of electric power demand quality, they also want more power. _____, transmission networks are being pressed closer to their operating limits.
 - a) On the other hand
 - b) As a result
 - c) For instance
 - d) In addition

6. In DC power transmission systems, electrical charge flows only in one direction (from producer to consumer), _____ in AC systems the flow of charge reverses itself many times each second.
- a) as
 - b) so
 - c) if
 - d) while
7. _____ a comprehensive theory is developed for high temperature superconductors, scientists will not be able to be sure of discovering a genuine new superconductor.
- a) Once
 - b) Unless
 - c) Even though
 - d) While
8. In an s-wave, the Cooper pair is so stable that a comparatively large amount of energy is needed to break it up. _____, no matter in which direction the Cooper pair travels through a superconductor, the energy needed to break it up remains the same.
- a) Moreover
 - b) Therefore
 - c) Similarly
 - d) Rather
9. A rechargeable battery is toxic, degrades substantially after a few years and, most importantly, must be carefully designed to avoid catastrophic overheating. _____, the exploding batteries in Samsung's Galaxy Note 7 smartphone caused the company to recall all the devices and cease production.
- a) Eventually
 - b) For example
 - c) In addition
 - d) On the contrary
10. Even if we have enough intelligence to design super-human artificial intelligence, the result may not be computers receiving super-human intellect _____ improving intelligence is far harder than just being intelligent.
- a) thus
 - b) but
 - c) despite
 - d) as

Relationships between Clauses and/or Sentences

1. Since President Clinton admitted to having an affair with Monica Lewinsky, three Republican lawmakers were forced to acknowledge adulterous affairs.

The first clause expresses the relationship of

- a. cause and effect b. time c. addition

2. The European Jews were the primary victim of the Holocaust. But Jews were not the only group singled out for persecution by the Nazi regime. As many as one-half million Gypsies, at least 250,000 physically or mentally disabled people, and more than three million Soviet prisoners-of-war fell victim to the Nazi genocide.

The relationship of the last sentence to the one before it is one of

- a. contrast b. cause and effect c. addition

3. Andrei Sakharov (1921-1989) was a Soviet physicist who spent two decades designing nuclear weapons. He was regarded as the father of the Soviet hydrogen bomb, contributing to the military might of the USSR. But gradually Sakharov became one of the regime's most courageous critics, a defender of human rights and democracy.

The relationship of the last sentence to the first two is one of

- a. contrast b. illustration c. addition

4. Famine in Africa cannot be due to overpopulation. There are actually three major contributing factors: drought, inefficient farming techniques and war.

The relationship of the second sentence to the first one is one of

- a. illustration b. contrast c. elaboration

5. Cantonese cuisine originates from the region around Canton in southern China.

Cantonese dishes rarely use "hot" spices like chili, unlike Szechuan cuisine.

The second clause in the last sentence expresses a relationship of

- a. comparison b. cause and effect c. purpose

The relationship of the second sentence to the first one is one of

- a. contrast b. elaboration c. result

6. A drug is a substance that can affect your body. Not all drugs are illegal. Drug like cannabis, ecstasy, and cocaine are illegal. Other drugs which are against the law include speed, heroine, LSD, and magic mushrooms. But many are legal, like alcohol, headache tablets, cigarettes, and glue.

The relationship of the last sentence to the second one is one of

- a. addition b. illustration c. contrast

The relationship of the third sentence to the one before it is one of

- a. contrast b. illustration c. addition

The relationship of the fourth sentence to the one before it is one of

- a. contrast b. illustration c. addition

7. A pig is a device that moves through an oil pipeline and breaks up congestion and sludge. This is why pigging is the term for the process that has been used for many years to clean large diameter pipelines in the oil industry. There are also "smart pigs" used to inspect pipelines for the purpose of preventing leaks, which can be explosive and dangerous to the environment.

The relationship of the second sentence to the first one is one of

- a. contrast b. elaboration c. addition

The relationship between the first sentence and the last one is one of

- a. illustration b. comparison c. addition

8. Six construction firms created a consortium called "Six Company, Inc." in order to submit a competitive proposal to build the Hoover Dam. As the lowest qualified bidder at \$48,890,955, Six Company was awarded the contract.

The relationship of the second sentence to the first one is one of

- a. comparison b. cause and effect c. time

9. The blast of the first atomic bomb was equivalent to 12,500 tons of TNT. By present standards, the bomb was a small one, and in today's arsenals it would be classed among the merely tactical weapons. However, it was still large enough to transform a city of some 340,000 people into hell in a matter of seconds.

The relationship of the last sentence to the second one is one of

- a. cause and effect b. contrast c. addition

10. The new employee had a good reason to be grateful to his boss for the opportunities which he had made available to him, and which subsequently led to his promotion.

The last clause expresses a relationship of

- a. addition b. comparison c. cause and effect

Passive Voice – An Exercise

1. The two conical shock waves that each aircraft forms and that spread out behind it, one starting at the nose and one at the tail, _____ the bow shock and tail shock waves.
 - a. is called
 - b. are called
 - c. were called
 - d. be called
2. Forty-two of seventy-nine nationally reported bridge failures in the U.S. between 1951 and 1988 _____ by collisions, almost half of which involved ships that rammed bridge supports.
 - a. are caused
 - b. was caused
 - c. were caused
 - d. will be caused
3. Initial efforts focused on the exterior of the tower, but in the next few months, the engineers plan to try other, more radical techniques to halt the tower's lean which _____ directly to the soil modifying the monument's footing.
 - a. will be applied
 - b. are applied
 - c. were applied
 - d. are being applied
4. There are still plenty of obstacles blocking the road that could, one day, lead to these applications, but more and more information _____ by chemists, physicists and engineers about the way the materials work and strategies for improving them.
 - a. are found out
 - b. was found out
 - c. have been found out
 - d. is being found out

5. The result is a sequence of alternating single and double bonds all the way down the carbon backbone, called "conjugated" sequence, in which the extra electrons can _____ to move along the polymer chain, forming an electric current.
- a. are induced
 - b. be induced
 - c. is induced
 - d. were induced
6. Some inspired lateral thinking led Zosel to the idea that a supercritical fluid might also dissolve and remove caffeine from coffee, and more recently, the attention of Britain's brewing industry _____ by the ability of supercritical fluids to dissolve things that will not dissolve in everyday solvents.
- a. is attracted
 - b. were attracted
 - c. has been attracted
 - d. will be attracted
7. Nowadays, one of the amazing properties of supercritical water, its ability to dissolve oxygen and so to support a flameless variety of combustion, _____ as a means of "clean" incineration of organic toxic waste.
- a. are explored
 - b. was explored
 - c. be explored
 - d. is being explored
8. Solid-state controllers could increase the overall capacity of today's transmission network by 20 percent or more: for example, the power transfer capacity of two 500-kilovolt lines serving Florida could _____ from 2,000 megawatts to 3,000 megawatts by the addition of thyristor-controlled series capacitors.
- a. is increased
 - b. was increased
 - c. be increased
 - d. will be increased

9. A second set of experiments, by Walter Hardy's group at the University of British Columbia in Vancouver, which involved measuring the size of magnetic fields superconductors expelled, _____ on another of their observable characteristics.
- a. was based
 - b. were based
 - c. is based
 - d. will be based
10. Any lithium ion battery needs tiny separators to keep its components apart; if these _____, they can fail, creating a short circuit that heats and damages other parts of the battery causing a runaway reaction.
- a. is damaged
 - b. was damaged
 - c. are damaged
 - d. are being damaged

Parts of Speech – An Exercise

1. If we feel our personal space has been invaded, we might avert our head and **place** an elbow between ourselves and the intruder.

The word **place** is used as

- a. a noun
- b. an adjective
- c. a verb
- d. an adverb

2. The key factor for **understanding** someone's message is to recognize a change or variation from the person's normal voice quality.

The word **understanding** is used as

- a. a noun
- b. an adjective
- c. a verb
- d. an adverb

3. At first, both the subjects who received cues to expect an important signal and those who did not receive any cues performed in a **similarly** effective manner.

The word **similarly** is used as

- a. a noun
- b. an adjective
- c. a verb
- d. an adverb

4. Unless the reliability of operational cueing systems is high, they may offer no real **benefit** while incurring considerable cost in sustained attention.

The word **benefit** is used as

- a. a noun
- b. an adjective
- c. a verb
- d. an adverb

5. Fears about the safety of the landmark became acute when a similarly **constructed** bell tower at the Cathedral of Pavia collapsed suddenly in 1989.

The word **constructed** is used as

- a. a noun
- b. an adjective
- c. a verb
- d. an adverb

6. All work is being done far from the monument itself because the engineers must guard against the possibility that altering the ground too **close** to the tower could eventually damage the building.

The word **close** is used as

- a. a noun
- b. an adjective
- c. a verb
- d. an adverb

7. Like conducting plastics, the new **light-emitting** materials the market is waiting for are conjugated polymers.

The word **light-emitting** is used as

- a. a noun
- b. an adjective
- c. a verb
- d. an adverb

8. The general rule is that like dissolves **like**: that is, most organic chemicals, which are nonpolar or "oily", are not soluble in polar solvents like water, but dissolve in solvents which are themselves oily.

The word **like** is used as

- a. a noun
- b. an adjective
- c. a verb
- d. an adverb

9. The engineers who manage the transmission of electric power must act with extreme caution, so they operate the **power** grid well below its theoretical maximum capacity.

The word **power** is used as

- a. a noun
- b. an adjective
- c. a verb
- d. an adverb

10. In AC systems, the flow of **charge** reverses itself many times each second; a terminal that had a positive potential a fraction of a second ago may now have a negative potential.

The word **charge** is used as

- a. a noun
- b. an adjective
- c. a verb
- d. an adverb

Modals – An Exercise

1. Prototype microalgae farms are at work in Australia. By mid-century, there _____ be up to 50 million hectares of them growing billions of tons of biomass.
 - a) can
 - b) must
 - c) could
 - d) would

2. What is really needed is a solution that sucks CO₂ out of the atmosphere while boosting food production. A good solution _____ be to bury biochar: agricultural waste like straw, manure and unused food that has been pyrolyzed to charcoal.
 - a) will
 - b) has to
 - c) is able to
 - d) would

3. Fortunately, we are living in the age of big data: decisions that used to be made on instinct _____ now be based on solid evidence.
 - a) must
 - b) should
 - c) might
 - d) can

4. Even after their worries subside, people with health anxiety _____ eventually begin worrying that they have not worried enough about their medical state, thus triggering the cycle anew.
 - a) will
 - b) ought to
 - c) have to
 - d) would

5. Individuals are encouraged to “observe” thoughts about illness or injury without letting them intrude emotionally. Or they _____ postpone their health worries until a scheduled time of day.
- a) must
 - b) might
 - c) are able to
 - d) do not have to
6. There are doubts over whether the periodic table is in the best possible configuration. Just as notes can be arranged in various ways to produce music, so the essence of the relationships between the elements _____ be depicted differently.
- a) would
 - b) should
 - c) could
 - d) must
7. The further down the table you go, the more of the positively charged protons an element’s nucleus contains. This creates a stronger pull on the orbiting electrons, meaning they _____ travel faster and faster.
- a) can
 - b) must
 - c) may
 - d) should
8. Most periodic tables separate the elements making up this f-block from the rest, putting it below the table, leaving a gap in group 3. However, there is a debate over which of the elements in the f-block _____ come first.
- a) should
 - b) could
 - c) might
 - d) would

9. Most organic chemicals, which are nonpolar or "oily", are not soluble in polar solvents like water, but _____ dissolve in solvents such as hexane, chlorine-containing hydrocarbons and ethers, which are themselves oily.

- a) may
- b) will
- c) would
- d) could

10. Zosel was familiar with the way chemists used supercritical fluids (SCFs) to purify mixtures of chemicals by dissolving out unwanted components. Some inspired lateral thinking led him to the idea that an SCF _____ also dissolve and remove caffeine from coffee.

- a) must
- b) will
- c) might
- d) can

Describing Graphs and Charts – Vocabulary

I. Verbs

rise, surge, decline, rocket, reduce, plummet, improve, slump, lose ground, plateau, decrease, recover, expand, drop, jump, stabilize, shoot up, go down, remain constant, fall, grow, level out, increase, soar, stay at the same level

For upward trends (i.e. when lines are going up on a graph)	For downward trends	When there is no change
1.	1.	1.
2.	2.	2.
3.	3.	3.
4.	4.	4.
5.	5.	5.
6.	6.	
7.	7.	
8.	8.	
9.	9.	
10.		
11.		

II. Adjectives

consistent, considerable, rapid, irregular, sharp, steady, sudden, progressive, constant, unimportant, uneven, marked, slight, invariable, steep, quick, gradual, minor, dramatic, unstable, significant, swift, taking a long time, inconsequential, fluctuating, continuous, negligible, prolonged

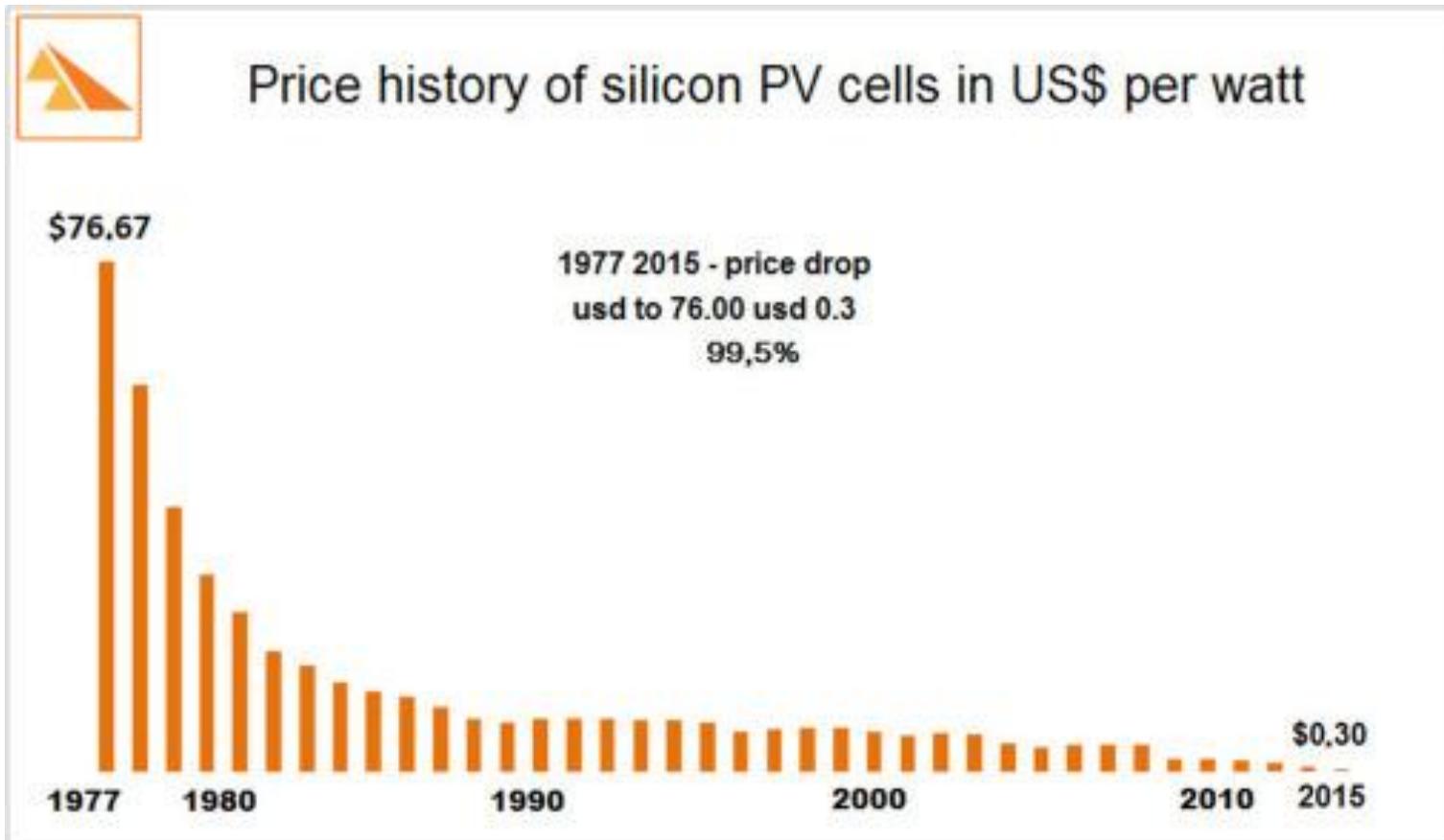
Fast	Slow	Large	Small	Changing	Unchanging
1.	1.	1.	1.	1.	1.
2.	2.	2.	2.	2.	2.
3.	3.	3.	3.	3.	3.
4.	4.	4.	4.	4.	4.
5.		5.	5.		5.
6.		6.			

Useful Vocabulary for Describing Graphs and Diagrams

Sentence	Noun	Verb	Adjective	Adverb
The short-term problems facing the company contributed to a _____ drop in production.	significance	signify	significant	significantly
Total profits of the corporation for the last quarter rose _____ compared with the previous one.	negligence	neglect	negligent	negligibly
As a result of growing environmental awareness, U.S. oil consumption was considerably _____ during the last decade.	reduction	reduced	reductive	reductively
According to the diagram, intelligent use of advertising has made an important _____ to the financial success of the firm.	contribution	contributed	contributive	contributively
We can observe _____ trends in some of the world's key gas producing countries.	expansion	expand	expansive	expansively
It is easy to detect that the output of the coal mines in the country _____ greatly in quantity over the period of the last 40 years.	variety	varied	various	variously

The graph in the yearly report demonstrates that the wages of private sector workers _____ rose.	graduation	graduate	gradual	gradually
The chart shows a considerable _____ in employment conditions in the industry.	improvement	improve	improving	improvingly
The production of smartphones in China was able to _____ its rocketing pace despite constant economic fluctuations.	maintenance	maintain	maintainable	maintainably
We witness a _____ increase in the general level of prices, which means that the economy of the country is suffering from inflation.	consistency	consist	consistent	consistently
The ongoing _____ of the supply to demand ratio reflected in the graph is an important indicator of the current state of the market.	stability	stabilize	stable	stably
The bar chart allows to conclude that the number of car accidents in the area has been growing _____ for the last five years.	exponent	expone	exponential	exponentially

Describing a Bar Graph – An Exercise



The graph is taken from Agmar Ferreira, Sheila S. Kunh, Katia C. Fagnani, Tiago A. De Souza, Camila Tonezer, Geocris Rodrigues Dos Santos, Carlos H. Coimbra-Araújo, "Economic Overview of the Use and Production of Photovoltaic Solar Energy in Brazil," *Renewable and Sustainable Energy Reviews*, Vol. 81, pp. 181-191, Jan. 2018. Accessed on: 26.11.2018. [Online]. Available: <https://doi.org/10.1016/j.rser.2017.06.102>

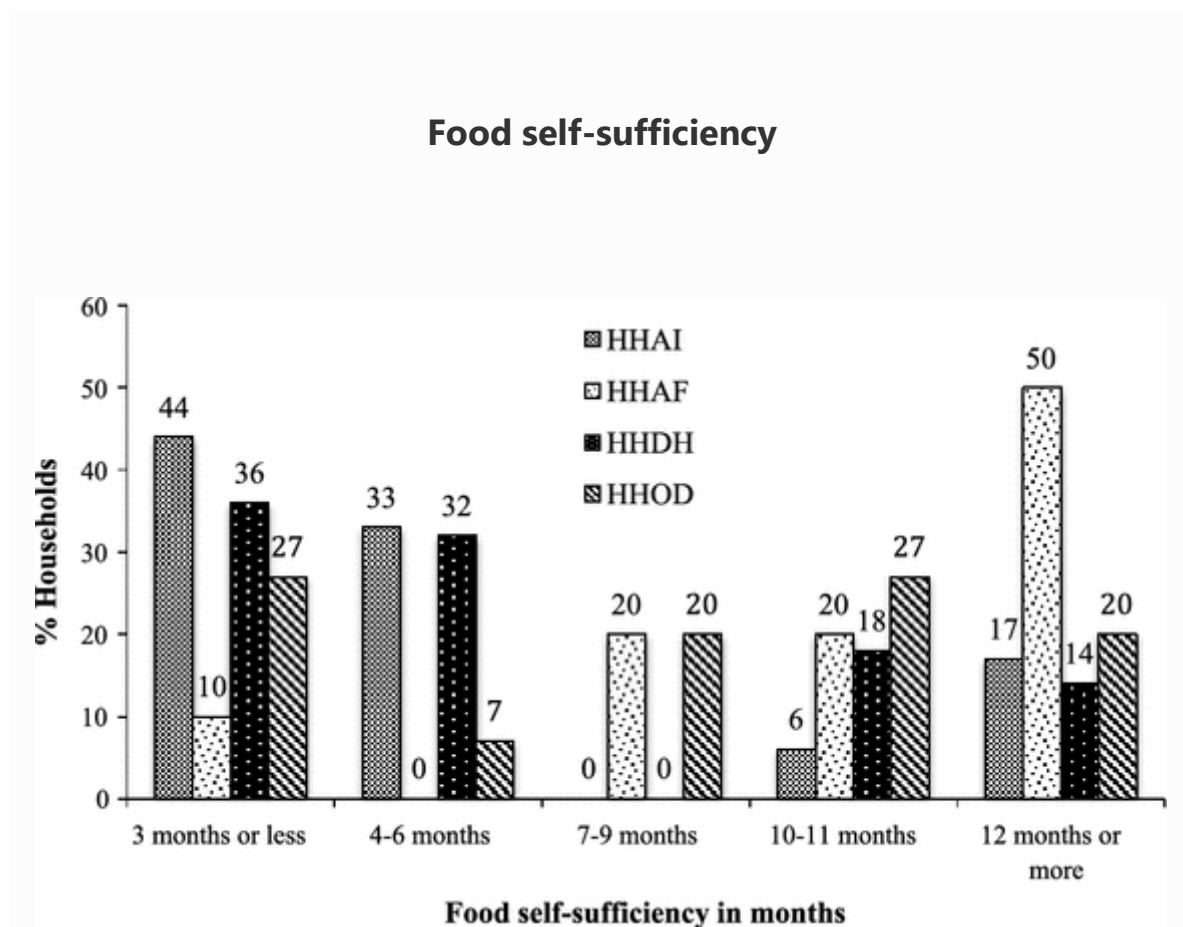
Describing a Bar Graph – An Exercise

Fill in the blanks with words from the list.

drop, coming down, low point, from, slowed down, detect, insignificant, subsequently, decrease, rise, remaining stable, gradual, dropped significantly, original, until, slight increase, occurred, fluctuation, followed by, one third

The bar graph shows the _____ of the unit cost of photovoltaic silicon cells _____ 1977 _____ 2015. It is possible to _____ that the price of the cells _____ between 1977 and 1980, _____ to approximately _____ of the _____ price of \$76.67 per watt. The price _____ during the next several years. Around 1990, there was a _____ in the cost of the cells, _____ its _____ during the next five years. _____, a minor _____ in the price ended by a minor _____ around 2000. During the next ten years, the changes in the cost of the cells were _____. From 2010 to 2015, there _____ a _____ price drop, until the cost of the cells reached its _____ of \$0.30 per watt.

Describing a Bar Graph – Exercise 2



Food available from the stock of self-production. *HHAI* Households based on agriculture and local informal sector off-farm income; *HHAF* Households based on agriculture and local formal sector off-farm income; *HHDH* Households based on agriculture and distant off-farm income, household head involved; *HHOD* Household based on agriculture and distant off-farm income, other household members involved

Source: Gartaula, H., Patel, K., Johnson, D. *et al.* From food security to food wellbeing: examining food security through the lens of food wellbeing in Nepal's rapidly changing agrarian landscape. *Agric Hum Values* **34**, 573–589 (2017). <https://doi.org/10.1007/s10460-016-9740-1>

Describing a Bar Graph – Exercise 2

Fill in the blanks using the following words.

approximately, indicates, between, off-farm, bar, correlation, additional, represents, the least, stability, a period of, by contrast, shorter, 50 percent, 14 percent, minority, observe, less, the most, tend, more

This _____ chart _____ the availability of self-produced food in agrarian households in Nepal throughout _____ one year. We can _____ that HHAF households are _____ self-sufficient, as _____ of them hold stocks of self-produced food for _____ twelve months or _____ and another 40 percent hold stocks that can last _____ 7 and 11 months. At the same time, only a _____ - 10 percent - of HHAF households hold stocks for 3 months or _____. _____, HHDH households are _____ self-sufficient, as their stocks of self-produced food last for a period of 12 months or more in only _____ of households, and an _____ 32 percent have stocks that will keep them going for a period of 4-6 months. For all households, income is based on both agriculture and _____ employment. There is a _____ between local off-farm income and food stock _____. The chart also _____ that households involved in the informal sector _____ to be able to hold on to their stocks for _____ periods.

Describing a Bar Graph – Exercise 3

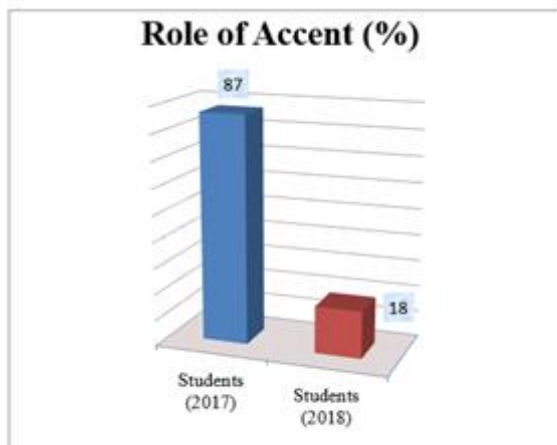


Figure 1

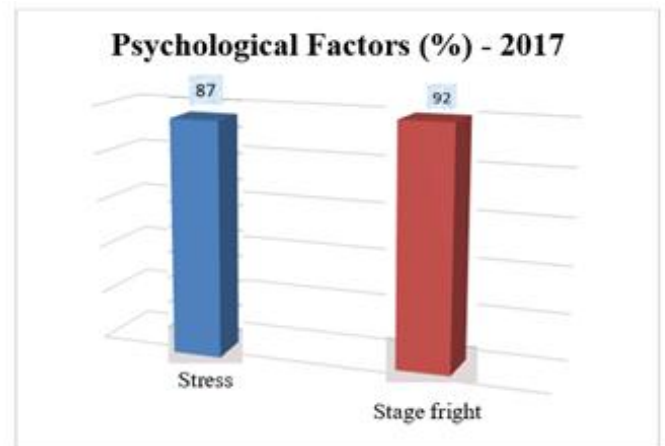


Figure 2

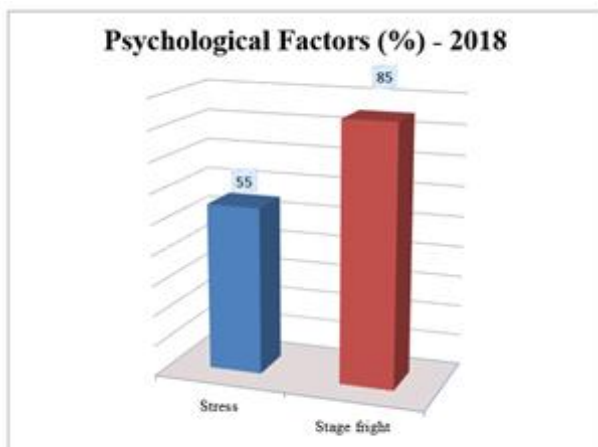


Figure 3

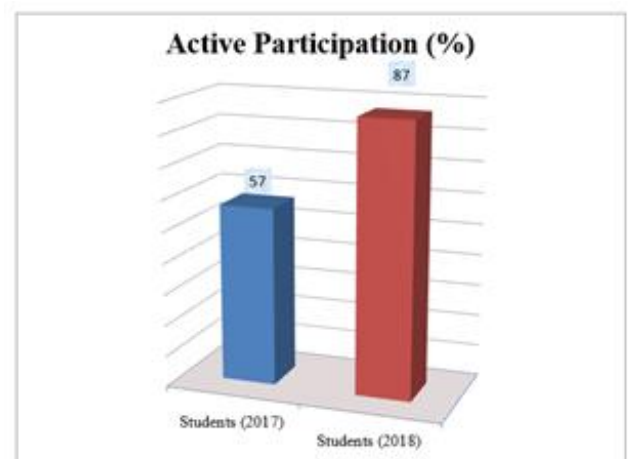


Figure 4

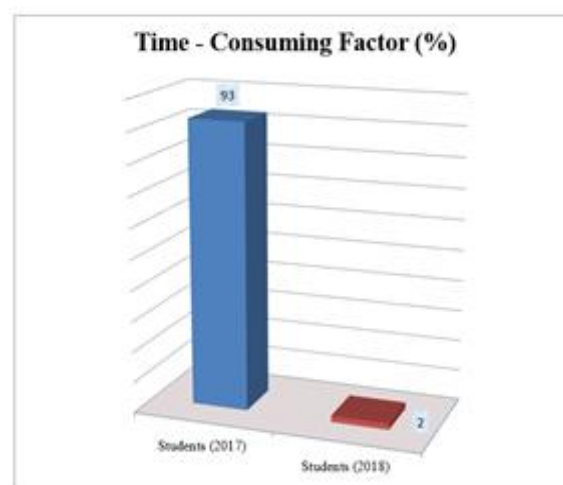


Figure 5

Describing a Bar Graph – Exercise 3

Fill in the blanks with words from the list below:

reduced, plunge, observe, in other words, increased, steep, positive development, decrease, in contrast to, correlation, also, significance, declined, show, explanation, noticeable, sharp drop, improvement, influence, less considerable, reduction in the parameters, however, considerable increase, in comparison with

In these bar charts we can see the _____ between the degree of importance attached to accent by students learning the English language and several psychological factors.

As we can _____ in Figure 1, there was a _____ in the number of the participants who accorded high value to the impact of accent in 2018: less than 20% of the responders saw it as a significant factor _____ 87% in 2017.

Interestingly, Figures 2 and 3 demonstrate that in 2018 there occurred a _____ in the percentage of students who reported experiencing stress related to their language studies. There was _____ a _____ though much _____ decline in the number of students who reported suffering from stage fright. These data point to a possible connection between the _____ importance of accent and the _____ in the psychological state of the students.

Another _____ can be seen in Figure 4: the chart illustrates a _____ in the students' active participation over the course of the second year of their studies (2018) _____ the previous academic year (2017). The data _____ an additional significant _____ of the importance attached to accent on the learning process: once the role or impact of accent _____ (2018), the students' active participation _____ (2018).

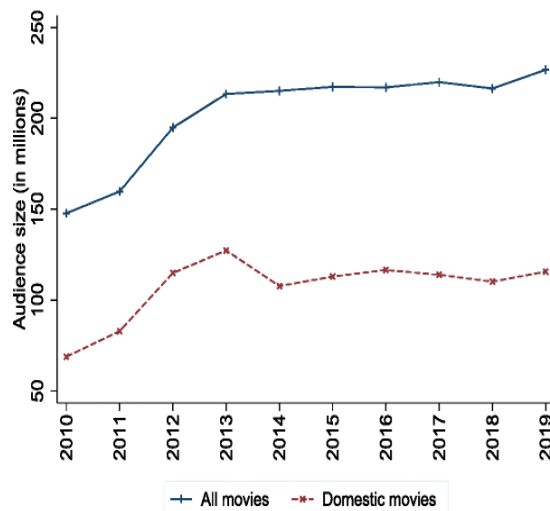
_____, there could be an additional factor which influenced the students' psychological state and performance: a higher level of language proficiency during the second year of studies could explain the _____ of stress and stage fright.

The same _____ could apply to the major _____ in the importance of the time-consuming factor which can be observed in Figure 5. _____, the students' improved command of academic English led to a _____ decline in the parameter's _____.

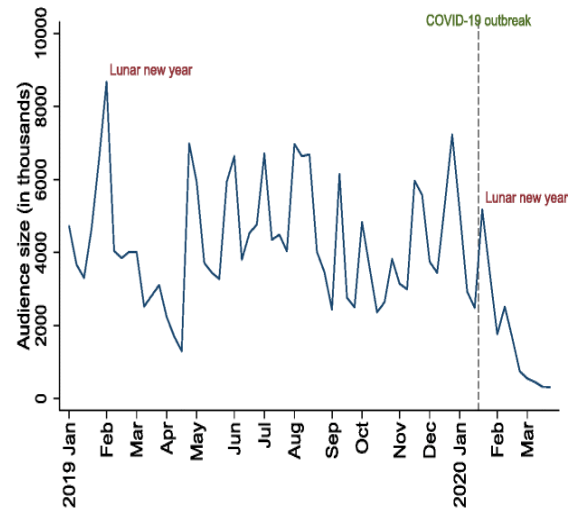
Analyzing a Line Graph – An Exercise

The Impact of COVID-19 on movie demand and price in South Korea

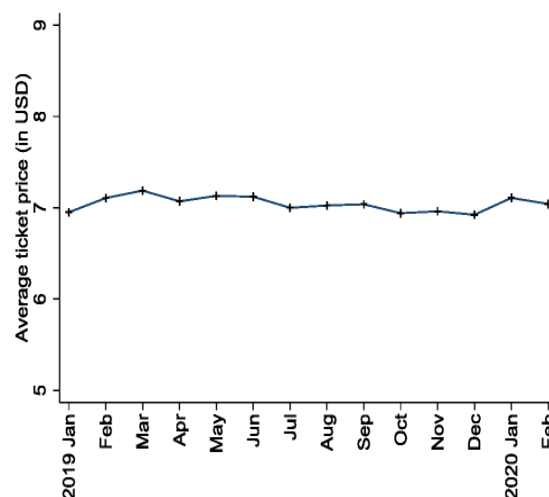
(a) Yearly audience size: 2010-2019



(b) Weekly audience size: 2019-2020



(c) Monthly ticket price: 2019-2020



Source: Kim, I.K. The impact of social distancing on box-office revenue: Evidence from the COVID-19 pandemic. *Quant Mark Econ* (2020). <https://doi.org/10.1007/s11129-020-09230-x>

Analyzing a Line Graph – An Exercise

Fill in the blanks with words from the list below:

same level, X-axis, lower than, fall, demonstrate, Y-axis, however, peaked, numerous fluctuations, about, line, stable, notice, all-time low, steep, insignificantly, rose slowly, climbed, relative stability, continuous line, dramatic decrease, dotted line, lunar new year, plunged

These _____ graphs _____ the impact of COVID-19 on movie demand and price. In graph a, the _____ represents audience size, while the _____ represents years. The _____ represents domestic movies, while the _____ represents all movies. From the outset, the demand for domestic movies was _____ the demand for movies in general. We can see that the demand for domestic as well as for all movies _____ between 2010 and 2011, then saw another increase between 2011 and 2012, and then another between 2012 and 2013. _____, between 2013 and 2014, while the demand for movies in general maintained _____, the demand for domestic movies saw a rather sharp decline. The demand for all types of movies increased _____ by the end of 2019. If we take a look at graph b, we will _____ that weekly audience size underwent _____ throughout 2019. In February 2019, for example, it _____ at around 900,000. This can be explained by the occurrence of the _____. By contrast, it only _____ to _____ 500,000 on the same occasion in January 2020, after having suffered a _____ fall. It then immediately experienced another _____ due to the COVID – 19 outbreak. It picked up slightly by mid-February, and then _____ to an _____. As for ticket prices, which are shown in graph c, we can see that these remained relatively _____ throughout 2019. Interestingly, ticket prices did not _____ with the outbreak of the crisis. Rather, the average price remained at approximately the _____.

Analyzing a Line Graph – Exercise 2

Gold to Oil Ratio



The graph is taken from Melike E. Bildirici, Fulya Ozaksoy Sonustun, "The Effects of Oil and Gold Prices on Oil-Exporting Countries," *Energy Strategy Reviews*, Vol.22, pp.290-302, Nov. 2018. Accessed on: 26.11.2018. [Online].

Available: <https://doi.org/10.1016/j.esr.2018.10.004>

Analyzing a Line Graph – Exercise 2

Fill in the blanks with words from the list below:

exponential growth, reaching, went down, tendencies, volatile, ratio, spiked, constantly fluctuated, deduce, peak, was succeeded by, rose sharply, depression, low point, climbed up, sharp decrease, plummeted, minor fluctuations, increase, high point

The line graph shows the gold to oil _____ through more than 40 years. It shows the correlation of the gold price per ounce and the oil price per barrel. Thus, it allows to _____ various _____ in the global oil market: the ratio's going up means a drop in the oil prices and vice versa. It is possible to see that from 1975 to 2015, the ratio _____, so we can conclude that the prices of oil were extremely _____. After _____ a _____ of 20 around 1980, the ratio _____, then _____, and then _____ again, a change signifying another rise in the oil prices. After coming down to a _____ of below 10 in the early 80s, the ratio _____ to approximately 16, which allows to conclude that the oil prices decreased. A period of _____ can be observed around 1985, when it _____ to 35, which means that the period was one of _____ in the oil prices. It _____ a period of _____ in the oil prices, during which they reached their highest point, reflected by the lowest gold to oil ratio. This lasted from 2005 to 2010 and was followed by a _____. After relatively _____, the gold to oil ratio reached its _____ of 40 in 2015.